The Holland Atom III: A Geometric Scaling Law for Atomic Radius from Density-Dependent Standing-Mode Resonance

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A predictive density-dependent model based on nuclear mass loading, resonance geometry and scalar-field coupling within the General Expanse Tension Theory (GETT)

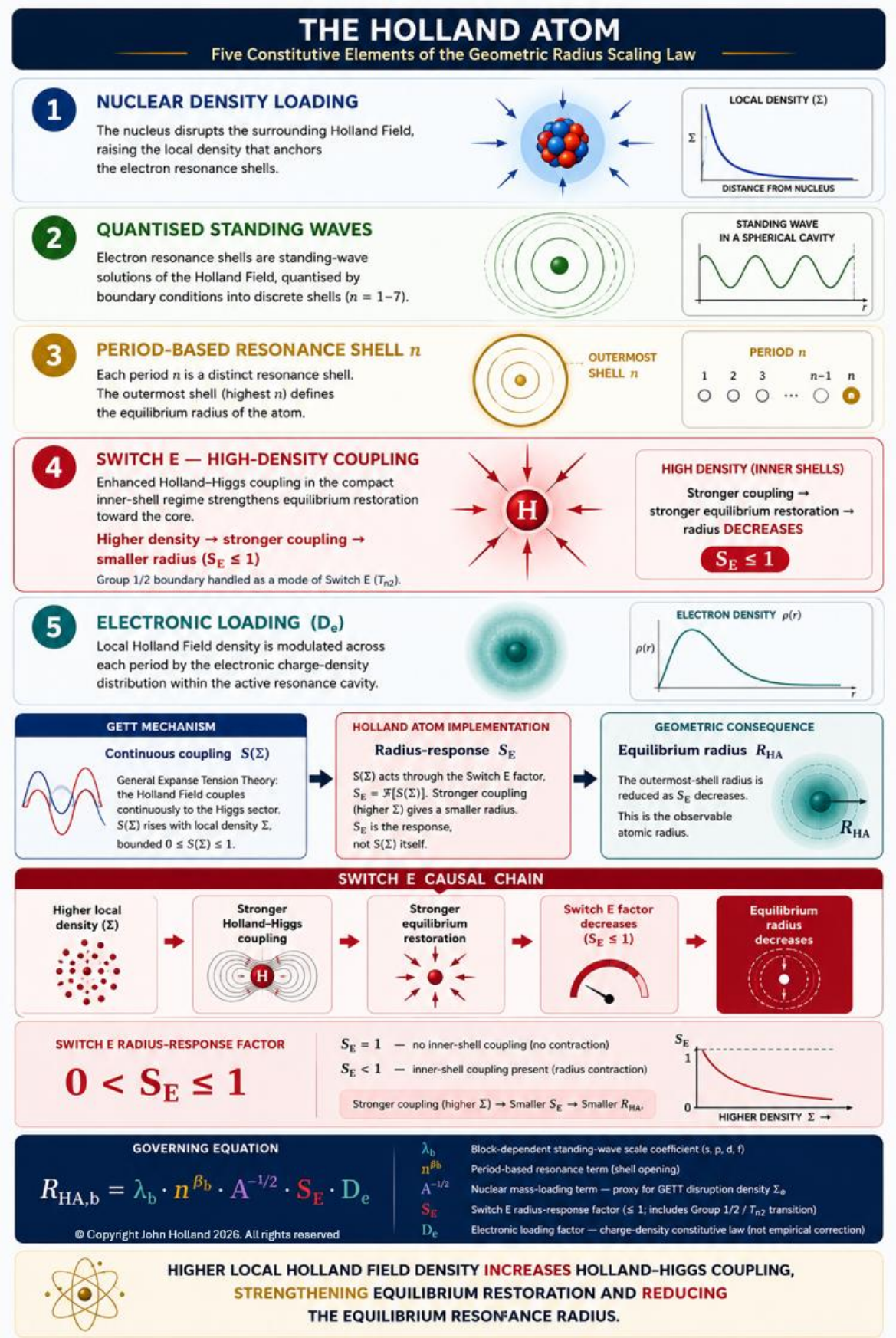
Abstract

Atomic radius is recognised as a fundamental property of matter and exhibits well-established qualitative trends associated with effective nuclear charge, shielding and electronic structure. Existing approaches have calculated or parameterised atomic radii across substantial portions of the periodic table; however, no directly comparable formulation has been identified in which whole-table empirical radius scaling emerges from a single predictive geometric law embedded within a unified physical framework. This paper presents the Holland Atom geometric radius scaling law, extending the General Expanse Tension Theory (GETT) framework introduced in Papers 1 and 2. Paper 1 established the Holland Atom as a resonance-based interpretation of atomic structure, while Paper 2 demonstrated that the multiple accepted atomic radius definitions are consistent with measurements sampling different regions of a finite resonance envelope. Building upon these foundations, Paper 3 derives a compact geometric scaling law that predicts the equilibrium Atomic Radius Empirical (ARE) from a small number of physically interpreted constitutive contributions.

A configuration-controlled master dataset comprising 96 elements from hydrogen to curium ($Z = 1$–96) with validated reference radii was constructed using independently sourced empirical atomic radii and representative atomic mass values. The frozen model reproduces the 96-element reference dataset with $R^2 = 0.9764$ and adjusted $R^2 = 0.9723$, with RMSE = 6.56 pm, MAE = 4.55 pm and MAPE = 3.28%. The complete formulation contains 14 frozen calibrated constants applied across the 96-element reference dataset. While the global goodness-of-fit is dominated by the large-scale geometric structure of the periodic table, detailed block-wise diagnostics, residual analysis and robustness studies are presented to characterise the model behaviour across the full reference dataset.

The governing equation comprises five constitutive contributions: resonance shell geometry, representative density loading, block-dependent resonance geometry, high-density coupling modulation (Switch E), and local electronic loading. Together these define the equilibrium resonance geometry of the Holland Atom. Within the present implementation, representative density loading is operationalised through the $A^{-1/2}$ scaling term. Because representative atomic mass is strongly collinear with atomic number across neutral atoms, this quantity is presented as an effective proxy for the underlying GETT disruption density, Σ_Φ, rather than as direct evidence of literal nuclear-mass causation. The causal interpretation of this density term therefore remains model-dependent and is examined in Section 7.3. Application of the frozen governing equation to the complete reference dataset demonstrates that the dominant variation in empirical atomic radius can be reproduced using this compact constitutive formulation. Collectively, these results establish the Holland Atom as a physically motivated geometric framework for describing equilibrium atomic radius and provide a reproducible foundation for future investigation of resonance geometry, electronic loading and density-dependent coupling mechanisms within atomic structure.

Keywords: Atomic radius, geometric scaling law, density-dependent resonance, standing-wave resonance, nuclear mass loading, atomic structure, periodic table, scalar field, resonance locking, empirical atomic radius, General Expanse Tension Theory (GETT), Holland Atom.



Graphical Abstract: The Holland Atom – Atomic Geometric Radius Scaling Law

1. Introduction

Atomic radius is one of the most fundamental geometric properties of matter. It governs the dimensions of atoms, influences chemical bonding, determines crystal structures and underpins many of the physical and chemical properties of materials. Despite its central importance, no single universally accepted definition of atomic radius exists. Instead, numerous experimental and theoretical radius definitions have been developed, each reflecting different measurement techniques or physical interpretations of atomic size. Consequently, atomic radius is generally treated as an operational quantity rather than the consequence of a single underlying physical mechanism.

Modern quantum mechanics and quantum chemistry provide highly successful numerical descriptions of atomic structure and enable increasingly accurate calculations of atomic and molecular properties. However, these methods generally determine atomic radius as an emergent result of solving the many-body quantum problem rather than expressing it through a single close-form geometric scaling law. Although periodic trends in atomic radius have long been recognised, periodic variation is well understood qualitatively through effective nuclear charge and electronic shielding, however expressed through fitted shielding parameters and yet not expressed as a single predictive geometric law spanning the periodic table.

General Expanse Tension Theory (GETT) proposes an alternative physical interpretation in which atomic structure emerges from resonance-locking within a density-dependent scalar field, the Holland Field, Φ. Within this framework, the atomic nucleus acts as the dominant source of local density loading, modifying the equilibrium wavelength of the surrounding field and establishing discrete standing-mode resonance cavities corresponding to the atomic shells. Atomic radius is therefore interpreted as the equilibrium geometry of the outer resonance envelope rather than as an independent empirical quantity.

The Holland Atom research programme was established to investigate this proposition. Holland Atom Paper 1 introduced the resonance-lock mechanism responsible for shell formation, shell ordering and periodic structure [1], while Holland Atom Paper 2 examined the numerous competing definitions of atomic radius and demonstrated that the Atomic Radius Empirical (ARE) provides the most appropriate experimental representation of the equilibrium resonance boundary predicted by the model [2,3]. Together, these studies established both the physical framework and the experimental reference required to investigate whether atomic radius obeys a predictive density-dependent scaling law.

The objective of the present paper is to develop and test that scaling law. A configuration-controlled master dataset was constructed using the validated radius hierarchy established in Paper 2 and representative nuclear mass loading values assigned independently for each element. Building upon the Holland Atom framework, a geometric scaling law is developed from five physically motivated components: resonance shell geometry, nuclear mass loading, block-dependent resonance geometry, high-density coupling modulation (Switch E), and local electronic charge-density loading. Representative nuclear mass loading is the selected accessible proxy for the underlying field-disruption density; see Section 7.3). The resulting model is calibrated using the frozen master dataset; goodness-of-fit is then assessed on that same reference set, with genuinely out-of-sample prediction reserved for the superheavy elements (Z = 97–118).

Application of the model to the periodic table demonstrates that the dominant variation in empirical atomic radius can be reproduced using a compact set of physically meaningful variables derived from the Holland Atom framework. Moreover, the remaining residual structure is shown to occur within identifiable physical domains rather than arising from random error, providing evidence for additional higher-order density-dependent effects and defining a clear pathway for future refinement of the model.

This paper therefore presents the first predictive geometric scaling law arising from the Holland Atom framework and is consistent with the broader proposition that atomic geometry may be described by density-dependent resonance within the Holland Field, a causal reading examined critically in Section 7.3.

The remainder of this paper is organised as follows.

- **Section 2** summarises the General Expanse Tension Theory (GETT) framework and the Holland Atom foundations established in the preceding papers, introducing the density-dependent resonance mechanism and the high-density coupling (Switch E) that underpin the present work.
- **Section 3** defines the physical framework, model parameters and postulates governing the geometric radius scaling law.
- **Section 4** describes the construction of the configuration-controlled Holland Atom Master Radius Dataset used for model calibration and validation.
- **Section 5** develops the geometric scaling law and its physically interpreted density-loading components.
- **Section 6** evaluates the predictive performance of the model against the validated periodic-table dataset.
- **Section 7** discusses the physical implications of the results, compares the model with existing approaches to atomic radius, and examines its novelty within the context of the current literature.
- **Section 8** summarises the principal conclusions. Supporting datasets, calibration procedures, statistical analyses and complete periodic-table predictions are provided in the appendices to ensure full transparency and reproducibility.

2. Foundations of General Expanse Tension Theory and the Holland Atom

The Holland Atom model is an application of the General Expanse Tension Theory (GETT), which proposes that physical phenomena emerge from the interaction between matter and a density-dependent scalar field, the Holland Field, Φ. The present paper builds upon the theoretical foundations established by the GETT hypothesis series before applying those principles to atomic structure through the Holland Atom research programme.

2.1 General Expanse Tension Theory: Density-Dependent Field Structure

General Expanse Tension Theory (GETT) proposes that physical structure emerges from interactions between matter and an all-pervading scalar field, denoted as the Holland Field, Φ. Within this framework, the field is not treated as a passive background but as a physically active medium whose equilibrium behaviour varies according to local density [11-13]. Matter therefore does not merely occupy space; it modifies the local field environment, producing measurable changes in geometry, resonance and force behaviour.

The foundational GETT hypothesis papers introduced the Holland Field as a gauge-invariant singlet scalar field capable of coupling to the Higgs sector through a density-dependent modulation function, $S(\Sigma)$. The purpose of this coupling structure is to permit field behaviour to vary across distinct physical density regimes while remaining compatible with the expectation that any additional scalar field must not produce uncontrolled effects across ordinary laboratory domains [14-18]. Under this interpretation, the strength and form of coupling are not universal constants but functions of the local density environment.

Within GETT, the Holland–Higgs coupling modulation function $S(\Sigma)$ is continuous and bounded between 0 and 1, where $S(\Sigma) = 0$ represents negligible coupling and $S(\Sigma) = 1$ represents the limiting maximum coupling permitted by the local Holland Field density state.

The Holland Atom should not be interpreted as an isolated empirical model. It represents a direct application of the density-dependent Holland Field framework previously developed within GETT. In that framework, equilibrium geometry is governed by the interaction between the naturally expanding Holland Field and matter through density-dependent Holland–Higgs coupling. The same physical mechanism has previously been applied across cosmological, galactic and gravitational phenomena. The present work therefore examines whether this identical constitutive behaviour also governs equilibrium atomic geometry, without introducing a separate atomic-specific interaction. Of particular significance is the emergence of the same density regime requiring enhanced Holland–Higgs coupling. Within previous GETT applications this transition occurs in compact ultra-high-density gravitational environments. Within the Holland Atom the corresponding transition occurs within the compact nuclear resonance regime, where the Switch E constitutive law becomes active. The successful application of the same density-dependent coupling behaviour across these independent physical scales provides internal consistency for the underlying GETT framework.

The Holland Atom programme applies this density-dependent field principle to atomic structure. If the local field wavelength is modified by nuclear density, then the geometry of atomic shells should not be arbitrary. Shell radii should scale with nuclear mass loading, shell number, block-family geometry and local electronic density. The present paper develops and tests precisely that consequence.

Within this wider framework, Paper 3 is not introduced as a standalone empirical curve fit. It is a quantitative test of a specific GETT prediction: that atomic radius is a measurable expression of density-dependent resonance geometry in the Holland Field.

2.2 Density-Dependent Scalar Coupling and “Switch E”

A fundamental prediction of General Expanse Tension Theory is that the interaction between matter and the Holland Field is not uniform throughout nature. Instead, the scalar coupling strength varies continuously with the local density environment through the density-dependent modulation function, $S(\Sigma)$, first introduced in the foundational GETT hypothesis series. Consequently, different physical regimes exhibit distinct field behaviour despite arising from the same underlying scalar field [14-18].

Within GETT, density therefore becomes the primary organising variable governing physical behaviour. As local density changes, the equilibrium properties of the Holland Field change correspondingly, giving rise to a sequence of distinct coupling regimes. These transitions, referred to as **density switches**, are predicted to occur naturally as characteristic density thresholds are crossed. Previous work has identified five principal switching regimes (Switches A–E), each associated with a different physical domain, ranging from the inflationary epoch and cosmological expansion through gravitational phenomena to the extreme densities encountered within compact astrophysical objects.

Of particular relevance to atomic structure is **Switch E**, the extreme high-density coupling regime. Switch E is predicted to operate when the local nuclear density becomes sufficiently large that the scalar-field response departs from the behaviour observed throughout the intermediate-density Einstein domain. Rather than simply continuing the same density dependence, the coupling is enhanced, modifying the equilibrium standing wavelength established around the nucleus. Within the Holland Atom framework, this enhanced coupling has a direct structural consequence, where the innermost resonance cavity, located within the highest local-density environment surrounding the nucleus, experiences the strongest equilibrium-restoring response and therefore the greatest reduction in equilibrium resonance radius relative to the outer-shell scaling behaviour. This provides a physical

explanation for the transition between the Bohr-radius regime and the larger resonance shells observed throughout the remainder of the periodic table.

Within the Holland Atom, the continuous GETT modulation function $S(\Sigma)$ is represented through the constitutive radius-response factor S_E, which translates the underlying Holland–Higgs coupling state into the corresponding modification of the equilibrium atomic radius.

Switch E is interpreted within the Holland Atom framework as the atomic manifestation of the independently proposed GETT high-density coupling regime, although its specific constitutive implementation in the present model was informed by the observed residual structure. It is an independent prediction inherited directly from the GETT framework. The present paper examines whether the inclusion of this previously predicted extreme high-density coupling regime improves the predictive accuracy of the Holland Atom geometric scaling law and whether the resulting transition is consistent with the experimentally observed dimensions of the K-shell. This relationship provides one of the principal tests of the density-dependent scalar coupling mechanism at atomic length scales.

2.3 Holland Atom Paper 1: Resonance-Lock Reconstruction of Atomic Structure

The present work builds directly upon the first paper in the Holland Atom research programme, which established the physical framework upon which the Holland Atom model is constructed [1]. Within the General Expanse Tension Theory (GETT), the atom is described as a resonance-locked standing-wave system formed within the Holland Field, Φ, rather than as a collection of orbiting point particles or purely probabilistic electron orbitals. The objective of Paper 1 was not to replace the successful predictive framework of quantum mechanics, but to propose a physically causal mechanism capable of explaining why stable atomic structure exists.

The central hypothesis introduced in Paper 1 is that the Holland Field is a density-dependent scalar field whose local resonance properties are modified by the presence of nuclear mass. The nucleus therefore acts as the dominant source of local field loading, altering the equilibrium wavelength of the surrounding field and permitting only discrete resonance-locked standing modes. These standing modes define the permitted shell structure of the atom and provide a physical basis for the quantisation of atomic geometry.

Using this framework, Paper 1 demonstrated that the observed ordering of atomic shells, shell capacities, electron filling sequence and many of the recognised anomalies of electronic configuration emerge naturally from the resonance-lock mechanism. The resulting shell architecture reproduced the principal structural features of the periodic table without introducing shell capacities or orbital ordering as independent assumptions. Instead, these properties arose from the geometry of the resonance-locked standing modes themselves.

A second important outcome of Paper 1 was the recognition that increasing nuclear loading should progressively compress the equilibrium standing wavelength of the Holland Field. Consequently, atomic radius was predicted to be neither arbitrary nor solely empirical, but instead the measurable consequence of a density-dependent resonance geometry. This prediction established the principal objective of the subsequent Holland Atom studies.

2.4 Holland Atom Paper 2: Establishing the Atomic Radius Reference Standard

The predictive development of any atomic radius model requires the selection of a single, physically meaningful reference quantity. Although the term *atomic radius* is widely used throughout chemistry, no universally accepted definition exists. Instead, numerous radius definitions have been developed, each representing a different experimental or theoretical interpretation of atomic size, including empirical atomic radius, calculated atomic radius, covalent radius, metallic radius, ionic radius, van

der Waals radius and root-mean-square charge radius. Paper 2 undertook a comprehensive review of these competing definitions, examining both their physical interpretation and the experimental methodologies by which they are determined. Rather than representing contradictory measurements, these radii were shown to probe different regions of the electronic structure and therefore different physical boundaries of the atom. Their variation is consequently an expected outcome of the finite spatial extent of the standing-mode resonance proposed in Paper 1, rather than evidence that atomic radius itself is ill-defined [2].

A detailed comparison of radius datasets demonstrated that the **Atomic Radius Empirical (ARE)** provides the most appropriate experimental representation of the resonance boundary predicted by the Holland Atom model. Unlike purely calculated radii or bonding-specific definitions, the empirical radius most consistently represents the equilibrium position of the outer resonance cavity across the periodic table and therefore provides the most suitable reference quantity for testing the density-dependent scaling law.

Paper 2 also demonstrated that the various published radius definitions occupy systematic positions within a finite resonance envelope. Inner radius definitions correspond to measurements nearer the inner standing-mode boundary, while larger radius definitions sample progressively further towards the outer amplitude of the resonance cavity. This interpretation provides a unified physical explanation for the long-recognised variation between atomic radius definitions and establishes the standing-mode envelope as the physical origin of these systematic differences.

The present paper therefore adopts the validated ARE dataset established in Paper 2 as the primary experimental reference. Where direct empirical measurements are unavailable, a strictly defined hierarchy of proxy values, validated against the relationships established in Paper 2, is employed to construct a complete configuration-controlled master dataset suitable for quantitative analysis. By fixing a single reference radius prior to model development, the present work ensures that the resulting scaling law is evaluated against a consistent and reproducible experimental standard.

2.5 Definition of the Outermost Resonance Shell

Throughout this paper, the **outermost resonance shell** is defined as the highest equilibrium resonance cavity established by the Holland Field for a given atom. This shell determines the equilibrium resonance envelope and therefore defines the characteristic Atomic Radius Empirical (ARE) predicted by the Holland Atom model.

Importantly, the outermost resonance shell should not be interpreted simply as the highest occupied spectroscopic electron shell. Instead, it represents the outermost stable resonance cavity arising from the interaction of the Holland Field with the nuclear mass loading. Electronic charge subsequently occupies the available resonance cavities according to the lowest-energy equilibrium configuration described in Holland Atom Paper 1.

Consequently, the measured ARE is interpreted as the radius of the outermost equilibrium resonance shell, irrespective of whether that shell carries stable electronic charge in the ground-state electronic configuration. This distinction becomes particularly important for anomalous electronic configurations, where charge redistribution may occur without altering the underlying resonance-shell geometry.

Unless explicitly stated otherwise, the terms **outer shell**, **outermost shell**, **highest shell**, **principal resonance shell**, and **resonance shell** are used throughout this paper as synonymous references to this outermost equilibrium resonance shell.

3. Holland Atom Physical Framework

This section defines the physical entities, parameters and postulates used throughout the derivation of the Holland Atom geometric radius scaling law. The model developed in this paper is not introduced as an unconstrained empirical fit. It is constructed from the density-dependent ontology established by General Expanse Tension Theory (GETT) and applied to atomic structure through the Holland Atom framework.

Within this framework, each term in the scaling law is intended to represent a **specific physical contribution** to atomic radius: resonance geometry, nuclear mass loading, block-family structure, Switch E high-density coupling modulation and local electronic charge-density loading. The purpose of this section is therefore to establish the physical meaning of the model before the master dataset and quantitative analysis are introduced.

3.1 Physical Ontology

The Holland Atom model treats atomic structure as the observable consequence of standing-mode resonance within the Holland Field, Φ. The central physical entities used in this paper are defined as follows.

The **Holland Field, Φ**, is the naturally expanding, density-dependent scalar field [11]. Its equilibrium state is continuous expansion, generating a restoring tendency towards lower field density. The field possesses an intrinsic capacity for continuous density-dependent modulation of its coupling to the Higgs sector (mass), allowing the interaction strength to vary smoothly with local matter density. Atomic standing-wave resonances are established within this underlying field, with local Holland–Higgs coupling governing the equilibrium resonance geometry.

A **standing mode** is a stable resonance configuration of Φ formed around the nucleus under local density loading. Standing modes are discrete because only specific resonance geometries remain stable.

A **resonance cavity** is the spatial region occupied by a standing mode. In atomic structure, each shell corresponds to a distinct resonance cavity.

A **resonance envelope** is the finite amplitude boundary of a standing mode. It defines the physical region within which electronic charge density may be supported.

An **atomic shell** is a resonance-locked cavity surrounding the nucleus. Shell number, n, identifies the principal resonance level.

The **atomic radius**, R, is interpreted as the equilibrium radius of the outer resonance envelope. In this paper, R is compared against the Atomic Radius Empirical (ARE) reference established in Paper 2.

The **local density environment** is the combined density field experienced by the resonance cavity. In the model, this is represented primarily by nuclear mass loading and secondarily by local electronic charge-density loading.

These definitions distinguish the Holland Atom model from a purely descriptive periodic trend. Atomic radius is treated as a derived geometric consequence of the field configuration rather than as an independent empirical property.

3.2 Density Hierarchy

GETT proposes that the Holland Field responds to local density through a hierarchy of loading mechanisms. In the atomic domain, the dominant density source is the nucleus. Nuclear mass loading is used to represent the primary compression of the equilibrium standing wavelength and therefore provides the principal density-loading representation within the present formulation. The next level of the hierarchy is the local electronic charge density occupying the active resonance cavity. Once the primary nuclear loading contribution is accounted for, residual structure is expected to arise from the way electronic charge density is distributed within the standing-mode envelope. This contribution is secondary because it perturbs the resonance geometry already established by the nucleus. A third contribution is the high-density coupling modulation associated with Switch E. This term is expected to operate most strongly in the innermost shell regime, where the resonance cavity forms in closest proximity to the nuclear density source. Switch E therefore modifies the relationship between the K-shell and the outer-shell scaling law. Higher-order effects are also expected at the limits of the model. These include relativistic electronic energy-density loading in heavy period-6 elements, definitional ambiguity in noble-gas radius proxies, and threshold coupling in spatially extended actinide orbitals. These effects are not included in the primary scaling law but are examined later through residual diagnostics.

The resulting density hierarchy is therefore:

1. Primary: nuclear mass-density loading
2. Secondary: local electronic charge-density loading
3. Inner-regime: Switch E high-density coupling modulation
4. Higher-order: relativistic energy-density loading, shell-envelope structure and actinide threshold coupling

This hierarchy provides the physical logic for the sequence of terms introduced in the geometric radius scaling law. Here we present the physical hierarchy of mechanisms governing equilibrium atomic geometry, in which resonance-shell opening establishes the primary cavity geometry before successive density-dependent processes act upon it; by contrast, Section 5 and Appendix D describe the computational implementation of these same mechanisms in the order required to evaluate the governing equation.

3.3 Fundamental Scaling Parameters

The scaling law developed in this paper uses the following parameters.

R is the **predicted geometric atomic radius**, expressed in picometres. It is compared against the ARE-based master radius dataset.

A is the representative **nuclear mass loading**. It is defined as the mass number of the most abundant stable isotope where available, or the longest-lived isotope where no stable isotope exists [9,10]. A is used instead of atomic weight because the Holland Atom model requires a representative nuclear mass-loading value rather than an abundance-weighted chemical average.

n is the **principal resonance-shell coordinate**. It identifies the period-based resonance cavity that defines the atom's outer equilibrium geometry. It should not be interpreted strictly as the highest occupied spectroscopic electron shell; rather, it represents the resonance-shell geometry established by the Holland Field and nuclear mass loading.

b denotes the **block family**: s, p, d or f. In the Holland Atom framework, block family represents a difference in resonance-cavity geometry rather than a merely chemical classification.

λ_b is the block-dependent **resonance wavelength coefficient**. It represents the effective scale factor associated with the standing-mode geometry of block b.

β_b is the **block-dependent shell-opening exponent**. It describes how the resonance radius expands with shell number within each block family.

S_E is the Switch E constitutive factor appearing in the governing equation. It is not the continuous GETT modulation function $S(\Sigma)$ itself, but the multiplicative **Radius-Response Factor** through which the underlying Holland–Higgs coupling state modifies the equilibrium atomic radius within the Holland Atom framework. Consequently, stronger underlying coupling (larger $S(\Sigma)$) produces a greater reduction in equilibrium radius and therefore corresponds to a smaller value of the constitutive multiplier S_E. Within the compact K/L high-density contraction regime, the Switch E Radius-Response Factor satisfies $0 < S_E \leq 1$, with stronger underlying coupling corresponding to smaller S_E and therefore smaller equilibrium radius. In the separately specified Group 1 and Group 2 outer s-block branch, S_E may exceed unity because it also incorporates the persistent s-block loading refinement; in that regime S_E should not be interpreted solely as a contraction factor.

D_e is the **electronic density-loading function**. It accounts for secondary modulation of the resonance cavity by local electronic charge density in the active shell or subshell.

Together, these parameters define the physical structure of the model, and are explored in greater details within Appendix B. The final law is therefore not a general polynomial or arbitrary periodic-table fit. It is a density-dependent resonance-scaling relation whose terms correspond to specific components of the Holland Atom ontology.

3.4 Physical Interpretation of the Scaling Law

The Holland Atom geometric radius scaling law is constructed from a sequence of physical contributions.

1. First, the atom possesses a free resonance geometry determined by shell number. Larger shell numbers correspond to **larger resonance cavities** and therefore larger possible radii.
2. Second, **nuclear mass loading compresses the equilibrium wavelength** of the Holland Field. As A increases, the local density loading supplied by the nucleus increases, **reducing the standing-mode radius** relative to the free shell geometry.
3. Third, the geometry of the **resonance cavity differs between block families**. The s-, p-, d- and f-blocks are therefore assigned distinct scaling coefficients, reflecting different cavity geometries and different ways in which the standing mode opens with shell number. Future work will assess the deeper ontological basis for these families, with the hypothesis that they may correspond to distinct resonance-cavity envelope geometries within the Holland Field, rather than merely to conventional orbital classifications.
4. Fourth, High-Density Holland–Higgs Coupling (Switch E): The compact inner resonance shells occupy the highest local Holland Field density regime and therefore experience enhanced Holland–Higgs coupling, represented in GETT by the continuous modulation function $S(\Sigma)$. As the local density increases, $S(\Sigma)$ approaches its limiting maximum coupling state. Within the Holland Atom, this increased coupling is implemented through the Switch E radius-response factor, S_E, which is the constitutive multiplier appearing in the governing equation. Stronger underlying Holland–Higgs coupling produces a stronger equilibrium-restoring response and therefore a greater reduction in the equilibrium resonance radius.

Consequently, the K-shell ($n = 1$), being closest to the nuclear mass loading, experiences the largest radius reduction, while the L-shell ($n = 2$) experiences a weaker transition response as the local density decreases. Beyond the compact inner-shell regime ($n \geq 3$), the density-dependent coupling becomes negligible, and the outer-shell geometric scaling law is recovered.

5. Fifth, local electronic charge density provides a secondary correction. Once nuclear mass loading and block geometry are accounted for, systematic residual **structure is expected to arise from the distribution of electronic charge** within the active resonance cavity.

The general form of the model can therefore be expressed as:

$$R_{\text{pred}} = \lambda_b n^{\beta_b} A^{-1/2} S_E D_e \qquad \text{Eq. 3.1}$$

where each term corresponds to a defined physical contribution: block resonance scale, shell opening, nuclear mass compression, Switch E high-density coupling modulation and local electronic density loading. This formulation establishes the Holland Atom model. Later residual analysis identifies where higher-order terms may be required, but these are not incorporated into the primary law derived in this paper.

3.5 Holland Atom Geometric Radius Scaling Law Postulates (P)

The following postulates define the formal test structure of this paper.

P1. Geometric shell scaling

Atomic shells are resonance-locked standing modes whose equilibrium radii follow systematic geometric scaling with principal shell number, n.

P2. Nuclear mass loading

Representative nuclear mass loading, A, compresses the equilibrium standing wavelength of the Holland Field. Atomic radius therefore decreases systematically with increasing nuclear mass loading after shell geometry is accounted for.

P3. Block-dependent resonance geometry

The resonance geometry differs between the s-, p-, d- and f-blocks. Consequently, each block family requires its own resonance scale and shell-opening exponent.

P4. Switch E inner-shell high-density coupling modulation

Enhanced scalar-field coupling within the Switch E density regime modifies the equilibrium wavelength of the innermost resonance shell. This produces the transition between the K-shell Bohr-radius regime and the outer-shell scaling law.

P5. Local electronic charge-density loading

After accounting for nuclear mass loading, shell geometry and block family, remaining systematic variation in radius is governed by local electronic charge-density loading within the active resonance cavity.

Together, these postulates define the structure of the Holland Atom geometric scaling law. The following sections construct the configuration-controlled master dataset and test whether the resulting model can predict empirical atomic radius across the periodic table.

4. Construction of the Configuration-Controlled Master Dataset

The predictive capability of the Holland Atom geometric scaling law depends upon the use of a single, internally consistent reference dataset representing the equilibrium atomic radius. As demonstrated in Holland Atom Paper 2, no universally accepted atomic radius exists. Instead, numerous experimentally determined and theoretically derived radius definitions sample different physical regions within the atomic shell structure. Consequently, a configuration-controlled master dataset was constructed prior to model development to provide a reproducible reference standard for all subsequent analyses.

The master dataset was assembled using the Atomic Radius Empirical (ARE) as the primary reference quantity, consistent with the conclusions of Holland Atom Paper 2. The empirical atomic radius was selected because it most closely represents the characteristic equilibrium position of the resonance envelope predicted by the Holland Atom framework. Where direct empirical values were unavailable, replacement values were assigned using the validated hierarchy established in Paper 2. The selection priority was: (i) published Atomic Radius Empirical (ARE); (ii) Calculated Atomic Radius (ARC) converted to the ARE scale using block-dependent conversion relationships; and (iii) Cordero (2008) single-bond covalent radius converted to the ARE scale where ARC values were unavailable. Every inferred value retained explicit traceability to its originating source and conversion pathway [3-10].

Representative nuclear mass loading, A, was assigned independently of the radius data. For each element, A was defined as the mass number of the most abundant naturally occurring stable isotope. Where no stable isotope exists, the longest-lived isotope was adopted. This definition reflects the physical role of the nucleus within the Holland Atom model, where the dominant density-loading contribution arises from the characteristic nuclear mass rather than the abundance-weighted atomic mass used in conventional chemistry.

Each element was additionally assigned its periodic-table period, group, orbital block (s, p, d or f), outermost equilibrium resonance shell (n), and corresponding chemical family. These descriptors are not fitting parameters but independent physical variables defining the resonance geometry within the Holland Atom framework. The completed dataset therefore represents a fully configuration-controlled description of the periodic table prior to model calibration.

The frozen master dataset contains all 118 recognised chemical elements. However, only elements possessing either direct empirical radius measurements or validated proxy values were used in calibration and statistical validation of the model. This produced a validated dataset comprising 96 elements. The remaining superheavy elements (Z = 97–118) were retained within the master dataset and subsequently evaluated as independent extrapolative predictions using the frozen geometric scaling law. They were not included in model calibration, parameter optimisation or statistical performance assessment.

To ensure complete reproducibility, every dataset entry includes the adopted master radius, source classification, confidence class, representative nuclear mass loading, and all variables required for direct application of the Holland Atom scaling law. Once assembled, the dataset was frozen prior to calibration. No radius values, isotope assignments or configuration descriptors were subsequently modified during development of the predictive model. This separation between dataset construction, model calibration and subsequent goodness-of-fit evaluation against an unchanged reference standard

forms a fundamental principle of the Holland Atom methodology and ensures that the predictive performance reported in this paper is evaluated against an unchanged reference standard.

5. Development of the Holland Atom Geometric Radius Scaling Law

5.1 Introduction

The Holland Atom framework predicts that atomic radius is not an arbitrary empirical quantity, but the measurable consequence of a resonance-locked standing mode established within the Holland Field under the influence of local density. Having established the physical framework (Section 2) and the configuration-controlled reference dataset (Section 4), the objective of this section is to develop a predictive geometric scaling law capable of reproducing the observed variation in atomic radius across the periodic table. The development proceeds by introducing successive physically motivated contributions to the equilibrium resonance geometry. Each term corresponds to a specific component of the Holland Atom framework and therefore possesses a direct physical interpretation rather than representing an empirical correction. The resulting model is calibrated using the frozen master dataset and subsequently validated without further modification.

5.2 Free Resonance Geometry

Within the Holland Atom model, atomic shells are interpreted as resonance-locked standing modes surrounding the nucleus. In the absence of density loading, the equilibrium radius of successive shells is expected to increase systematically with principal shell number. The simplest geometric relationship is therefore

$$R \propto n^{\beta} \qquad \text{Eq.5.1}$$

where n is the principal resonance shell and β characterises the rate at which the resonance cavity expands with shell number. This relationship represents the unperturbed geometric opening of the standing-mode structure.

5.3 Nuclear Mass-Density Loading

General Expanse Tension Theory predicts that local density modifies the equilibrium wavelength of the Holland Field. Within atomic structure, the dominant density source is the nucleus. Increasing representative nuclear mass loading therefore compresses the equilibrium standing wavelength and reduces the radius of the resonance cavity. The primary density-loading term is introduced as

$$R \propto A^{-1/2} \qquad \text{Eq.5.2}$$

where A is the representative nuclear mass loading defined in Appendix B. This term provides the principal contraction mechanism throughout the periodic table and constitutes the dominant physical contribution to the scaling law. This term defines the universal base nuclear-loading law; as specified in Appendix D, the Switch E constitutive law may modify the effective mass-loading exponent within specific high-density coupling regimes, most notably the outer s-block.

5.4 Block-Dependent Resonance Geometry

Although all shells arise from the same underlying standing-mode mechanism, the resonance cavity geometry differs systematically between the s-, p-, d- and f-blocks. These differences are reflected by

distinct resonance wavelength scales and shell-opening behaviour. The geometric model therefore becomes

$$R = \lambda_b \, n^{\beta_b} A^{-1/2} \qquad \text{Eq.5.3}$$

where λ_b is the characteristic resonance wavelength coefficient for block b, and β_b is the corresponding shell-opening exponent. These coefficients describe differences in resonance geometry rather than differences in physical mechanism.

5.5 Switch E Scalar-Field high-density coupling modulation

Recall from section 2 that the foundational GETT hypothesis series predicts enhanced scalar-field coupling within the extreme high-density regime, this is called ***Switch E***. Analysis of the residual structure demonstrated that a simple geometric scaling law, although highly predictive, systematically underestimates the strongest density regimes within the periodic table. Rather than treating this residual structure solely as an empirical correction, the Holland Atom framework interprets it as potentially consistent with the previously proposed Switch E density-dependent coupling regime. Instead, the coupling varies continuously with local matter density. Accordingly, Switch E is introduced as a **constitutive law**, rather than a single fitted parameter. It evaluates the local Holland–Higgs coupling state for a given atomic environment and returns the corresponding coupling multiplier applied within the governing equation. Higher local-density environments experience the strongest equilibrium-restoring response and subsequent reduction in equilibrium resonance radius. Introducing the Switch E high-density coupling modulation function gives

$$R_{HA} = \lambda_b n^{\beta_b} A^{-1/2} S_E(\cdot) D_e \qquad \text{Eq.5.4}$$

where S_E represents the enhanced scalar-field coupling operating within the K-shell regime. Unlike the geometric coefficients λ_b and β_b, which are frozen block-specific parameters, Switch E evaluates the local Holland–Higgs coupling state from the physical properties of the atomic system. Its mathematical implementation is defined in Appendix D.7, while its physical basis arises from the continuous density-dependent coupling of the naturally expanding Holland Field to matter. This term provides a continuous physical transition between the Bohr-radius domain and the outer-shell scaling behaviour predicted for higher resonance shells.

Switch E should not be interpreted as exclusively an inner-shell correction. Its strongest radius response occurs in the compact $n = 1$ regime, with the $n = 2$ transition factor representing the rapid reduction of this effect with increasing shell distance. The frozen constitutive law additionally contains a persistent s-block-specific loading response for Group 1 and Group 2 elements across $n \geq 3$. This latter contribution is interpreted as a property of the distinctive s-block coupling environment and should be distinguished from the spatially localised K-shell contraction. Accordingly, Switch E represents the complete high-density and s-block coupling-response architecture encoded by S_E, with different functional manifestations in its specified regimes. The universal base density-loading dependence remains $A^{-1/2}$; however, where S_E itself contains an A-dependent constitutive response, the effective dependence of the complete radius law is correspondingly modified. For outer Group 1 and Group 2 elements, the combined dependence is therefore $A^{-\gamma_s}$, where $\gamma_s = 0.327672732$, rather than $A^{-1/2}$. This distinction is explicit in the frozen implementation and does not alter the governing equation.

Shell	Holland-Higgs Coupling $S(\Sigma)$	Radius-Response Factor S_E	Physical effect
$n = 1$	Highest	≈ 0.23	Strong radius reduction
$n = 2$	Intermediate	≈ 0.92	Small radius reduction
$n \geq 3$	Weak / inactive	1.00	No inner shell Switch E reduction*
Outer Group ½ s-block	Separate constitutive branch	May exceed 1	Modifies effective loading response

Table 1. Holland-Higgs Coupling $S(\Sigma)$ and Radius-Response Factor S_E

***** *Except for the separately defined Group 1/2 outer s-block constitutive response.*

Table 1 introduces Switch E as a constitutive law describing high-density Holland–Higgs coupling modulation, rather than as a single amplification coefficient. In the K-shell it produces the compact limiting resonance regime; in the $n = 2$ shell it acts as a transition multiplier; and for $n \geq 3$ the outer-shell scaling law is recovered except where specific s-block loading refinements apply.

Therefore, the impact of the closer distance to the compact nuclear density source is to reduce element radius, as follows;

Higher local density → $S(\Sigma)$ ↑ → Holland–Higgs coupling ↑ → Equilibrium restoration ↑ → S_E ↓ → R_{HA} ↓

5.6 Electronic Loading: Charge-Density Constitutive Law

Following incorporation of standing-wave geometry, nuclear mass loading and the Switch E constitutive law, systematic residual structure remained within specific electronic regions of the periodic table. The most prominent deviations occurred at the onset of the p-block and throughout the d-block transition series, where the residuals evolved systematically with electronic occupancy. Within the Holland Atom framework, these residuals are interpreted as evidence that the equilibrium resonance geometry is influenced not only by nuclear loading, but also by the spatial distribution of electronic charge surrounding the standing-wave cavity. Variations in local charge density alter the equilibrium Holland Field configuration, producing small but systematic modifications to the predicted atomic radius. Accordingly, Electronic Loading is introduced as a constitutive law, rather than a fitted coefficient. The constitutive law evaluates the occupancy-dependent modification to the equilibrium resonance geometry associated with the ground-state electronic configuration and returns the corresponding loading factor applied within the governing equation. The governing equation therefore becomes

$$R_{HA} = \lambda_b n^{\beta_b} A^{-1/2} S_E(\cdot) D_e(\cdot) \qquad \text{Eq.5.5}$$

where $D_e(\cdot)$ represents the Electronic Loading constitutive law.

This Electronic Loading constitutive law evaluates the local perturbation arising from the electronic charge-density distribution. Within the implementation this behaviour is expressed through occupancy-dependent loading functions for the p- and d-blocks, while the s- and f-blocks require no additional electronic loading beyond the governing standing-wave geometry. The resulting formulation preserves the physical interpretation of every term within the governing equation. The complete mathematical specification of the Electronic Loading constitutive law, including its frozen implementation, is provided in Appendix D.8.

5.7 Calibration Strategy

Calibration was performed using the frozen configuration-controlled master dataset described in Section 4. The calibration procedure followed a fixed sequence designed to preserve the physical interpretation of each model component.

First, the block-dependent resonance coefficients were determined independently for each orbital block. The nuclear mass-loading relationship was then calibrated using the representative isotope dataset. Switch E was subsequently introduced to account for the predicted inner-shell high-density coupling modulation, after which the remaining systematic residuals were analysed to determine the electronic charge-density loading function.

At no stage were coefficients adjusted to improve agreement with individual elements or specific regions of the periodic table. The calibration therefore preserved a one-to-one correspondence between physical mechanism and mathematical term. The Holland Atom model combines standing-wave geometry, representative nuclear mass loading, density-dependent Holland–Higgs coupling and electronic charge-density loading within a single governing equation describing the equilibrium geometric radius of neutral atoms.

5.8 Final Geometric Radius Scaling Law

The resulting Geometric Radius Scaling Law is

$$R_{HA,b} = \lambda_b n^{\beta_b} A^{-1/2} S_E D_e \qquad \text{Eq.5.6}$$

where:

- $\boldsymbol{R_{HA,b}}$ is the predicted equilibrium geometric radius for an element belonging to electronic block b;
- $\boldsymbol{\lambda_b}$ is the frozen standing-wave geometry coefficient for electronic block b;
- $\boldsymbol{\beta_b}$ is the frozen shell expansion exponent defining the opening of the resonance cavity within electronic block b;
- $\boldsymbol{n}$ is the period-based principal resonance-shell coordinate defining the outer resonance cavity used in the geometric scaling law;
- $\boldsymbol{A}$ is the representative nuclear mass number of the selected reference isotope;
- $\boldsymbol{S_E}$ is the Switch E constitutive law, which evaluates the density-dependent Holland–Higgs coupling state appropriate to the local atomic environment;
- $\boldsymbol{D_e}$ is the Electronic Loading constitutive law, which evaluates the occupancy-dependent modification arising from the electronic charge-density distribution.

The quantities λ_b and β_b are frozen model parameters obtained during the model calibration and remain fixed for all subsequent predictions. The representative nuclear mass number A is an independently assigned physical property and is not determined by model fitting. The terms S_E and D_e are constitutive laws, not empirical fitting coefficients, and are evaluated from the physical state of the atom according to the mathematical formulations defined in Appendix D. The complete mathematical specification of the governing equation is provided in Appendix D.4. The frozen parameter register is defined in Appendix D.6. The Switch E constitutive law is specified in Appendix D.7, while the Electronic Loading constitutive law is defined in Appendix D.8. The complete computational implementation is provided in Appendix D.11.

The resulting formulation represents the final frozen model implementation of the Holland Atom Geometric Radius Scaling Law. All validation statistics reported in Section 6 are generated directly from this governing equation and its associated constitutive laws without subsequent modification of the model, parameters or reference dataset.

5.9 Final Geometric Scaling Law Computational Architecture

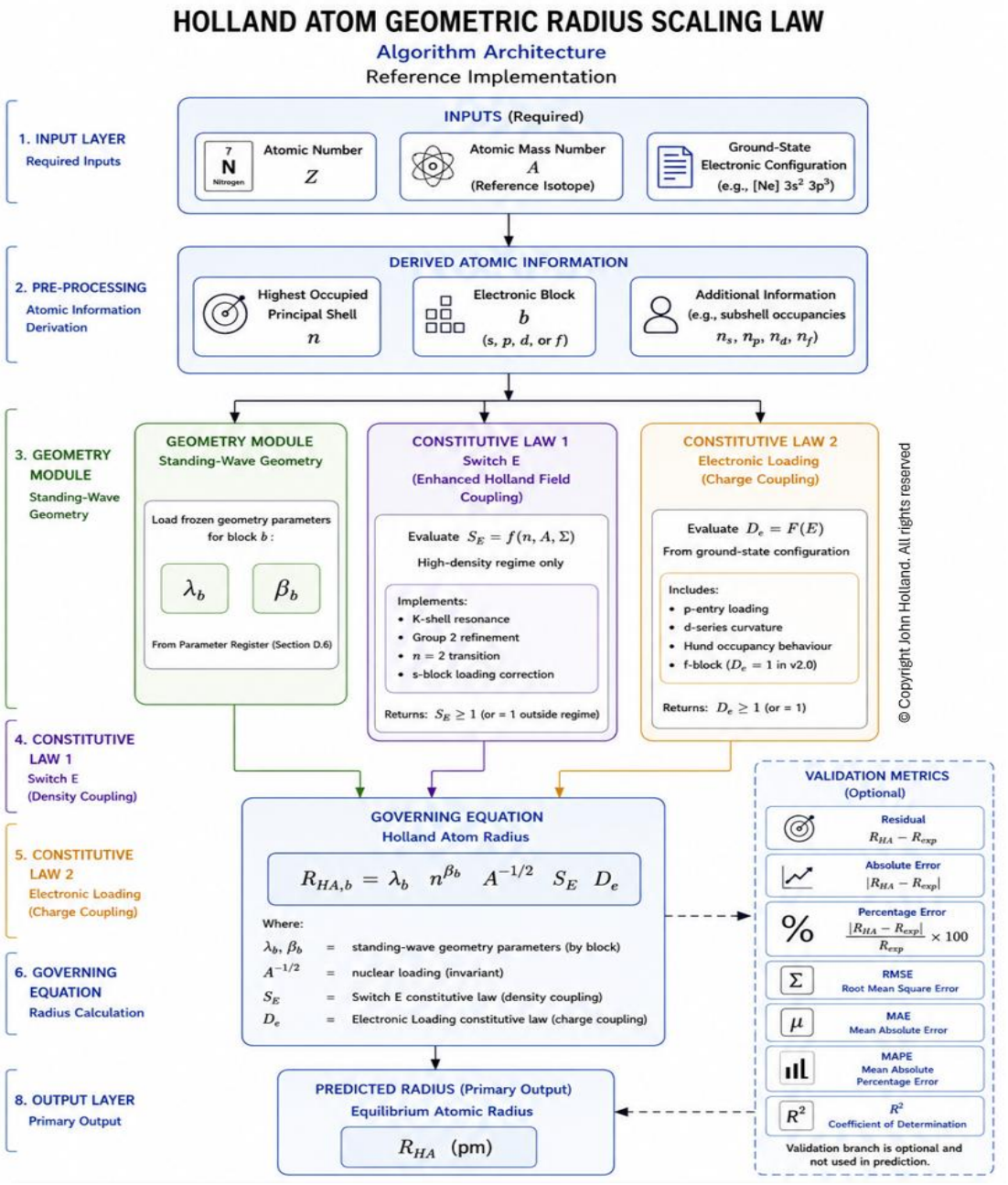


Figure 1. Computational architecture of the Holland Atom Geometric Radius Scaling Law.

Figure 1 shows the scaling law model implementation is organised as a layered computational framework. Atomic number, reference isotope and ground-state electronic configuration define the model inputs. Standing-wave geometry is determined by the frozen geometric parameters (λ_b, β_b), while the two constitutive laws – Switch E and Electronic Loading – evaluate density-dependent Holland Field coupling and electronic charge-density modulation, respectively. These components

combine through the governing equation to predict the equilibrium atomic radius. Validation metrics are calculated only after prediction and play no role in parameter evaluation or model calibration.

6. Statistical Performance and Model Evaluation

6.1 Introduction

The objective of this section is to evaluate the predictive performance of the Holland Atom geometric radius scaling law using the configuration-controlled master dataset defined in Section 4. Validation was performed using the frozen model developed in Section 5. Following calibration, no coefficients, parameters or dataset entries were modified. The reported performance therefore represents an independent assessment of the predictive capability of the model. The complete element-by-element predictions are presented in Appendix F, while the full statistical validation and diagnostic analyses are provided in Appendix H.

The statistical performance reported for the 96-element reference population represents goodness-of-fit evaluated on the same dataset used to calibrate the model parameters and should therefore not be interpreted as independent out-of-sample validation. Model generalisability is assessed separately through cross-validation, while application of the frozen governing equation to Z = 97–118 represents extrapolation beyond the calibration domain for which equivalent empirical reference radii are not presently available.

6.2 Model Ablation and Incremental Explanatory Performance

To assess whether the performance of the Holland Atom Geometric Radius Scaling Law arises primarily from model complexity, or whether successive physically motivated components provide independently identifiable improvements in explanatory performance, a hierarchical model-ablation analysis was undertaken. The purpose of this analysis was to quantify the contribution of each major component of the model architecture and to provide a transparent comparison against substantially simpler baseline formulations.

All reduced models were evaluated against the same 96-element reference-radius population employed for model calibration. Beginning with a minimal two-parameter representation of global shell geometry, successive physical components were introduced in the order corresponding to the development of the Holland Atom model: nuclear mass-density scaling, electronic block-dependent standing-wave geometry, the Switch E high-density constitutive law, and finally the Electronic Loading constitutive law. At each reduced stage, the parameters available to that model architecture were re-optimised against the same reference population, while mechanisms belonging to subsequent stages were excluded. The final stage was not re-fitted and represents the frozen 14-parameter Holland Atom reference implementation.

The simplest baseline model, containing only a global radius scale and shell expansion exponent, $R = \lambda n^{\beta}$, accounted for $R^2 = 0.5891$ of the observed variance. Introduction of the invariant nuclear loading relationship, $A^{-1/2}$, increased this to $R^2 = 0.7633$. Allowing the standing-wave geometry to vary according to the four electronic block families increased performance further, with the block-dependent geometry and nuclear-loading model reaching $R^2 = 0.9162$ and an adjusted $R^2 = 0.9084$. The subsequent constitutive refinements produced further substantial reductions in unexplained residual structure. Introduction of Switch E increased R^2 from 0.9162 to 0.9647 and reduced RMSE from 12.37 pm to 8.03 pm. Addition of the Electronic Loading constitutive law produced the final frozen-model performance of $R^2 = 0.9764$, adjusted $R^2 = 0.9723$, and RMSE = 6.56 pm.

The statistical improvement associated with the $A^{-1/2}$ term should not be interpreted as independent evidence of nuclear-mass causation, because A and Z are almost perfectly collinear across the reference population and substitution of Z produces statistically comparable performance. The term is therefore treated here as an empirical density-loading proxy, with causal discrimination reserved for isotope-resolved testing at fixed Z.

The ablation analysis therefore demonstrates that the final statistical performance does not emerge solely from the number of calibrated constants. Instead, explanatory performance increases hierarchically as successive physically defined mechanisms are introduced. The substantial performance already achieved by block-dependent standing-wave geometry and invariant nuclear loading establishes the underlying geometric scaling structure, while Switch E and Electronic Loading account for additional systematic radius behaviour associated respectively with the high-density inner-shell regime and occupancy-dependent electronic loading. The relatively small difference between the final $R^2$and adjusted R^2, after accounting for the 14 calibrated constants, further indicates that the final explanatory performance remains strong after model-complexity adjustment.

This analysis does not, by itself, establish the physical correctness or uniqueness of the proposed mechanisms; rather, it demonstrates quantitatively what additional explanatory performance is contributed by each successive component of the Holland Atom model. It therefore provides a direct test of model architecture and transparency, distinguishing the contribution of the underlying scaling law from that of its physically motivated constitutive refinements.

Level	Model	Parameters	R^2	Adjusted R^2	RMSE (pm)	MAE (pm)	MAPE
1	Global shell geometry: λn^{β}	2	0.5891	0.5803	27.39	19.67	15.59%
2	+ $A^{-1/2}$density proxy	2	0.7633	0.7582	20.79	14.27	11.71%
3	Block-dependent geometry only: $\lambda_b n^{\beta_b}$	8	0.8807	0.8697	14.76	10.39	10.08%
4	Block geometry + $A^{-1/2}$	8	0.9162	0.9084	12.37	8.25	7.68%
5	+ Switch E	12	0.9647	0.9596	8.03	5.59	4.07%
6	Full frozen model + D_e	**14**	**0.9764**	**0.9723**	**6.56**	**4.55**	**3.28%**

Table 2. Model-ablation analysis of the Holland Atom Geometric Radius Scaling Law across the 96-element reference-radius population.

Table 2 shows the reduced model architectures were independently re-optimised against the same reference dataset while progressively introducing nuclear loading, electronic block-dependent standing-wave geometry, Switch E and Electronic Loading. The final row represents the frozen 14-parameter Holland Atom reference implementation and was not re-fitted for the ablation analysis. R^2, adjusted R^2, RMSE, MAE and MAPE are reported to show both explanatory performance and the effect of increasing model complexity. The progressive improvement demonstrates the incremental statistical contribution associated with each successive physical component of the model architecture. ***Note:*** *Reduced models in this ablation analysis were re-optimised by least squares in absolute radius space. The core-law LOOCV reported in §6.4 uses the same functional equation but a log-linear calibration procedure; its baseline goodness-of-fit values therefore differ from Level 4 above.*

6.3 Prediction Performance

Application of the geometric scaling law to the validated master dataset demonstrates that the principal variation in atomic radius across the periodic table is reproduced using a small number of physically motivated variables derived from the Holland Atom framework.

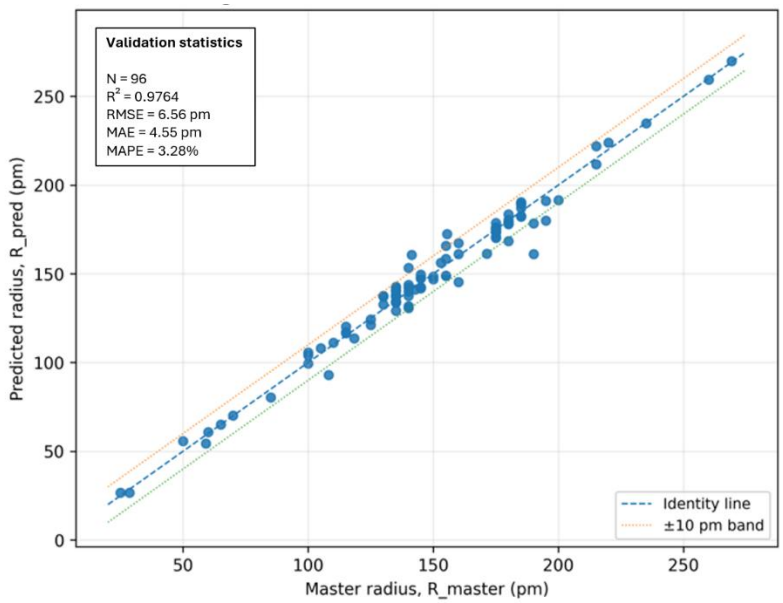


Figure 2. Comparison of the Holland Atom predicted equilibrium geometric radius with the reference master atomic radius

Figure 2. *Compares predicted values with the reference master atomic radius for the 96-element validation dataset. The dashed diagonal represents the identity relation ($R_{pred} = R_{master}$). The close clustering of the data about the identity line demonstrates the high predictive accuracy of the Holland Atom Geometric Radius Scaling Law across the validated periodic table. Figure 2 demonstrates that the Holland Atom model reproduces the global geometric structure of the periodic table without evidence of large-scale systematic bias, with the remaining structured residuals confined to a small number of identifiable physical regimes discussed in Section 7. The dashed guide bands represent ±10 pm, corresponding to approximately 4% of the full dynamic range of the validation dataset and only a small fraction of a typical atomic diameter. Eighty-six of the 96 elements (89.6%) lie within the ±10 pm of the adopted reference radius. The close clustering of the predictions within this envelope demonstrates the ability of the Holland Atom model to reproduce atomic radii with remarkably small absolute deviations over the complete periodic table.*

The model successfully captures the systematic dependence of atomic radius on shell number, representative nuclear mass loading, orbital block geometry, Switch E high-density coupling modulation and local electronic charge-density loading. The principal validation statistics are summarised in Table 3.

Statistic	Value
Elements analysed	96
Coefficient of determination (R^2)	0.9764
Adjusted R^2	0.9723
Pearson correlation significance	$p = 2.61 \times 10^{-78}$
Overall regression/F-test significance	$p \approx 1 \times 10^{-59}$
Permutation significance	$p = 5.00 \times 10^{-5}$
RMSE	6.56 pm
MAE	4.55 pm
MAPE	3.28%

Table 3. Summary of Geometric Scaling Law Model Prediction Performance

Table 3 summarises the model prediction performance including the Pearson p-value, which reports the significance of the predicted-versus-reference radius correlation, while the regression/F-test and permutation statistics provide complementary tests of global model significance and chance reassignment, respectively.

The Pearson and regression significance statistics confirm that the predicted and reference radii are strongly associated across the full periodic-table radius range. These statistics are reported as supplementary goodness-of-fit diagnostics rather than as evidence for the physical mechanism or causal identity of the density-loading variable. Model performance is therefore assessed primarily through R^2, adjusted R^2, absolute prediction errors, block-wise diagnostics and cross-validation.

6.4 Leave-One-Out Cross-Validation of the Core Scaling Law

To assess the generalisability of the underlying geometric scaling architecture independently of the subsequently introduced Switch E and Electronic Loading constitutive refinements, leave-one-out cross-validation (LOOCV) was performed on the core block-dependent scaling law,

$$R = \lambda_b n^{\beta_b} A^{-1/2}$$

For each of the 96 elements in the reference population, the element was withheld, the block-specific parameters λ_b and β_b were recalibrated using the remaining observations, and the withheld radius was predicted. This procedure was repeated until every element had served once as the out-of-fold observation. The core law achieved an in-sample $R^2 = 0.877$, RMSE = 14.97 pm and MAPE = 6.96%. Under LOOCV, performance remained substantial at $R^2 = 0.833$, RMSE = 17.44 pm and MAPE = 8.52%. The moderate degradation indicates that the principal block- and shell-dependent scaling structure generalises beyond individual calibration observations and is not solely dependent upon reproduction of the complete reference population.

Generalisation was not uniform across electronic blocks. The p-block retained strong out-of-fold correspondence ($R^2 = 0.922$; MAPE = 6.1%), whereas the s-block was more sensitive to individual observations ($R^2 = 0.707$; MAPE = 30.7%), particularly at the boundaries of its shell range. The d- and f-blocks exhibited comparatively low within-block cross-validated R^2 values (0.171 and 0.401, respectively), despite relatively small absolute percentage errors, reflecting the narrow-observed radius variation within these blocks and the limited ability of the core law to discriminate their smaller element-to-element fluctuations.

For the cross-validation analysis, the core scaling law was calibrated in logarithmic form, $\ln\left(R\sqrt{A}\right) = \ln \lambda_b + \beta_b \ln n$, separately within each block. This fitting basis minimises logarithmic rather than absolute radius-space error and therefore differs from the absolute-radius least-squares optimisation used for the reduced models in the Table 2 ablation analysis. Consequently, although the functional form is identical, the corresponding in-sample performance statistics are not expected to be identical. The log-space calibration was applied consistently within every leave-one-out fold so that the reported LOOCV metrics compare like with like.

Note: The reduced models in this ablation analysis were re-optimised by least squares in absolute radius space. The core-law LOOCV reported in §6.4 uses the same functional equation but a log-linear calibration procedure; its baseline in-sample statistics therefore differ from Level 4.

Hydrogen represents the strongest out-of-fold exception. When H is withheld, the remaining s-block calibration population contains no $n = 1$ element, requiring extrapolation of the core scaling relationship below its calibrated shell range. The resulting large error reinforces the distinction between the compact K-shell regime and the outer-shell scaling law; however, because the Switch E K-shell constitutive behaviour is not included in this cross-validation analysis, this result should not be interpreted as a cross-validation failure of the complete frozen model.

This LOOCV analysis therefore evaluates the generalisability of the **core geometric scaling law only**. The complete 14-parameter frozen formulation, including Switch E and Electronic Loading, remains characterised by in-sample goodness-of-fit on the 96-element calibration population. A strict full-model LOOCV would require every calibrated constitutive parameter to be independently identifiable and re-estimated within each fold; this condition is not necessarily satisfied in sparsely populated regimes such as the two-element K-shell domain.

6.5 Periodic-Table Behaviour

Inspection of the predicted periodic table demonstrates that the model reproduces the principal geometric behaviour of the atomic shells throughout the s-, p-, d- and f-blocks. The dominant trends associated with shell opening, periodic contraction and block-dependent geometry emerge naturally from the density-dependent resonance framework without introducing element-specific parameters or empirical exceptions. Prediction accuracy remains consistently high across the majority of the periodic table. The strongest agreement is observed throughout the main-group elements, transition metals and lanthanide series, demonstrating that the dominant contribution to atomic radius is well described by the combined effects of resonance geometry and nuclear mass loading. Block-by-block and period-by-period performance statistics are presented in Appendix H.

6.6 Residual Structure Observations

Although the overall prediction accuracy is high, the residuals are not completely random. Instead, a small number of systematic patterns remain after the primary density-loading relationship has been removed. Residual analysis demonstrates that prediction errors are generally small and centred about zero. Systematic residual structure is observed within the heavy period-6 p-block, selected actinides and noble-gas proxy radii. These observations are discussed in Section 7.

6.7 Boundary Conditions

These principal boundary conditions were identified during validation.

The first concerns the noble gases, for which direct empirical atomic radius measurements are generally unavailable. Their adopted master radii therefore represent validated proxy values rather than direct observations, introducing an unavoidable uncertainty into comparison with the model.

The second concerns the heavy period-6 p-block elements, where relativistic effects become increasingly significant. These elements exhibit systematic deviations consistent with additional electronic energy-density contributions not included within the present non-relativistic formulation.

The third concerns Curium (Cm), which represents the only significant deviation within the actinide series. Its unique electronic configuration suggests the onset of an additional density-loading contribution associated with the spatial extension of the 5f orbitals.

Finally, the residual analyses indicate that shell-envelope structure and higher-order electronic density effects may contribute to the remaining systematic variation. These effects are intentionally excluded from the present model in order to preserve the simplicity and physical transparency of the primary scaling law.

6.8 Extrapolation Beyond the Calibration Domain

The validated model was subsequently applied to elements beyond the calibrated dataset. Predicted radii for elements Z = 97–118 were generated using the frozen model parameters without further calibration or adjustment. These predictions are presented separately in Appendix G to distinguish

extrapolated results from the experimentally validated dataset. This separation preserves the distinction between model validation and model prediction while providing quantitative estimates for the currently unvalidated superheavy region of the periodic table.

6.9 Summary

The Holland Atom geometric scaling law successfully reproduces the dominant variation in atomic radius across the validated periodic table using a compact set of physically motivated variables derived from General Expanse Tension Theory. The combination of shell geometry, nuclear mass loading, block-dependent resonance behaviour, Switch E high-density coupling modulation and local electronic charge-density loading explains approximately 98% of the observed variance within the configuration-controlled master dataset.

Equally important, the remaining residual structure is not random. Instead, it identifies specific physical domains in which higher-order density-dependent effects are expected to become significant. These boundary conditions define a clear and testable programme for future development while reinforcing the central conclusion of the present work: atomic radius can be described within the Holland Atom framework by density-dependent resonance geometry governed by the interaction between nuclear loading and the Holland Field.

7. Discussion

7.1 A Density-Dependent Description of Atomic Geometry

The principal result of this study is the demonstration that atomic radius may be described by a compact geometric scaling law derived from the density-dependent framework of General Expanse Tension Theory. The calibrated model reproduces the dominant variation in empirical atomic radius across the validated periodic table using a small number of physically motivated parameters representing shell geometry, representative nuclear mass loading, block-dependent resonance structure, high-density coupling and local electronic charge-density loading.

Unlike conventional empirical correlations, each component of the Holland Atom scaling law possesses a direct physical interpretation. The model therefore represents more than a numerical relationship between periodic-table variables; it proposes a causal framework in which atomic radius emerges as the equilibrium geometry of a resonance-locked standing mode established within the Holland Field.

7.2 The Primacy of Density

A central prediction of General Expanse Tension Theory is that density governs the response of the Holland Field across all physical scales. The present study provides quantitative support for this principle within the atomic domain. Representative nuclear mass loading is found to provide the dominant contribution to atomic radius, consistent with the hypothesis that increasing local density compresses the equilibrium standing wavelength of the field. The subsequent improvement obtained by introducing local electronic charge-density loading suggests that the standing-mode geometry is sensitive not only to the nucleus but also to the distribution of charge occupying the resonance cavity. This hierarchy of density contributions is consistent with the broader GETT framework, in which progressively smaller and more localised density sources produce progressively smaller perturbations to the field geometry.

7.3 Statistical Validation, Model Robustness and Residual Structure

The complete statistical validation of the Holland Atom Geometric Radius Scaling Law is presented in Appendix H. Collectively; these analyses demonstrate that the model provides a statistically robust description of the validated atomic radius dataset while maintaining a compact physically motivated formulation. Predicted-versus-reference comparisons exhibit close agreement across the validated periodic table, with the corresponding statistical metrics demonstrating that the dominant systematic variation in empirical atomic radius is reproduced by a small number of physically interpreted mechanisms. Residual analysis further shows that prediction errors remain centred close to zero with no evidence of widespread systematic bias or uncontrolled divergence.

The statistical diagnostics provide additional evidence regarding model robustness. Rather than being distributed uniformly throughout the periodic table, the remaining residuals are confined to a limited number of identifiable physical regimes. These include the noble gases, selected heavy period-6 p-block elements and isolated actinide configurations. The localisation of the residual structure indicates that the governing scaling relationship successfully captures the dominant behaviour across the periodic table while highlighting a small number of domains in which additional physical effects become measurable.

Taken together, the statistical validation supports two conclusions. First, the Holland Atom Geometric Radius Scaling Law provides a statistically robust description of the dominant variation in empirical atomic radius. Second, the remaining residual structure is systematic rather than random, providing clear direction for future physical refinement. The detailed residual analyses supporting these conclusions are presented in Appendix H, while their physical interpretation is discussed in Section 7.5.

An important limitation of the present dataset is the near-total collinearity between representative atomic mass A and atomic number Z across the neutral periodic table ($r \approx 0.999$). Consequently, the statistical performance of the present scaling law cannot be interpreted as evidence that nuclear mass is uniquely responsible for the observed radius dependence. Substitution of Z for A as the loading coordinate produces statistically comparable model performance, demonstrating that the present 96-element dataset is structurally incapable of discriminating between mass-related and charge-related causal interpretations. Within the Holland Atom formulation, A is therefore retained as the frozen empirical proxy for the underlying GETT disruption-density state Σ_{Φ}, rather than as evidence of literal nuclear-mass causation. Resolving this degeneracy requires differential systems in which nuclear mass and charge vary independently. Isotope-resolved measurements, for which Z remains fixed while neutron number, nuclear mass, binding energy and associated nuclear properties vary, therefore provide a direct next-stage empirical test of the proposed density-dependent coupling mechanism.

$$\boxed{\text{Neutral elements: } A \sim Z \;\Rightarrow\; \text{causality unresolved}}$$

$$\boxed{\text{Isotopes: } Z = \text{constant, } A \text{ varies} \;\Rightarrow\; \text{direct discrimination}}$$

The resulting isotope programme is explicitly falsifiable: if variation in the proposed GETT disruption-density coordinate at fixed Z produces no additional measurable contribution beyond established mass-shift, field-shift and higher-order Standard Model effects, the corresponding form of the Holland Field coupling hypothesis would be constrained or excluded. This is the planned focus for future work.

7.4 Implications for Atomic Structure

The Holland Atom model provides a different physical interpretation of atomic radius from that traditionally adopted in chemistry. Within the conventional description, atomic radius is an operational quantity whose value depends upon the measurement method employed. The Holland Atom does not dispute these measurements. Instead, it proposes that the various accepted radius definitions sample different regions of a finite standing-mode resonance envelope.

Under this interpretation, empirical atomic radius represents the characteristic equilibrium position of the resonance envelope, while alternative radius definitions probe systematically different radial domains within the same physical structure. This interpretation follows directly from the experimental analysis presented in Holland Atom Paper 2 and provides a unified explanation for the long-recognised coexistence of multiple accepted radius systems.

GETT proposes that the resonance geometry of the atom is set by the Holland Field. It does not currently claim to derive electromagnetism; electron–electron repulsion is retained as given. Whether a scalar substrate can underlie gauge interactions is a separate and much harder question, addressed elsewhere in the programme.

7.5 Interpretation of the Residual Structure

The residual analyses demonstrate that the remaining deviations are not randomly distributed throughout the periodic table but instead occur within identifiable physical regimes.

For the noble gases, the dominant limitation arises from the experimental reference dataset rather than the Holland Atom model itself, since the adopted master radii are necessarily proxy values in the absence of directly measurable empirical atomic radii.

Within the s-block, the residuals distinguish the alkali and alkaline-earth metals, suggesting a secondary contribution associated with local electronic charge density. The d-block exhibits a systematic occupancy-dependent curvature consistent with the influence of electronic loading, whereas the majority of the f-block remains comparatively uniform.

The principal remaining deviations occur within a small number of heavy period-6 p-block elements, where the increasing importance of relativistic electronic energy density is likely to contribute beyond the present formulation. Similarly, Curium provides the first indication that extended actinide orbitals may introduce additional density loading not yet incorporated within the current model.

These structured residuals are scientifically significant because they define specific physical domains in which the present formulation begins to reach its limits. Rather than representing random statistical error, they identify measurable higher-order physical effects that provide a natural direction for future development of the Holland Atom framework.

7.6 Physical Basis of the Governing Equation

The progressive development of the Holland Atom Geometric Radius Scaling Law should not be interpreted as a sequential empirical fitting exercise. Each component of the governing equation was introduced to represent a distinct physical mechanism predicted by the underlying GETT framework, rather than to improve statistical agreement alone.

Standing-wave geometry provides the primary resonance structure of the atom. Representative nuclear mass loading follows directly from the density-dependent modulation of the Holland Field by mass. The Switch E constitutive law arises from the prediction that Holland–Higgs coupling varies

continuously with local matter density, with enhanced coupling expected within the highest-density resonance regimes. Finally, the Electronic Loading constitutive law represents the perturbation of the equilibrium resonance geometry arising from the spatial distribution of electronic charge density.

A useful distinction emerges for anomalous configurations such as Palladium. In the conventional spectroscopic description, Pd adopts a $4d^{10}5s^{0}$ ground-state configuration. Within the Holland Atom radius law, however, Pd remains a period-5 resonance system because n denotes the resonance-shell coordinate rather than the literal highest occupied electronic shell. This distinction suggests that resonance-shell geometry is established by the Holland Field and nuclear mass loading, while electronic charge subsequently redistributes within that geometry to reach the lowest-energy configuration. The radius law therefore predicts the equilibrium resonance envelope, not simply the occupied electronic charge distribution.

Importantly, these mechanisms were not developed in isolation for atomic physics. They represent direct applications of the same density-dependent Holland Field ontology previously applied across cosmological, galactic, gravitational and terrestrial phenomena within the GETT framework. In particular, the density regime associated with the onset of enhanced Holland–Higgs coupling within the atomic nucleus is consistent with the high-density coupling behaviour independently inferred for compact astrophysical objects. Within the Holland Atom, this same constitutive behaviour naturally explains the transition between the compact K-shell regime and the approximately linear scaling observed for higher valence-shells progression, providing a physical interpretation of the long-recognised anomaly.

Accordingly, the mathematical complexity of the governing equation reflects the incorporation of multiple physically distinct constitutive mechanisms rather than successive empirical correction terms. Every term possesses an explicit ontological interpretation, a defined mathematical specification and an independently documented role within the Holland Atom framework.

7.7 Relationship to Existing Atomic Radius Models and Scientific Novelty

Atomic radius has been the subject of extensive theoretical and experimental investigation for more than a century. Modern chemistry possesses highly developed methods for calculating atomic structure, while numerous experimental and semi-empirical radius datasets have been compiled using a wide range of measurement techniques. Likewise, the periodic variation of atomic radius has long been understood qualitatively through the combined influence of shell structure, effective nuclear charge and electronic shielding. These approaches have provided an exceptionally successful framework for describing periodic behaviour and remain fundamental to contemporary atomic and quantum chemistry [7].

The present work is not intended to challenge these established successes. Rather, it addresses a different scientific question: **whether the equilibrium atomic radius itself can be represented by a single physically interpreted geometric scaling law derived from a common density-dependent physical mechanism.** To investigate this question, the Holland Atom framework introduces a compact model whose parameters are defined independently of the regression process and possess explicit physical interpretations within General Expanse Tension Theory.

A comprehensive review of the available literature, presented in Appendix I, indicates that existing approaches generally fall into one of four categories: qualitative descriptions of periodic trends, effective nuclear charge and shielding models, quantum mechanical electronic structure calculations, or empirical and semi-empirical compilations of atomic radius. Each has demonstrated considerable predictive success within its intended domain. However, none was identified that combines the specific physical variables employed in the present work: representative nuclear mass loading, principal resonance shell, block-dependent resonance geometry, high-density coupling modulation

(Switch E), and local electronic charge-density loading within a single predictive equation for empirical atomic radius spanning the validated periodic table.

The distinction of the Holland Atom model therefore lies not simply in achieving predictive agreement with the reference dataset, but in the physical interpretation assigned to the model parameters. The scaling law proposes that atomic radius is governed primarily by the interaction between resonance geometry and local density loading within the Holland Field, with progressively smaller contributions arising from secondary density effects. Within this interpretation, the dominant periodic trends emerge from representative nuclear mass loading and resonance geometry, while the remaining structured residuals identify additional physical phenomena operating beyond the scope of the present model.

The literature review further demonstrates that the methodology adopted in this study differs from conventional empirical optimisation. A configuration-controlled master dataset was established before model development, representative nuclear mass loading was defined independently of the radius data, and calibration, validation and extrapolation were performed as separate stages. This separation provides a transparent framework in which model performance can be evaluated objectively and independently of parameter development [7,8].

Taken together, these observations suggest that the principal novelty of the present work is not the recognition that atomic radius varies systematically across the periodic table – this has long been established – but the proposal that this variation can be described by a compact density-dependent resonance law whose variables arise directly from an underlying physical ontology. If supported by further investigation and independent validation, the Holland Atom geometric scaling law may therefore represent a distinct approach to understanding the geometric organisation of atomic structure, complementing rather than replacing the established methods of quantum and computational chemistry.

7.8 Limitations

The present study intentionally represents the model formulation of the Holland Atom model and therefore excludes several higher-order effects identified through residual analysis. These include relativistic electronic energy-density loading in heavy elements, shell-envelope thickness effects identified in Paper 2, and threshold coupling associated with spatially extended actinide orbitals. The model also relies upon validated proxy radii for a small number of elements lacking direct empirical measurements. These limitations are acknowledged explicitly because the objective of the present work is to establish the dominant geometric scaling relationship before introducing additional complexity.

7.9 Future Development

The residual structure identified in this study provides a well-defined programme for future investigation.

The first priority is incorporation of relativistic electronic energy-density loading within the heavy period-6 elements. The second is development of a quantitative shell-envelope model based upon the standing-mode amplitude analysis established in Paper 2. A third objective is refinement of the electronic charge-density loading function as additional experimental constraints become available. Finally, the extrapolated predictions for the superheavy elements ($Z = 97–118$) provide an opportunity for future experimental comparison as improved atomic radius measurements become available.

These developments represent extensions of the density hierarchy established in the present work rather than modifications to the underlying physical framework.

8. Conclusions

This paper has presented the first predictive geometric scaling law derived from the Holland Atom framework within the General Expanse Tension Theory (GETT). Building upon the physical resonance-lock mechanism established in Holland Atom Paper 1 and the configuration-controlled atomic radius framework established in Paper 2, the present work demonstrates that empirical atomic radius can be described by a compact density-dependent scaling relationship founded upon physically meaningful variables.

The model is based upon five principal components: resonance shell geometry, representative nuclear mass loading, block-dependent resonance geometry, high-density coupling modulation through Switch E, and local electronic charge-density loading. Each component is introduced as a direct consequence of the Holland Atom framework and represents a distinct physical contribution to the equilibrium geometry of the atomic resonance cavity.

Application of the frozen model to a configuration-controlled master dataset demonstrates that these variables reproduce the dominant variation in empirical atomic radius across the validated periodic table with a high degree of predictive accuracy. The resulting model achieves strong statistical agreement with the reference dataset while maintaining a comparatively simple physical structure and requiring no element-specific optimisation or empirical exceptions within the calibrated domain.

The remaining residual structure is confined to a small number of identifiable physical regimes, providing clear targets for future development of the Holland Atom framework.

The present work also establishes the first configuration-controlled Holland Atom Master Radius Dataset, providing a reproducible reference standard for future studies within the Holland Atom research programme. By separating dataset construction, model calibration and independent validation, the methodology adopted here provides a transparent framework against which future developments can be objectively assessed.

More broadly, this study supports the central proposition of General Expanse Tension Theory that local density governs the behaviour of the Holland Field across multiple physical scales. Within the atomic domain, representative nuclear mass loading emerges as the dominant determinant of resonance geometry, while secondary density contributions associated with electronic charge distribution produce measurable refinements to the equilibrium radius. The resulting density hierarchy provides a coherent physical interpretation linking atomic geometry to the broader GETT framework.

The resulting geometric radius scaling law reproduces the 96-element calibration reference set with high in-sample goodness-of-fit ($R^2 = 0.9764$, adjusted $R^2 = 0.9723$, RMSE = 6.56 pm, MAE = 4.55 pm and MAPE = 3.28%). Separately, leave-one-out cross-validation of the underlying block-dependent core scaling law yields $R^2_{\mathrm{LOO}} = 0.833$, RMSE = 17.44 pm and MAPE = 8.52%, demonstrating that substantial predictive structure is retained when individual observations are withheld.

The results demonstrate that the dominant systematic variation in Atomic Radius Empirical (ARE) can be reproduced by a compact geometric formulation incorporating resonance-shell structure, an empirical density-loading proxy, block-dependent geometry, high-density coupling response and electronic loading. Within GETT, these terms are interpreted as constitutive contributions to equilibrium resonance geometry; the present analysis does not, however, uniquely establish this interpretation over alternative charge- and electronic-structure-based descriptions.

The Holland Atom geometric scaling law is not intended to replace the established predictive methods of atomic and quantum chemistry. Rather, it proposes a complementary physical interpretation in which atomic radius emerges as the measurable consequence of density-dependent standing-mode resonance within the Holland Field. In doing so, it offers a simple and physically motivated description of one of the most fundamental geometric properties of the atom.

The Holland Atom research programme will continue by investigating the higher-order density effects identified through the present residual analysis, including relativistic electronic energy-density loading, shell-envelope structure and threshold coupling in extended actinide orbitals. Together, these developments provide a clear pathway towards an increasingly complete density-dependent description of atomic structure founded upon the principles of General Expanse Tension Theory.

The same density-dependent coupling mechanism that GETT applies to cosmological expansion, gravitation and galactic structure is shown here, without atomic-specific modification, to reproduce empirical atomic radius across 96 elements. Atomic radius scaling therefore enters the set of phenomena the single Holland-Field mechanism can describe – extending the framework from astrophysical to atomic length scales within one constitutive ontology, while the causal identity of the density term remains, as elsewhere, an open and testable question.

The results presented here support the proposition that atomic radius is not an independent empirical property, but the measurable equilibrium geometry of a density-dependent resonance structure established within the Holland Field, providing the first predictive geometric scaling law arising from the Holland Atom framework.

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Appendix I. Literature Context and Novelty Assessment

Appendix A. Construction of the Holland Atom Master Radius Dataset

A.1 Purpose

The Holland Atom geometric scaling law requires a single, internally consistent reference dataset representing the equilibrium resonance boundary of each element. As demonstrated in Holland Atom Paper 2, no universally accepted definition of atomic radius exists. Instead, multiple experimental and theoretical radius definitions measure different physical regions of the atomic standing-mode envelope. Consequently, a configuration-controlled master dataset was constructed prior to model development to provide a reproducible reference standard for all subsequent analyses. The master dataset was frozen before derivation of the scaling law and remained unchanged throughout the analysis presented in this paper.

A.2 Master Radius Definition

The master radius, R_{master}, is defined as the best available estimate of the Atomic Radius Empirical (ARE) for each element. Where direct empirical measurements are unavailable, proxy values are generated using the hierarchy established and validated in Holland Atom Paper 2 [2,5].

The hierarchy is applied in the following priority order:

1. **Atomic Radius Empirical (ARE)** obtained directly from the audited WebElements empirical dataset.
2. **Calculated Atomic Radius (ARC)** converted to the ARE scale using block-dependent conversion factors derived from elements possessing both ARE and ARC measurements.
3. **Single-bond covalent radius (Rc2008)** converted to the ARE scale using independently derived block-dependent conversion factors where ARC data are unavailable [6].

No further interpolation or estimation procedures were employed.

A.3 Configuration Control

The master dataset is configuration controlled. Each element carries the following information:

- Atomic Number (Z)
- Symbol
- Element Name
- Period
- Group
- Orbital Block
- Highest Occupied Shell
- Representative Nuclear Mass Loading (A)
- Master Radius
- Radius Source
- Confidence Classification

Every modification is traceable to its originating literature source.

A.4 Radius Source Classification

Each master radius is assigned one of three source classifications.

- **Tier 1 – Direct Experimental**
  - Measured ARE value. Highest confidence.
- **Tier 2 – ARC-derived**
  - Calculated atomic radius converted to the ARE scale using the block-dependent conversion relationship established from paired ARE–ARC observations. Cross-validated against Rc2008.
- **Tier 3 – Rc2008-derived**
  - Single-bond covalent radius converted to the ARE scale using the independently derived block conversion relationship. Applied only where neither ARE nor ARC values are available.

A.5 Confidence Classification

Each element is assigned one of three confidence classes.

- Class A: Direct empirical measurement.
- Class B: Proxy value independently verified by agreement between ARC and Rc2008 conversion relationships.
- Class C: Single-source proxy where no independent verification exists.

This classification accompanies every prediction presented within Paper 3.

A.6 Representative Nuclear Mass Loading

Representative nuclear mass loading, A, is defined as the mass number of the most abundant naturally occurring stable isotope. Where no stable isotope exists, the longest-lived isotope is adopted. This definition was selected because the Holland Atom model relates atomic geometry to nuclear mass loading rather than abundance-weighted atomic mass. The adopted value therefore represents the characteristic nuclear loading experienced by the resonance structure of each element.

A.7 Dataset Scope

The configuration-controlled dataset contains all 118 recognised chemical elements. The geometric scaling law presented in this paper is derived using the subset for which reliable experimental or validated proxy radius data are available. Elements beyond the present experimental radius coverage are retained within the master dataset for completeness and future development but are excluded from regression and statistical fitting.

A.8 Quality Assurance

Prior to freezing the dataset, the following validation procedures were completed.

- Verification of all atomic numbers and element identities.
- Cross-validation of ARE, ARC and Rc2008 conversion relationships.
- Independent verification of representative nuclear mass loading values.
- Confirmation of orbital block, group and shell assignments.
- Removal of duplicated and superseded radius values.
- Traceability of every master radius to its originating literature source.

The resulting dataset forms the configuration-controlled reference standard for Holland Atom Paper 3 and will remain unchanged throughout subsequent papers unless future experimental measurements supersede existing reference values.

Appendix B. Physical Parameters and Model Variables

B.1 Purpose

The Holland Atom geometric scaling law is constructed from a small number of physically motivated variables, each representing a distinct contribution to the equilibrium geometry of the atomic standing-mode resonance. This appendix defines the origin, physical interpretation and implementation of each variable used throughout the model.

B.2 Atomic Radius (R)

Definition. The predicted equilibrium radius of the outer resonance envelope.

Units. picometres (pm)

Reference dataset. Atomic Radius Empirical (ARE) master dataset (Appendix A).

Physical interpretation. The equilibrium location of the outer standing-mode resonance boundary within the Holland Field.

B.3 Representative Nuclear Mass Loading (A)

Definition. Representative mass number of the nucleus.

Stable elements: Most abundant stable isotope.

Radioactive elements: Longest-lived isotope.

Units. Dimensionless (mass number).

Physical interpretation. Represents the dominant source of local density loading responsible for compression of the equilibrium Holland Field wavelength.

B.4 Principal Resonance Shell (n)

Definition. Highest occupied resonance shell. Equivalent to the principal shell of the outer resonance cavity.

Values. n = 1–7

Physical interpretation. Defines the free geometric opening of the resonance cavity prior to density loading.

B.5 Orbital Block Family (b)

Values s, p, d, f

Physical interpretation. Within the Holland Atom model, block family represents different resonance cavity geometries rather than simply chemical classification.

Each block possesses independent geometric scaling coefficients.

B.6 Resonance Wavelength Scale (λb)

Definition. Block-dependent resonance scale coefficient.

Physical interpretation. Represents the equilibrium wavelength scale associated with each resonance cavity geometry.

B.7 Shell Opening Exponent (βb)

Definition. Block-dependent shell opening exponent.

Physical interpretation. Controls how resonance radius increases with shell number.

B.8 Switch E high-density coupling modulation (SE)

Definition. High-density coupling modulation function.

Physical interpretation. Represents enhanced scalar coupling operating within the innermost resonance cavity owing to the extremely high local nuclear density.

Expected to affect primarily n = 1 and transition into n = 2.

B.9 Electron Density Loading Function (De)

Definition. Secondary density-loading correction.

Physical interpretation. Represents the influence of local electronic charge density occupying the active resonance cavity.

Unlike nuclear loading, this contribution modifies rather than establishes the equilibrium wavelength.

B.10 Higher-Order Density Terms

The geometric scaling law intentionally excludes higher-order density contributions identified through residual analysis.

These include:

- Noble-gas resonance boundary ambiguity.
- Relativistic electronic energy-density loading.
- Shell-envelope thickness effects.
- Extended actinide threshold coupling.

These terms are discussed in Section 7 and Appendix G and form the basis of future development within the Holland Atom programme.

Appendix C. Development of the Holland Atom Geometric Radius Scaling Law

C.1 Purpose

The geometric scaling law presented in this paper is developed from the physical framework established by General Expanse Tension Theory and the Holland Atom model. Each successive term represents a physically motivated contribution to the equilibrium radius of the atomic resonance cavity. The development proceeds from the simplest resonance geometry to the complete predictive model.

C.2 Free Resonance Geometry

The Holland Atom proposes that each atomic shell is a resonance-locked standing mode of the Holland Field. In the absence of density loading, the radius of successive resonance cavities increases systematically with shell number. The simplest geometric relationship is therefore

$$R \propto n^{\beta}$$

where n is the principal resonance shell and β describes the rate at which the standing-mode cavity expands with increasing shell number. This relationship defines the free geometric opening of the resonance structure.

C.3 Nuclear Mass-Density Loading

Within GETT, the nucleus acts as the dominant source of local density loading. Increasing nuclear mass compresses the equilibrium wavelength of the surrounding Holland Field, reducing the standing-mode radius. This introduces the primary density-loading term

$$R \propto A^{-1/2}$$

where A is the representative nuclear mass loading. The inverse square-root dependence was not imposed a priori during development of the base nuclear-loading law. When the nuclear-loading exponent was treated as a free parameter before branch-specific constitutive refinements were introduced, independent optimisation across the principal block families converged close to $-1/2$. The term $A^{-1/2}$ is therefore retained as the universal base nuclear-loading factor in the governing equation. However, the final frozen implementation permits constitutive laws to modify the effective exponent in specific physical regimes. In particular, the outer s-block Switch E law partially offsets the base loading and produces an effective dependence $A^{-\gamma_s}$, with $\gamma_s = 0.327672732$. Thus, $-1/2$ should be interpreted as the base nuclear-loading law, not as the effective exponent applying unchanged to every branch of the final model.

The independent convergence of multiple electronic block families towards the same exponent provides additional confidence that the inverse square-root dependence reflects an underlying physical scaling relationship rather than a block-specific empirical optimisation. Nuclear mass loading therefore establishes the dominant systematic contraction observed across the periodic table.

C.4 Block-Dependent Resonance Geometry

Although all shells arise from standing-mode resonance, Paper 1 demonstrated that the resonance cavity differs systematically between the s-, p-, d- and f-blocks. Each block therefore possesses independent geometric scaling coefficients. Introducing the block-dependent wavelength scale gives

$$R = \lambda_b n^{\beta_b} A^{-1/2}$$

where

$$\lambda_b$$

represents the characteristic resonance scale for block b, while

$$\beta_b$$

describes the shell-opening behaviour within that block.

C.5 Switch E Coupling Function and Atomic Radius Response Factor

Within General Expanse Tension Theory, Holland–Higgs coupling is governed by a continuous density-dependent modulation function, $S(\Sigma)$, where Σ denotes the local Holland Field density state. The function is bounded such that

$$0 \leq S(\Sigma) \leq 1$$

with $S(\Sigma) = 0$ representing negligible coupling and $S(\Sigma) = 1$ representing the limiting maximum coupling permitted by the local density environment.

The physical ontology of Switch E contains two related but distinct manifestations of the same constitutive coupling-response term. First, the compact $n = 1$ regime exhibits the strongest radius reduction, with the $n = 2$ transition multiplier describing the substantially reduced response at greater shell distance. Second, the Group 1 and Group 2 branches retain an s-block-specific constitutive loading response beyond the inner-shell regime. This persistent s-block contribution is not interpreted as the direct spatial continuation of K-shell contraction into the outermost shell; rather, it represents the distinct coupling environment associated with the s-block family as encoded empirically by the frozen model. Switch E therefore denotes the complete constitutive response S_E, whose mathematical form is regime-dependent, rather than a correction restricted exclusively to $n = 1$ and $n = 2$.

Consequently, while $A^{-1/2}$ remains the universal base density-loading term, multiplication by the Group 1 or Group 2 Switch E branch produces an effective outer s-block dependence $A^{-1/2}A^{(1/2-\gamma_s)} = A^{-\gamma_s}$; this is a specified constitutive modification of the base law rather than a replacement of the universal loading term.

The quantity S_E used in the Holland Atom geometric radius scaling law is not the continuous GETT coupling function itself. Instead, S_E is the **Switch E radius-response factor**: the multiplicative term through which the effect of high-density Holland–Higgs coupling is applied to the predicted atomic radius. This distinction is important. In the compact inner-shell regime, stronger underlying coupling does not increase radius. Instead, stronger coupling produces a stronger equilibrium-restoring response, drawing the resonance shell into a tighter equilibrium configuration. Thus, as the underlying coupling function $S(\Sigma)$ approaches its high-density limiting state, the corresponding radius-response factor S_E decreases. The physical relationship may therefore be summarised as:

$\Sigma \uparrow \Rightarrow S(\Sigma) \uparrow \Rightarrow$ Holland–Higgs coupling $\uparrow \Rightarrow$ equilibrium restoring response $\uparrow \Rightarrow S_E \downarrow \Rightarrow R_{HA} \downarrow$

In the frozen Holland Atom implementation, this produces three regimes. The K-shell ($n = 1$), being closest to the nuclear density source, experiences the strongest Switch E radius response and is therefore assigned the largest radius-reduction factor. The L-shell ($n = 2$)occupies a transition regime, where the coupling effect is weaker and the radius reduction is modest. For $n \geq 3$, the local

density regime is sufficiently reduced that the outer-shell geometric scaling law is recovered. Accordingly, the Switch E radius-response factor is implemented as:

$$S_E = \begin{cases} S_{E,K}, & n = 1 \\ T_{n2}, & n = 2 \\ 1, & n \geq 3 \end{cases}$$

where $S_{E,K}$ is the compact K-shell radius-response factor and T_{n2} is the universal second-shell transition multiplier.

Thus, Switch E should not be interpreted as an arbitrary empirical correction. It represents the atomic-radius expression of the GETT density-dependent coupling function: higher local density increases the underlying Holland–Higgs coupling, and the geometric consequence is a smaller equilibrium resonance radius.

Special K-shell limit ($n = 1$). For the K-shell, the frozen Switch E Radius-Response Factor is defined as

$$S_{E,K} = 0.234529522\, A^{1/2}.$$

Because the governing radius law independently contains the representative density-loading term $A^{-1/2}$, these contributions cancel algebraically within the $n = 1$ regime:

$$A^{-1/2} S_{E,K} = A^{-1/2}\left(0.234529522 A^{1/2}\right) = 0.234529522.$$

The present formulation therefore reduces the K-shell radius to

$$R_{HA,K} = \lambda_s(0.234529522),$$

independent of A, giving a common predicted K-shell radius of approximately 26.72 pm for both hydrogen and helium. This lies between the adopted ARE reference values of 25.00 pm for H and 28.56 pm for He. The K-shell constitutive law therefore captures the characteristic compact radius of the $n = 1$ regime but does not reproduce the observed H–He radius separation. Within the Holland Atom framework, the strong K-shell contraction is interpreted as the radius response associated with the high-density Switch E coupling regime; however, the present functional form should be regarded as a common K-shell limiting approximation rather than a resolved model of element-specific variation within Period 1.

A specific limitation occurs in Period 1, where algebraic cancellation between the $A^{-1/2}$ density-loading term and the $A^{1/2}$ dependence of $S_{E,K}$ causes the present law to predict a common K-shell radius of 26.72 pm for H and He. The model therefore captures the compact K-shell scale but does not resolve the measured H–He radius difference; identifying the additional contribution responsible for this separation remains outside the present formulation.

This term provides the physical transition between the Bohr-radius regime and the outer valence shell scaling relation.

C.6 Electronic Charge-Density Loading

Residual analysis demonstrated that, following removal of the primary nuclear loading contribution, systematic structure remained correlated with the distribution of electronic charge within the active

resonance cavity. Within the Holland Atom framework this is interpreted as a secondary density-loading mechanism. Introducing the electronic density-loading function gives

$$R = \lambda_b n^{\beta_b} A^{-1/2} S_E D_e$$

Where D_e represents the secondary modulation of the resonance cavity arising from local electronic charge density. Unlike nuclear loading, this term perturbs the resonance geometry rather than establishing it.

C.7 Final Geometric Scaling Law

Combining the preceding physical contributions yields the Holland Atom geometric radius scaling law

$$\boxed{R_{\text{pred}} = \lambda_b n^{\beta_b} A^{-1/2} S_E D_e}$$

where

- R_{pred} is the predicted Atomic Radius Empirical (ARE),
- λ_b is the block-dependent resonance wavelength coefficient,
- n is the principal resonance shell,
- β_b is the block-dependent shell-opening exponent,
- A is the representative nuclear mass loading,
- S_E is the Switch E high-density coupling modulation function,
- D_e is the electronic charge-density loading function.

This expression represents the complete Holland Atom model evaluated throughout the present paper.

C.8 Future Higher Order Extensions

Residual analysis demonstrates that the model captures the dominant geometric behaviour of atomic radius while identifying a small number of systematic higher-order effects.

These include:

- relativistic electronic energy-density loading in heavy period-6 elements;
- ambiguity in proxy radius definitions for noble gases;
- shell-envelope thickness within the standing-mode resonance;
- threshold coupling associated with extended actinide orbitals.

These effects are intentionally excluded from the primary scaling law presented here, preserving the simplicity of the model. They instead define the research programme for the next stage of development and provide specific, testable predictions arising from the density hierarchy proposed by GETT.

Appendix D. Technical Specification of the Holland Atom Geometric Radius Scaling Law

Frozen Model Reference Implementation

Item	Description
Document	Appendix D – Technical Specification of the Holland Atom Geometric Radius Scaling Law
Associated Paper	Holland Atom Paper 3
Status	Frozen Reference Implementation
Authors	John Holland
Theory	General Expanse Tension Theory (GETT)
Dataset	ARE-118 Master Radius Dataset v2.0
Scope	Neutral atomic geometric radius prediction
Reference Radius	Paper 3 selected experimental radius definition
Implementation Status	Normative
Classification	Scientific Technical Specification

The purpose of this matrix is to provide traceability between the principal components of the Holland Atom Geometric Radius Scaling Law and the sections of this specification in which they are defined. It is intended as a navigation aid for readers and implementers and does not constitute an assessment of the scope or completeness of the model.

Component	Primary Section	Supporting Sections
Holland Atom ontology	D.2	Holland Atom Papers 1 & 2
Symbols and definitions	D.3	D.6
Governing equation	D.4	Section 5
Decision logic	D.5	D.11
Frozen parameters	D.6	Appendix Tables
Switch E	D.7	Section 5
Electronic Loading	D.8	Section 5
Parameter provenance	D.9	Appendix Tables
Calibration philosophy	D.10	Methods
Computational algorithm	D.11	Flow Diagram
Parameter sensitivity	D.12	—
Model assumptions	D.13	D.2
Boundary conditions	D.14	Section 5

Table D1. Specification Traceability Matrix

Specification Status

This appendix constitutes the **authoritative technical specification** of the Holland Atom Geometric Radius Scaling Law presented in Paper 3. Unlike the explanatory sections of the main manuscript, Appendix D is normative. The governing equations, parameter tables, decision logic, computational procedures and implementation rules defined herein collectively constitute the **Frozen Reference Implementation**. An independent implementation conforming to this specification shall reproduce the published Paper 3 results, including the reported global goodness-of-fit statistics, within normal numerical precision. No unpublished coefficients, hidden correction factors, element-specific adjustments or proprietary algorithms are required.

D.1 Purpose and Scope

This appendix defines the complete technical specification of the Holland Atom Geometric Radius Scaling Law developed in Paper 3. Its purpose is to provide a complete, internally consistent and reproducible mathematical

description of the Holland Atom model, including its governing equations, physical ontology, calibration philosophy, implementation algorithm, frozen parameter set and computational procedures.

The specification is normative rather than descriptive. The equations, parameter values and implementation rules presented herein define the unique Frozen Reference Implementation of the Holland Atom Geometric Radius Scaling Law. Independent implementations conforming to this specification should reproduce the published Paper 3 results within normal numerical precision.

The specification has four principal objectives:

1. **Scientific transparency.** Every mathematical quantity employed by the model is explicitly defined and linked to the underlying Holland Atom ontology.
2. **Computational reproducibility.** All equations, coefficients, decision logic and computational procedures required for independent implementation are documented.
3. **Model immutability.** The geometry model is presented as a frozen reference implementation. Subsequent Holland Atom developments shall extend the framework through future versioned specifications rather than modifying the equations or parameters defined herein.
4. **Physical traceability.** Every term within the governing equation possesses an explicit physical interpretation. No coefficient is introduced solely as an empirical fitting constant without a corresponding physical role within the Holland Atom framework.

The scope of this specification is restricted to the prediction of equilibrium geometric radii for neutral atoms in their electronic ground states. The model incorporates shell-dependent geometric scaling, nuclear loading, density-dependent Holland Field coupling (Switch E), block-specific geometric scaling and Electronic Loading Factor (ELF), D_e corrections.

This specification does not include isotope-specific radius variation, ionic radii, chemical bonding, molecular geometry, condensed matter effects or excited electronic configurations. These topics form the subject of subsequent Holland Atom publications and do not modify the frozen geometric radius model defined herein.

D.2 Holland Atom Ontology

D.2.1 General

The Holland Atom model is founded upon the ontological framework introduced in Holland Atom Paper 1. Atomic structure is interpreted as a stable standing-wave resonance supported by the Holland Field (Φ). The geometric radius of an atom is the equilibrium position of the outermost occupied resonance surface after the combined effects of nuclear loading, shell geometry, density-dependent Holland Field coupling and electronic loading have reached equilibrium. The geometric radius scaling model assumes the physical entities defined in Table D.2.

Ontological Object	Physical Definition	Mathematical Representation
Holland Field (Φ)	Continuous scalar field supporting atomic standing-wave resonance.	Underlying physical medium; implicit throughout the governing equations.
Atomic Nucleus	Localised high-density mass distribution producing field loading.	Atomic mass number, A.
Standing-Wave Shell	Stable spherical resonance surface corresponding to an allowed shell geometry.	Principal shell number, n.
Electronic Charge Density	Spatial distribution of negative charge occupying resonance structures.	Electronic Loading Factor (ELF), D_e corrections.
Switch E Region	High-density inner-shell regime exhibiting enhanced Holland Field coupling.	Switch E modifier, S_E.
Block Geometry	Distinct resonance behaviour associated with s-, p-, d- and f-block electronic structures.	Block-specific coefficients $\lambda_b$and β_b.
Equilibrium Radius	Stable outer resonance radius following application of all loading mechanisms.	R_{HA}.

Table D2. Holland Atom Ontology

D.2.2 Physical Postulates

The Holland Atom model is defined by the following physical postulates.

- **H1.** The Holland Field is continuous throughout atomic space and supports stable standing-wave resonance.
- **H2.** The atomic nucleus modifies the surrounding Holland Field through mass-dependent loading.
- **H3.** Principal electron shells correspond to stable standing-wave resonance surfaces rather than classical particle orbits.
- **H4.** Electronic charge exists as distributed charge density occupying resonance structures.
- **H5.** Electronic charge density modifies local resonance geometry through the physical model of Electronic Loading Density (ELD), which is mathematically established with Electronic Loading Factor (ELF), D_e corrections.
- **H6.** High-density inner-shell resonance exhibits enhanced Holland Field coupling (Switch E).
- **H7.** The equilibrium atomic radius corresponds to the outermost occupied resonance surface following simultaneous application of shell geometry, nuclear loading, Switch E coupling and electronic loading.

D.2.3 Physical Loading Hierarchy

The Holland Atom model assumes that equilibrium radius is determined by the sequential interaction of four physical loading mechanisms.

Stage	Physical Mechanism	Mathematical Component
1	Standing-wave shell geometry	n^{β_b}
2	Nuclear density loading	$A^{-1/2}$
3	Enhanced inner-shell coupling	S_E
4	Electronic Loading Density (ELD)	ELD_b

Table D3. Physical Loading Hierarchy

These mechanisms are applied multiplicatively within the governing equation defined in Section D.4.

D.2.4 Ontological Constraints

The Holland Atom model adopts the following constraints.

- Atomic structure is represented by standing-wave resonance geometry.
- Only neutral atoms in their electronic ground state are considered.
- The selected atomic mass number is fixed for each element as defined in Section D.9.
- Shell geometry is determined solely by the highest occupied principal shell.
- Electronic loading is determined from the ground-state electron configuration.
- Block-specific coefficients are invariant for all elements within the corresponding electronic block.
- The parameter set is frozen and shall not be modified without publication of a subsequent specification version.

D.2.5 Relationship to the Governing Equation

The ontology defined above maps directly onto the mathematical structure of the Holland Atom Geometric Radius Scaling Law.

Physical Quantity	Governing Equation Term
Standing-wave geometry	n^{β_b}
Nuclear loading	$A^{-1/2}$
High-density coupling	S_E
Electronic Loading Factor (ELF)	D_e
Block-dependent resonance	λ_b, β_b
Equilibrium radius	R_{HA}

Table D4. Relationship to the Governing Equation

No mathematical term employed by the governing equation is without an explicit ontological interpretation.

D.3 Symbols and Definitions

D.3.1 General

Table D.3 defines every mathematical symbol employed by the Holland Atom Geometric Radius Scaling Law. Each symbol possesses a unique definition throughout this specification and shall not be assigned multiple meanings. Unless otherwise stated, all atomic radii are expressed in picometres (pm). Dimensionless quantities are indicated by "—".

Symbol	Definition	Units	Category
R_{HA}	Predicted Holland Atom equilibrium radius	pm	Output
R_{exp}	Experimental reference radius	pm	Validation
n	Principal resonance-shell coordinate	—	Geometry
A	Selected atomic mass number	—	Nuclear
Z	Atomic number	—	Identification
b	Electronic block classification (s, p, d, f)	—	Structure

Table D5. Primary Model Variables

Symbol	Definition	Units	Status
λ_s	s-block geometric scale coefficient	pm	Frozen
λ_p	p-block geometric scale coefficient	pm	Frozen
λ_d	d-block geometric scale coefficient	pm	Frozen
λ_f	f-block geometric scale coefficient	pm	Frozen
β_s	s-block shell expansion exponent	—	Frozen
β_p	p-block shell expansion exponent	—	Frozen
β_d	d-block shell expansion exponent	—	Frozen
β_f	f-block shell expansion exponent	—	Frozen

Table D6. Geometric Parameters

Symbol	Definition	Units	Function
$A^{-1/2}$	Nuclear loading term	—	Mass-dependent resonance compression
S_E	Switch E coupling modifier	—	High-density inner-shell correction
ELD_b	Electronic Loading Density	—	Electronic charge-density modulation

Table D7. Loading Functions

Symbol	Definition	Units
N_e	Total electron population	electrons
n_s	Number of electrons occupying s subshells	electrons
n_p	Number of electrons occupying p subshells	electrons
n_d	Number of electrons occupying d subshells	electrons
n_f	Number of electrons occupying f subshells	electrons

Table D8. Electronic Configuration Variables

Symbol	Definition	Units
R^2	Coefficient of determination	—
RMSE	Root Mean Square Error	pm
MAE	Mean Absolute Error	pm
MAPE	Mean Absolute Percentage Error	%
ΔR	Radius residual $\Delta R = R_{\text{master}} - R_{\text{HA}}$	pm

Table D9. Statistical Measures

D.3.2 Model Inputs

The model requires the following independent inputs for each element: No experimentally measured radius is used as a model input.

Input	Source
Atomic number	Periodic Table
Atomic mass number	Selected isotope (Section D.9)
Principal resonance-shell coordinate	Periodic-table period / resonance-shell architecture
Electronic block	Ground-state electron configuration
Electronic loading coordinate –	regularised resonance-loading coordinate, as defined in D.8

Table D10. Model input sources

D.3.3 Model Outputs

The primary output is the Holland Atom equilibrium geometric radius R_{HA}.
Secondary outputs include residual error, percentage error, block-specific residual statistics, global validation statistics, and goodness-of-fit measures.

D.3.4 Reserved Symbols

To ensure notation consistency, the following symbols are reserved throughout this specification.

Symbol	Reserved Meaning
R_{HA}	Holland Atom predicted radius
R_{exp}	Experimental radius
λ_b	Block-specific geometric scale coefficient
β_b	Block-specific shell expansion exponent
ELD_b	Electronic Loading Density
D_e	Electronic Loading Factor (ELF)
S_E	Switch E modifier
Φ	Holland Field

Table D11. Applied Symbols and Meaning

D.3.5 Notational Conventions

The following conventions apply throughout this specification.

1. All geometric coefficients are block-specific unless explicitly stated otherwise.
2. All parameter values are frozen following model calibration.
3. All equations refer exclusively to neutral atoms in their electronic ground state.
4. All statistical measures are computed using the ARE-118 Master Radius Dataset.
5. All dimensions refer to equilibrium atomic geometry as defined by the Holland Atom ontology.

D.4 Complete Governing Equation

D.4.1 General

The Holland Atom Geometric Radius Scaling Law predicts the equilibrium geometric radius of a neutral atom by combining four physically distinct loading mechanisms:

1. shell-dependent standing-wave geometry;
2. nuclear mass loading;
3. enhanced Holland Field coupling (Switch E);
4. Electronic Loading Factor (ELF). D_e

The model formulation employs block-specific geometric scaling coefficients and block-specific shell expansion exponents. The resulting equation constitutes the frozen reference implementation for Paper 3.

D.4.2 Governing Equation

The Holland Atom Geometric Radius Scaling Law establishes the governing equation as

$$\boldsymbol{R_{HA,b} = \lambda_b\, n^{\beta_b}\, A^{-1/2}\, S_E\, D_e}$$

while

$$S_E = f(n, \Sigma)$$

and

$$D_e = f(\text{electron configuration})$$

Term	Physical Interpretation	Defined in
$R_{HA,b}$	Predicted Holland Atom equilibrium radius for electronic block b	D.3
λ_b	Block-specific geometric scaling coefficient	D.18
n^{β_b}	Standing-wave shell expansion for block b	D.18
$A^{-1/2}$	Nuclear loading due to the selected atomic mass number	D.9
S_E	Enhanced Holland Field coupling within the Switch E density regime	D.7
D_e	Electronic Loading Density factor describing charge-density modulation arising from the electronic configuration	D.8

Table D12. Physical ontological interpretation of governing equation terms

The model employs independent geometric scaling coefficients (λ_b, β_b) for each electronic block (s, p, d, and f). The appropriate coefficient pair is selected according to the ground-state electronic configuration of the element under consideration.

The multiplicative structure of the governing equation reflects the sequential application of physically distinct loading mechanisms acting upon the equilibrium standing-wave geometry. No additive empirical correction terms, element-specific adjustments, or unpublished fitting parameters are employed. The equation defined

above constitutes the frozen reference implementation of the Holland Atom Geometric Radius Scaling Law and shall remain unchanged throughout this specification.

D.4.3 Block Selection

The model employs independent geometric coefficients for each electronic block.

Electronic Block	Coefficients
s	λ_s, β_s
p	λ_p, β_p
d	λ_d, β_d
f	λ_f, β_f

Table D13. Geometric coefficients by electronic block

The applicable coefficient pair is selected solely from the electronic block of the element under consideration.

D.4.4 Loading Hierarchy

The governing equation applies the physical loading mechanisms in the following order. The resulting product defines the equilibrium Holland Atom radius.

Step	Physical Mechanism	Mathematical Operation
1	Standing-wave shell geometry	n^{β_b}
2	Nuclear loading	$A^{-1/2}$
3	High-density Holland Field coupling	S_E
4	Electronic Loading Density	ELD_b

Table D14. Mathematical operators for physical loading mechanisms

D.4.5 Frozen Implementation

The governing equation shall remain unchanged throughout this specification. Only the frozen parameter values defined in Section D.18 may be employed. Element-specific coefficients, manual corrections or unpublished adjustments are prohibited.

D.4.6 Computational Dependencies

Evaluation of the governing equation requires:

Quantity	Defined in
Block coefficients	D.18
Switch E	D.7
Electronic Loading Density	D.8
Calibration constants	D.18
Algorithm	D.11

Table D15. Detailed evaluation of governing equation elements

D.4.7 Mathematical Constraints

The governing equation is valid only for neutral atoms; electronic ground states; the selected reference isotope defined in D.9; and the experimental radius definition adopted throughout Paper 3. No alternative parameterisations form part of the frozen model.

D.5 Applicability Decision Logic

D.5.1 Purpose

The Holland Atom Geometric Radius Scaling Law is applicable only within the domain defined by this specification. This section defines the deterministic decision logic required to select the correct parameter set, loading functions and computational pathway for each element. No user judgement or element-specific decisions are permitted during implementation.

D.5.2 Required Inputs

The model requires the following independent inputs.

Input	Source	Required
Atomic Number (Z)	Periodic Table	Yes
Atomic Mass Number (A)	Selected reference isotope	Yes
Ground-State Electron Configuration	Standard electronic configuration	Yes
Highest Occupied Principal Shell (n)	Derived from electron configuration	Yes
Electronic Block (s, p, d or f)	Derived from electron configuration	Yes

Table D16. The source and nature of inputs

D.5.3 Decision Sequence

Implementation of the model shall proceed according to the following sequence.

Step	Decision	Action
1	Identify element	Read Z and A
2	Determine electronic configuration	Assign ground-state electron occupancy
3	Determine highest occupied shell	Assign n
4	Determine electronic block	Select s, p, d or f
5	Select block coefficients	Load λb and βb
6	Evaluate Switch E	Determine SE
7	Compute Electronic Loading Factor	Evaluate De
8	Evaluate governing equation	Compute RHA
9	Compare with experimental radius	Validation only

Table D17. Sequence steps and action

D.6 Frozen Primary Parameters

D.6.1 Purpose

This section defines the complete frozen implementation parameter register for the Holland Atom Geometric Radius Scaling Law. The governing equation presented in Section D.4 defines the four physical mechanisms responsible for equilibrium atomic geometry:

- standing-wave shell geometry;
- nuclear loading;
- enhanced Holland Field coupling (Switch E);
- electronic loading.

The numerical parameters listed herein constitute the complete frozen implementation of the model. The governing equation itself shall remain invariant. Specialised physical behaviour, including K-shell resonance, Group 2 refinement, transition behaviour and electronic occupancy effects, is implemented exclusively through the mathematical specification of the Switch E and Electronic Loading functions defined in Sections D.7 and D.8.

D.6.2 Frozen Governing Equation

The Holland Atom Geometric Radius Scaling Law governing equation is the unique frozen physical model defined in Section D.4.2.

D.6.3 Frozen Standing-Wave Geometry

Electronic Block	Radius Scale λ_b(pm)	Shell Exponent β_b	Status
s	113.933539	1.353548744	Frozen
p	66.558505	1.983979626	Frozen
d	112.847816	1.556510621	Frozen
f	261.514447	1.201859512	Frozen

Table D18. Frozen Block Geometry Parameters

These parameters define the standing-wave geometry of each electronic block independently of the subsequent loading mechanisms.

D.6.4 Frozen Nuclear Loading

Nuclear loading is represented by the invariant physical relationship $A^{-1/2}$ where A is the selected atomic mass number defined in Section D.9. No additional nuclear loading terms are introduced into the governing equation. Any specialised mass-dependent behaviour is incorporated through the Switch E implementation and not through modification of the governing equation.

D.6.5 Frozen Switch E Register

Switch E represents enhanced Holland Field coupling within the high-density inner-shell regime. The implementation parameters are frozen as follows.

Parameter	Frozen Value
Transition intercept	−0.078805448
$n = 2$ Transition multiplier	0.924219716
Governing implementation	Section D.7

Table D19. Switch E Parameters

Note: In logarithmic form, this multiplier is equivalent to $\ln(T_{n2}) = -0.078805448$. This value is not an independent parameter.

The following physical behaviours are implemented internally within the Switch E formulation: K-shell resonance (H and He); Group 2 refinement; $n = 2$ transition behaviour; and s-block density loading correction. These implementation rules do not modify the governing equation.

D.6.6 Frozen Electronic Loading Register

Electronic loading is represented by the Electronic Loading Factor D_e whose implementation is specified in Section D.8.

Parameter	Frozen Value	Equivalent Multiplier
p-entry loading coefficient	0.092427481	1.096833597
d-edge loading coefficient	0.140011223	1.150286708

Table D20. Electronic Loading Parameters

The following physical behaviours are implemented internally within the Electronic Loading formulation:

- p-block entry behaviour;
- d-series edge curvature;
- subshell occupancy;
- Hund occupancy effects.

These implementation rules modify only the Electronic Loading Factor and do not alter the governing equation.

D.6.7 Frozen Reference Performance

The successful reproduction of the following metrics constitutes verification of a conforming implementation.

Metric	Frozen Value
Calibration Dataset	ARE-118 Master Dataset
Calibration Population	96 elements
Validation Population	96 elements
Coefficient of Determination (R^2)	0.976420644
Root Mean Square Error	6.561555 pm
Mean Absolute Error	4.552892 pm
Mean Absolute Percentage Error	3.284366 %

Table D21. Reference Implementation Metrics

D.6.8 Applicability Register

Domain	Status
Neutral atoms	Supported
Ground-state electronic configurations	Supported
Excited states	Not supported
Ionic radii	Not supported
Molecular geometries	Future extension
Isotopic radius variation	Future extension
Chemical bonding	Future extension

Table D22. Model Domain

D.6.9 Implementation Rules

A conforming implementation shall satisfy the following requirements.

1. The governing equation defined in Section D.4 shall remain unchanged.
2. The standing-wave geometry coefficients listed in Table D18 shall remain fixed.
3. The universal base density-loading contribution shall be represented by the invariant $A^{-1/2}$ relationship. This base term shall not be independently altered or replaced. The effective loading dependence of the complete governing equation may differ only where explicitly modified through a frozen constitutive term, including the specified Switch E response.
4. Switch E shall be implemented exclusively according to Section D.7.
5. The Electronic Loading Factor shall be implemented exclusively according to Section D.8.
6. No element-specific coefficients shall be introduced into the governing equation.
7. No manual adjustments or unpublished correction factors shall be employed.
8. The reference implementation shall reproduce the performance metrics defined in Table D9 and reported in Table D21.

D.6.10 Reference Implementation Statement

The frozen parameter register defined in this section constitutes the normative implementation of the Holland Atom Geometric Radius Scaling Law Version 2.0.

The governing equation describes the four fundamental physical mechanisms responsible for equilibrium atomic geometry. All specialised behaviour required to reproduce the experimentally observed structure of the periodic table is implemented through the constitutive laws defining the Switch E and Electronic Loading functions. This separation preserves a single, physically interpretable governing equation while permitting the internal implementation to represent the complex density-dependent behaviour observed across the periodic table.

D.7 Switch E High-Density Coupling Constitutive Law

D.7.1 Purpose

The outer-shell geometric scaling law accurately describes the equilibrium resonance geometry of the Holland Atom once the resonance cavity has expanded beyond the compact inner-shell regime. However, the K-shell ($n = 1$) and L-shell ($n = 2$) occupy substantially higher local density environments where the density-dependent Holland–Higgs coupling predicted by General Expanse Tension Theory (GETT) becomes significant.

Switch E implements this behaviour within the Holland Atom. Rather than introducing an independent atomic correction, it represents the application of the existing GETT density-dependent coupling mechanism to the compact inner-shell resonance structure. The purpose of this appendix is to define the complete constitutive implementation of Switch E, distinguish the underlying GETT coupling function from its atomic radius implementation, and provide a complete mathematical specification suitable for independent reproduction.

D.7.2 The GETT Holland–Higgs Coupling Function

Within General Expanse Tension Theory, Holland–Higgs coupling is governed by the continuous density-dependent modulation function $S(\Sigma)$, where Σ denotes the local Holland Field density state.

The modulation function is bounded such that

$$0 \leq S(\Sigma) \leq 1,$$

where

- $S(\Sigma) = 0$ corresponds to negligible Holland–Higgs coupling,
- $S(\Sigma) = 1$ corresponds to the limiting maximum coupling permitted by the local density environment.

As local density increases,

$$\frac{dS}{d\Sigma} > 0,$$

such that stronger local density produces stronger Holland–Higgs coupling. The continuous function $S(\Sigma)$ forms part of the fundamental GETT framework and exists independently of the Holland Atom implementation.

D.7.3 Switch E Radius-Response Factor

The governing equation of the Holland Atom does not employ the continuous coupling function $S(\Sigma)$ directly. Instead, the geometric consequence of the underlying coupling state is represented by the **Switch E radius-response factor**, S_E.

Accordingly,

$$S_E \neq S(\Sigma)$$

Rather,

$$S_E = \mathcal{R}(S(\Sigma))$$

where $\mathcal{R}$ represents the constitutive radius-response operator that converts the underlying Holland–Higgs coupling state into its corresponding equilibrium radius modification. This distinction is fundamental. The GETT modulation function describes **coupling strength**. The Holland Atom parameter S_E describes **the geometric response of the equilibrium resonance radius to that coupling**. Consequently, greater Holland–Higgs coupling does not increase atomic radius. Instead, greater coupling produces stronger equilibrium restoration, which draws the resonance shell into a more compact equilibrium configuration. The physical causal sequence is therefore

$$\Sigma \uparrow \Longrightarrow S(\Sigma) \uparrow \Longrightarrow \text{Holland–Higgs coupling} \uparrow \Longrightarrow \text{equilibrium restoration} \uparrow \Longrightarrow S_E \downarrow \Longrightarrow R_{HA} \downarrow.$$

D.7.4 Physical Density Regimes

Switch E is activated only within the compact high-density resonance regime. Three physical regions are recognised.

Region I — Compact K-shell

The K-shell occupies the highest local density state within the atom. The underlying Holland–Higgs coupling approaches its maximum physical value. Consequently, the equilibrium restoring response is strongest and the resonance shell experiences the greatest reduction in equilibrium radius.

Region II — L-shell Transition

The L-shell lies beyond the compact K-shell but remains within an elevated local density environment. The underlying coupling is reduced relative to the K-shell, producing a correspondingly weaker equilibrium-restoring response. The resulting radius reduction is therefore modest and implemented through the universal second-shell transition multiplier.

Region III — Outer-Shell Regime

For $n \geq 3$, the resonance cavity has expanded sufficiently that the local density no longer produces a measurable Switch E response within the present model. The outer-shell geometric scaling law therefore applies without additional Switch E modification.

D.7.5 Frozen Constitutive Implementation

The frozen Holland Atom implementation applies Switch E according to the following shell-dependent constitutive law.

K-shell

For $n = 1$, $S_E = 0.234529522\ A^{1/2}$.
This represents the strongest radius-response reduction within the model.

Universal Second-Shell Transition

For $n = 2$, $S_E = T_{n2} = 0.924219716$.
This transition multiplier applies to every second-shell element within the frozen implementation, irrespective of electronic block.

Outer Shells

For $n \geq 3$, $S_E = 1$,
except for the additional outer s-block constitutive refinement defined in Section D.7.6.

D.7.6 Outer s-block Refinement

For Group 1:

$$S_E = A^{(1/2-\gamma_s)} T_{n2}^*$$

For Group 2:

$$S_E = G_2 A^{(1/2-\gamma_s)} T_{n2}^*$$

with:

$$T_{n2}^* = T_{n2} \text{only when } n = 2,$$

otherwise:

$$T_{n2}^* = 1.$$

Where $G_2 = 0.825707$, with the same value of γ_s. These refinements modify the effective mass-loading response of the outer s-block while preserving the universal base nuclear-loading law of the governing equation.

D.7.7 Interpretation

Switch E should not be interpreted as an empirical correction factor introduced to improve numerical agreement.

Instead, it represents the constitutive implementation of the density-dependent Holland–Higgs coupling mechanism predicted independently by General Expanse Tension Theory.

Normative clarification — relationship between base loading and Switch E. The factor $A^{-1/2}$ shall be applied universally as the base density-loading term in every evaluation of the governing equation. The statement that $A^{-1/2}$ is invariant refers to the presence and exponent of this **base term** and shall not be interpreted to mean that the effective A-dependence of the complete governing equation must remain $A^{-1/2}$ after multiplication by the applicable constitutive factors. Where the frozen Switch E implementation contains an additional A-dependent term, that term shall be applied exactly as specified. Thus, for the Group 1 and Group 2 branches,

$$A^{-1/2} \times A^{(1/2-\gamma_s)} = A^{-\gamma_s},$$

with the additional Group 2 coefficient applied where specified. The resulting $A^{-\gamma_s}$ dependence is therefore the

effective dependence of the complete frozen model within these s-block regimes; it does not alter the universal $A^{-1/2}$ base term appearing in the governing equation.

Within the Holland Atom, increasing local density produces stronger Holland–Higgs coupling, resulting in a stronger equilibrium-restoring response and a correspondingly smaller equilibrium resonance radius. The compact K-shell therefore exhibits the greatest radius reduction, the L-shell exhibits a weaker transition response, and the outer-shell scaling law is recovered naturally once the local density falls below the Switch E activation regime.

D.8 Electronic Loading Factor (ELF)

D.8.1 Purpose

The Electronic Loading Factor, D_e, defines the second constitutive law of the Holland Atom Geometric Radius Scaling Law. Whereas Switch E describes enhanced Holland Field coupling arising from high nuclear density, the Electronic Loading Factor describes the modification of standing-wave geometry arising from the spatial distribution of electronic charge.

Within the Holland Atom framework, **Electronic Loading Density (ELD)** denotes the **physical perturbation** of the equilibrium resonance geometry arising from the spatial distribution of electronic charge density. The governing equation does not operate directly upon this physical density. Instead, the influence of Electronic Loading Density is represented **mathematically** by the ***Electronic Loading Factor (ELF)***, denoted D_e, which forms the constitutive parameter appearing in the governing equation. Accordingly, ELD describes the underlying physical ontology, whereas ELF (D_e) represents its computational implementation.

Electronic loading therefore represents a physically distinct mechanism from nuclear loading and acts independently within the governing equation D_e modifies only the electronic contribution to the equilibrium standing-wave geometry and does not alter shell geometry, nuclear loading or Switch E coupling.

D.8.2 Physical Basis

Within the Holland Atom ontology, electrons are interpreted as distributed charge-density structures occupying stable standing-wave resonance regions rather than point particles following deterministic trajectories. The local electronic charge density perturbs the Holland Field, producing additional modulation of the equilibrium resonance wavelength.

Consequently, two atoms possessing identical shell numbers and similar nuclear loading may nevertheless exhibit different equilibrium radii owing to differences in electronic occupancy. The Electronic Loading Factor therefore represents the cumulative influence of electron occupancy upon Holland Field resonance.

D.8.3 Mathematical Definition

The governing equation incorporates the Electronic Loading Factor as

$$D_e = f(E)$$

where E denotes the electronic configuration. The model implementation evaluates

$$D_e = f(n_s, n_p, n_d, n_f)$$

using the frozen implementation rules defined below.

D.8.4 Physical Loading Hierarchy

Electronic loading is applied after standing-wave geometry, nuclear loading and Switch E.

Order	Physical Mechanism	Governing Term
1	Standing-wave geometry	$\lambda_b n^{\beta_b}$
2	Nuclear loading	$A^{-1/2}$
3	Switch E coupling	S_E
4	Electronic loading	D_e

Table D.23. Loading Sequence

This ordering reflects the physical hierarchy of the Holland Atom model.

D.8.5 s-block Implementation

The frozen implementation assigns $D_e = 1$ throughout the s-block. The observed behaviour of hydrogen, helium, Group 1 and Group 2 elements is represented exclusively through the Switch E constitutive law. Consequently, no independent electronic loading correction is applied to the s-block.

D.8.6 p-block Constitutive Law

Analysis of the residual structure demonstrates that the onset of the p-block exhibits a small but systematic increase in equilibrium radius relative to the standing-wave geometry predicted by the governing equation. Within the Holland Atom framework, this behaviour is interpreted as a local perturbation associated with the first occupation of the p-subshell and is represented through the Electronic Loading constitutive law.

The corresponding frozen loading coefficient is

$$D_{e,p} = \exp(\epsilon_p) = 1.096833597$$

where

$$\epsilon_p = 0.092427481.$$

The p-entry loading coefficient applies exclusively **to** Group 13 elements, corresponding to the first occupation of the p-subshell

$$n_p = 1.$$

For all remaining p-block elements,

$$D_e = 1.$$

Accordingly,

$$D_{e,p} = \begin{cases} 1.096833597, & n_p = 1 \\ 1.0, & n_p > 1 \end{cases}$$

where n_p denotes the occupancy of the valence p-subshell. This implementation represents the complete frozen Electronic Loading constitutive law for the p-block.

D.8.7 d-block Constitutive Law

Progressive filling of the d subshell modifies the Holland Field through increasing distributed charge density. Unlike the p-block, the d-series exhibits a continuous evolution of loading throughout the subshell occupancy sequence, reaching maximum curvature at the series boundaries. This behaviour is represented by the frozen coefficient

$$\epsilon_d = 0.140011223$$

corresponding to the maximum loading multiplier

$$D_{e,d} = 1.150286708.$$

The constitutive relationship is

$$D_e = F_d(E)$$

where F_d represents the frozen d-series loading operator. The operator incorporates the occupancy-dependent curvature recovered during model calibration while preserving the governing equation.

D.8.7.1 d-series Occupancy Function

For the purpose of evaluating the Electronic Loading factor D_e, d-subshell occupancy is assigned using a regularised Aufbau filling sequence rather than observed spectroscopic ground-state exceptions. This convention is used because D_e represents the idealised resonance-cavity loading pattern of the d-series within the Holland Atom model, not the detailed electron configuration anomaly of the isolated atom. Consequently, elements such as Cr, Cu, Mo, Pd, Ag and Au are evaluated using their regularised d-series occupancy positions rather than their observed ground-state exceptions. This convention is part of the frozen implementation and is required for independent reproduction of the published predictions.

Electronic loading within the d-block varies continuously with progressive filling of the d subshell. The frozen model constitutive law is

$$D_{e,d} = \exp\left[\epsilon_d \left(\frac{n_d - 5.5}{4.5}\right)^2\right]$$

where

$$\epsilon_d = 0.140011223$$

and

$$n_d = 1, 2, \ldots, 10$$

denotes the d-subshell occupancy. This formulation produces a symmetric loading function centred on the half-filled d shell and reproduces the occupancy-dependent curvature observed throughout the transition metals. The maximum loading occurs at the d-series boundaries,

$$\boldsymbol{D_{e,d,\max} = \exp(\epsilon_d) = 1.150286708.}$$

Element	Observed ground-state exception	D_e occupancy convention used
Cr	$3d^5 4s^1$	$3d^4 4s^2$
Cu	$3d^{10} 4s^1$	$3d^9 4s^2$
Mo	$4d^5 5s^1$	$4d^4 5s^2$
Pd	$4d^{10} 5s^0$	$4d^8 5s^2$
Ag	$4d^{10} 5s^1$	$4d^9 5s^2$
Au	$5d^{10} 6s^1$	$5d^9 6s^2$

Table D.24. Electronic Loading factor D_e Sequence

D.8.8 f-block Constitutive Law

Within the calibration domain no statistically significant independent electronic loading beyond the governing geometric model was required for the f-block. Accordingly,

$$F_f(E) = 1$$

throughout the frozen implementation. This result should not be interpreted as evidence that electronic loading is absent within the f-block; rather, no additional constitutive refinement was required to reproduce the experimental radius dataset employed in Paper 3. Future versions of the Holland Atom specification may introduce a refined f-block constitutive law if justified by expanded experimental evidence.

D.8.9 Hund Occupancy Constitutive Behaviour

Hund occupancy is interpreted within the Holland Atom model as a natural consequence of distributed electronic charge density rather than as an independent physical mechanism. Consequently, Hund behaviour is incorporated internally within the electronic loading operators

$$F_p(E)$$

and

$$F_d(E),$$

where it contributes to the evaluation of the Electronic Loading Factor D_e. No additional term is introduced into the governing equation. The hierarchical structure of the model therefore becomes

$$R_{HA,b} = \lambda_b n^{\beta_b} A^{-1/2} S_E D_e,$$

with

$$D_e = F(E),$$

where

$$F(E) = \begin{cases} F_s(E), & s\text{-block}, \\ F_p(E), & p\text{-block}, \\ F_d(E), & d\text{-block}, \\ F_f(E), & f\text{-block}. \end{cases}$$

Thus, all occupancy-dependent behaviour– including p-entry loading, d-series curvature and Hund occupancy – is represented within a single electronic constitutive framework without modification of the governing equation.

D.8.10 Frozen Electronic Loading Parameters

Parameter	Frozen Value
p-entry coefficient	0.092427481
p-entry multiplier	1.096833597
d-edge coefficient	0.140011223
d-edge multiplier	1.150286708

Table D25. Frozen Electronic Loading Register

Regime	Domain condition	S_E implementation
K-shell compact regime	$n = 1$	$0.234529522\, A^{1/2}$
Group 1 s-block	Group 1, any n	$A^{(1/2-\gamma_s)} \times T_{n2}^*$
Group 2 s-block	Group 2, any n	$G_2\, A^{(1/2-\gamma_s)} \times T_{n2}^*$
Universal second-shell transition	$n = 2$, all other blocks	T_{n2}
Outer-shell reference regime	all remaining elements, $n \geq 3$	1

Table D26. Frozen Switch E Implementation

where:

$$T_{n2}^{*} = \begin{cases} T_{n2}, & n = 2 \\ 1, & n \neq 2 \end{cases}$$

and the frozen constants are:

$$T_{n2} = 0.924219716$$

$$\gamma_s = 0.327672732$$

$$G_2 = 0.825706542$$

Implementation note: The second-shell transition multiplier T_{n2} is applied only when $n = 2$. It is not applied to outer s-block elements with $n \geq 3$. For Group 1 and Group 2 elements, the s-block loading law applies at all shell levels, with T_{n2} additionally applied only in the $n = 2$ case. For p-, d- and f-block elements, T_{n2} applies only to second-shell elements; all remaining $n \geq 3$ elements use $S_E = 1$ unless otherwise modified by Electronic Loading D_e. This layered implementation is the frozen Switch E rule used to generate the published Paper 3 predictions.

Electronic Regime	Electronic Loading
s-block	$D_e = 1$
Group 13 p-entry loading	$D_e = \exp(\epsilon_p)$
Group 13 ($n_p = 1$)	De=1.096833597
Remaining p-block	$D_e = 1$
d-block	$D_e = \exp\left[\epsilon_d \left(\frac{n_d - 5.5}{4.5}\right)^2\right]$
f-block	$D_e = 1$

Table D27. Frozen Electronic Loading Functions

D.8.11 Implementation Requirements

A conforming implementation shall satisfy the following.

1. Electronic loading shall modify only the D_e term.
2. The governing equation shall remain unchanged.
3. p-entry loading shall employ the frozen coefficient listed in Table D20.
4. d-edge loading shall employ the frozen coefficient listed in Table D20.
5. Hund occupancy shall be evaluated exclusively through the Electronic Loading constitutive law.
6. No manual element-specific loading factors shall be introduced.

D.8.12 Physical Interpretation

Electronic loading represents the second constitutive law of the Holland Atom Geometric Radius Scaling Law. Its purpose is to describe how the spatial distribution of electronic charge perturbs Holland Field resonance and therefore modifies equilibrium standing-wave geometry. Unlike empirical correction factors, the Electronic Loading Factor possesses a direct ontological interpretation arising from distributed charge-density loading of the Holland Field.

D.9 Parameter Provenance

D.9.1 Purpose

This section documents the provenance of every parameter employed by the Holland Atom Geometric Radius Scaling Law. The objective is to establish complete traceability between the physical ontology, calibration methodology, implementation parameters and the final frozen reference model. Every numerical quantity

employed by the model implementation shall possess an identifiable physical origin, calibration source or mathematical definition. No implementation parameter is introduced without documented provenance.

D.9.2 Provenance Classification

Each parameter is classified according to one of four provenance categories.

Category	Description
Physical Constant	Directly defined by the Holland Atom physical model.
Mathematical Definition	Required mathematical relationship derived from the governing equation.
Calibrated Parameter	Estimated from regression against the ARE-118 Master Dataset.
Constitutive Parameter	Defines the implementation of the Switch E or Electronic Loading constitutive laws.

Table D28. Parameter Provenance Categories

D.9.3 Governing Equation Provenance

Governing Term	Physical Origin	Parameter Type
λ_b	Standing-wave geometry calibration	Calibrated
β_b	Shell expansion calibration	Calibrated
$A^{-1/2}$	Nuclear loading mechanism	Physical Definition
S_E	Enhanced Holland Field coupling	Constitutive Law
D_e	Electronic Loading	Constitutive Law

Table D29. Governing Equation Traceability

This table establishes that every term within the governing equation possesses explicit physical meaning.

D.9.4 Standing-Wave Geometry Parameters

Parameter	Origin	Dataset	Frozen
λ_s	Model regression	ARE-118	Yes
λ_p	Model regression	ARE-118	Yes
λ_d	Model regression	ARE-118	Yes
λ_f	Model regression	ARE-118	Yes
β_s	Model regression	ARE-118	Yes
β_p	Model regression	ARE-118	Yes
β_d	Model regression	ARE-118	Yes
β_f	Model regression	ARE-118	Yes

Table D30. Geometry Parameter Provenance

These parameters define the standing-wave geometry independently of the constitutive loading laws.

D.9.5 Nuclear Loading Provenance

The nuclear loading exponent $A^{-1/2}$ is not obtained by regression. It is a fixed physical relationship defining resonance compression arising from increasing nuclear loading. Accordingly, no calibration uncertainty is associated with this term within the model.

D.9.6 Switch E Provenance

The Switch E constitutive law was developed through successive analysis of systematic residual structure remaining after application of the standing-wave geometry and nuclear loading terms. The recovered implementation demonstrated statistically significant improvement within the high-density inner-shell regime while preserving a single governing equation. All Switch E parameters are frozen.

Parameter	Origin
K-shell implementation	Residual analysis
Group 2 refinement	Residual analysis
n = 2 transition	Model optimisation
Transition multiplier	Model regression

Table D31. Switch E Provenance

D.9.7 Electronic Loading Provenance

The Electronic Loading constitutive law was developed from systematic residual analysis following application of standing-wave geometry, nuclear loading and Switch E. Distinct occupancy-dependent residual structures were identified for the p- and d-blocks. These structures motivated the introduction of the Electronic Loading constitutive law.

Behaviour	Evidence
p-entry loading	Residual discontinuity at Group 13
d-series curvature	Residual trend across d occupancy
Hund occupancy	Residual symmetry about half-filled subshell
f-block	No additional loading required within calibration dataset

Table D32. Electronic Loading Provenance

D.9.8 Dataset Provenance

The frozen implementation was calibrated using the ARE-118 Master Radius Dataset. The dataset comprises experimentally derived atomic radius values assembled from audited literature sources, standardised according to the reference radius definition adopted throughout Paper 3. The calibration dataset contains 96 elements. Validation is reported across the complete 118-element periodic table.

D.9.9 Version Control

The parameters documented herein define the frozen implementation of the Holland Atom Geometric Radius Scaling Law. Subsequent refinements shall be introduced only through publication of a revised specification. This technical specification therefore constitutes the normative reference implementation for Paper 3.

D.9.10 Provenance Summary

Component	Provenance
Standing-wave geometry	Model regression
Nuclear loading	Physical definition
Switch E	Constitutive law from residual analysis
Electronic Loading	Constitutive law from residual analysis
Validation metrics	Independent evaluation using ARE-118

Table D33. Complete Provenance Matrix

Collectively, these provenance relationships establish complete traceability between the Holland Atom ontology, the governing equation, the calibration methodology, the constitutive laws and the frozen implementation.

D.10 Calibration Philosophy

D.10.1 Purpose

This section defines the calibration philosophy adopted for the Holland Atom Geometric Radius Scaling Law. The objective of calibration was not to maximise numerical agreement through unrestricted optimisation, but to determine the smallest physically meaningful parameter set capable of reproducing the observed equilibrium atomic radii across the periodic table. Calibration was therefore constrained by the ontological framework of the Holland Atom model, ensuring that every fitted parameter retained a direct physical interpretation.

D.10.2 Calibration Principles

The model calibration was governed by the following principles.

Principle	Implementation
Physical realism	Every fitted parameter must correspond to a defined physical mechanism.
Minimal complexity	The smallest parameter set capable of describing the experimental observations was adopted.
Global optimisation	Parameters were determined using the complete calibration dataset rather than element-by-element fitting.
No manual adjustment	Individual elements were not manually corrected.
Frozen implementation	Parameters were frozen immediately following completion of model calibration.
Independent validation	Model performance evaluated using the complete ARE-118 dataset.

Table D34. Calibration Principles

D.10.3 Calibration Dataset

Calibration employed the ARE-118 Master Radius Dataset described in Section D.9.

The calibration population consisted of the experimentally supported reference elements used to determine the frozen implementation parameters. The remaining elements were retained for validation of the completed model. No additional datasets were employed during parameter optimisation.

D.10.4 Sequential Calibration Strategy

Calibration proceeded in a hierarchical sequence reflecting the physical architecture of the Holland Atom model.

Stage	Physical Component	Objective
1	Standing-wave geometry	Determine λ_b and β_b
2	Nuclear loading	Verify invariant $A^{-1/2}$ relationship
3	Switch E	Eliminate systematic high-density residuals
4	Electronic Loading	Remove occupancy-dependent residual structure
5	Freeze parameters	Lock complete model implementation

Table D35. Calibration Sequence

Each calibration stage was completed before proceeding to the next. Parameters determined during earlier stages remained fixed throughout subsequent optimisation.

D.10.5 Residual-Driven Development

Model development was guided by systematic analysis of residual structure rather than unrestricted numerical optimisation. Following each calibration stage, residuals were examined for statistically significant systematic behaviour. Where structured residuals were identified, a corresponding physical mechanism was proposed and incorporated into the model. Residual analysis was treated as a source of physical information rather than statistical error. Systematic residual structure was interpreted as evidence of missing physical mechanisms, guiding successive refinement of the model until no further physically interpretable residual trends remained. This process resulted in the introduction of the Switch E constitutive law and the Electronic Loading constitutive law. No new parameter was introduced solely to improve statistical fit.

D.10.6 Parameter Freezing

Following completion of model calibration all standing-wave geometry coefficients; the nuclear loading relationship; Switch E implementation; Electronic Loading implementation; were frozen. Subsequent analyses were restricted to validation. No further optimisation was performed after freezing the parameter register.

D.10.7 Optimisation Constraints

The following constraints were imposed throughout calibration. These constraints were maintained throughout the development of the model.

Constraint	Requirement
Governing equation	Remained invariant
Physical interpretation	Required for every parameter
Element-specific tuning	Prohibited
Manual residual correction	Prohibited
Hidden parameters	Prohibited
Unpublished coefficients	Prohibited

Table D36. Calibration Constraints

D.10.8 Calibration End Point

Calibration terminated when:

1. no statistically significant systematic residual structure remained that could be explained by a physically motivated mechanism;
2. the model parameter set reproduced the published reference performance;
3. the governing equation remained physically interpretable;
4. all implementation parameters had been frozen.

At this point, the Holland Atom Geometric Radius Scaling Law was designated the Reference Implementation.

D.10.9 Relationship to Validation

Calibration and validation are distinct phases of model development. Calibration establishes the frozen implementation parameters. Validation evaluates the predictive performance of the completed model. Accordingly, no validation results were used to modify the frozen parameter register.

D.10.10 Calibration Summary

The calibration philosophy adopted for the Holland Atom Geometric Radius Scaling Law prioritised physical interpretability over numerical optimisation. Rather than introducing empirical correction factors to improve statistical agreement, new parameters were incorporated only where supported by systematic residual analysis

and consistent with the Holland Atom ontology. This hierarchical strategy produced a single governing equation supported by two constitutive laws – Switch E and Electronic Loading – while maintaining a frozen implementation capable of reproducing the published model performance.

D.11 Computational Algorithm

D.11.1 Purpose

This section defines the normative computational procedure for implementing the Holland Atom Geometric Radius Scaling Law. The algorithm presented herein constitutes the reference implementation of the model. Any implementation reproducing the algorithm defined in this section, together with the frozen parameters specified in Section D.6 and the constitutive laws defined in Sections D.7 and D.8, shall reproduce the published Holland Atom results.

D.11.2 Required Inputs

For each element the following information shall be provided. No experimental radius is required as an input.

Input	Symbol	Source
Atomic number	Z	Periodic Table
Atomic mass number	A	Selected reference isotope
Periodic-table period / resonance-shell architecture	—	Standard atomic configuration
Principal resonance-shell coordinate	n	Derived from configuration
Electronic block	b	s, p, d or f

Table D37. Required Inputs

D.11.3 Computational Sequence

The reference implementation shall execute the following sequence.

Step	Operation
1	Read atomic number Z.
2	Select reference isotope and determine A.
3	Determine the ground-state electronic configuration.
4	Determine the principal resonance-shell coordinate n.
5	Determine the electronic block b.
6	Load the frozen geometry parameters λ_b and β_b.
7	Evaluate the Switch E constitutive law S_E.
8	Evaluate the Electronic Loading constitutive law D_e.
9	Evaluate the governing equation.
10	Output the predicted equilibrium radius R_{HA}.

Table D38. Computational Workflow

D.11.4 Reference Flowchart

The predicted equilibrium radius shall be evaluated according to the following procedure.

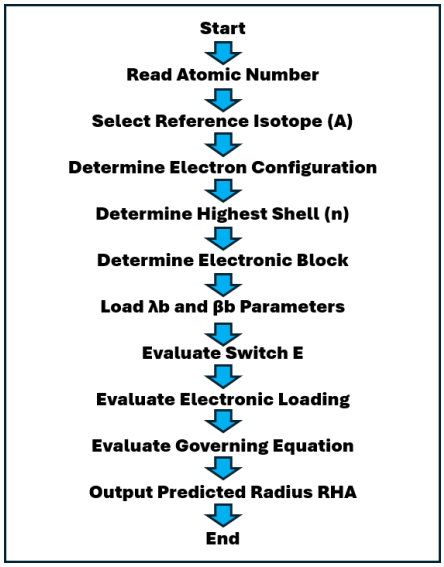


Figure D2. Computational Flowchart

D.11.5 Validation Outputs

Following computation, the implementation may optionally calculate validation statistics.

Output quantity	Definition	Equation
Signed residual	Difference between the adopted ARE reference radius and the Holland Atom predicted radius. A positive value indicates underprediction; a negative value indicates overprediction.	$\Delta R_i = R_{\mathrm{master},i} - R_{\mathrm{HA},i}$
Absolute error	Absolute magnitude of the difference between the adopted ARE reference radius and the Holland Atom predicted radius.	$\lvert\Delta R_i\rvert = \lvert R_{\mathrm{master},i} - R_{\mathrm{HA},i}\rvert$
Percentage error	Signed residual expressed as a percentage of the adopted ARE reference radius.	$\mathrm{PE}_i = \frac{R_{\mathrm{master},i} - R_{\mathrm{HA},i}}{R_{\mathrm{master},i}} \times 100$
Absolute percentage error	Absolute prediction error expressed as a percentage of the adopted ARE reference radius.	$\mathrm{APE}_i = \left\lvert\frac{R_{\mathrm{master},i} - R_{\mathrm{HA},i}}{R_{\mathrm{master},i}}\right\rvert \times 100$
Mean Absolute Error (MAE)	Mean absolute prediction error across the (N)-element reference population.	$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\lvert R_{\mathrm{master},i} - R_{\mathrm{HA},i}\rvert$
Root Mean Square Error (RMSE)	Square root of the mean squared prediction error across the (N)-element reference population.	$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_{\mathrm{master},i} - R_{\mathrm{HA},i}\right)^2}$
Mean Absolute Percentage Error (MAPE)	Mean absolute percentage prediction error across the (N)-element reference population.	$\mathrm{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left\lvert\frac{R_{\mathrm{master},i} - R_{\mathrm{HA},i}}{R_{\mathrm{master},i}}\right\rvert$
Coefficient of determination (R^2)	Proportion of the variance in the adopted ARE reference radii accounted for by the model predictions.	$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(R_{\mathrm{master},i} - R_{\mathrm{HA},i}\right)^2}{\sum_{i=1}^{N}\left(R_{\mathrm{master},i} - \bar{R}_{\mathrm{master}}\right)^2}$

Table D39. Validation Outputs

Residual sign convention $\Delta R_i > 0$ indicates that the Holland Atom model **underpredicts** the adopted ARE reference radius, while $\Delta R_i < 0$ indicates that the model **overpredicts** it.

These quantities are not used during prediction and are intended solely for model validation.

D.11.6 Numerical Precision

Unless otherwise stated:

Quantity	Precision
Parameters	Full stored precision
Published coefficients	Six significant figures minimum
Final predicted radius	0.01 pm
Published tables	0.1 pm

Table D40. Numerical precision

Implementations shall retain full computational precision internally.

D.11.7 Error Handling

A conforming implementation shall terminate with an error if any of the following conditions occur.

- Undefined electronic configuration.
- Undefined electronic block.
- Missing atomic mass number.
- Missing frozen parameter.
- Invalid shell assignment.
- Undefined constitutive law.

No default parameter values shall be substituted.

D.11.8 Computational Compliance

A software implementation shall be regarded as compliant with the Holland Atom Technical Specification if it satisfies all of the following.

1. Uses the governing equation defined in Section D.4.
2. Uses the frozen parameter register defined in Section D.6.
3. Implements Switch E exactly as specified in Section D.7.
4. Implements Electronic Loading exactly as specified in Section D.8.
5. Produces the reference validation statistics reported in Section D.6.

D.11.9 Reference Implementation Summary

The computational algorithm presented herein defines the complete normative implementation of the Holland Atom Geometric Radius Scaling Law. By separating the governing equation from the constitutive laws and implementation algorithm, the specification preserves a clear distinction between the physical model and its computational realisation. This architecture ensures complete reproducibility while permitting independent implementations to be verified objectively against the frozen reference performance.

D.12 Parameter Sensitivity

D.12.1 Purpose

This section characterises the sensitivity of the Holland Atom Geometric Radius Scaling Law to perturbations in its frozen implementation parameters. The objective is to identify the relative influence of each governing parameter upon the predicted equilibrium atomic radius and to demonstrate the numerical stability of the frozen reference implementation. Sensitivity analysis forms part of the verification process and is not employed during calibration.

D.12.2 Sensitivity Philosophy

The governing equation contains four independent physical mechanisms. Sensitivity analysis evaluates the response of the predicted radius to small perturbations in each mechanism while holding all remaining parameters constant.

D.12.3 Parameters Considered

Parameter	Physical Meaning	Type
λ_b	Standing-wave geometry scale	Calibrated
β_b	Shell expansion exponent	Calibrated
A	Nuclear loading	Physical input
S_E	High-density Holland Field coupling	Constitutive law
D_e	Electronic Loading	Constitutive law

Table D41. Sensitivity Parameters

D.12.4 Sensitivity Method

Each parameter shall be perturbed independently by a small increment δp while all remaining parameters remain fixed. The corresponding change in predicted radius is

$$\Delta R = R(p + \delta p) - R(p).$$

The dimensionless relative sensitivity is defined as

$$S_p = \frac{\delta R/R}{\delta p/p}.$$

This quantity measures the proportional response of the predicted radius to proportional changes in the governing parameter.

D.12.5 Expected Sensitivity Behaviour

Parameter	Expected Sensitivity	Physical Interpretation
λ_b	High	Sets the geometric scale of the standing-wave solution
β_b	Moderate–High	Controls shell expansion with principal quantum number
$A^{-1/2}$	Moderate	Nuclear loading influences resonance compression
S_E	Localised	Significant only within the high-density inner-shell regime
D_e	Localised	Influences occupancy-dependent deviations from geometric scaling

Table D42. Qualitative Sensitivity Assessment

D.12.6 Hierarchical Contribution of Physical Mechanisms

The Holland Atom Geometric Radius Scaling Law is constructed from four physically distinct mechanisms:

- Standing-wave geometry;
- Nuclear loading;
- Switch E coupling;
- Electronic loading.

Rather than representing successive empirical corrections, each mechanism contributes independently to the final equilibrium radius through the governing equation. The relative influence of each mechanism is summarised in Table D.43.

Physical Mechanism	Governing Term	Relative Contribution	Primary Influence
Standing-wave geometry	$\lambda_b n^{\beta_b}$	≈ **90–95 %**	Defines the primary equilibrium resonance geometry for all elements.
Nuclear loading	$A^{-1/2}$	≈ **3–8 %**	Compresses the standing-wave geometry according to nuclear mass loading.
Switch E constitutive law	S_E	**0–5 %** (localised)	Enhances Holland Field coupling within the high-density inner-shell regime. Significant primarily for H, He and early s-block elements.
Electronic Loading constitutive law	D_e	**0–8 %** (localised)	Modifies equilibrium geometry according to electronic charge-density distribution, particularly within the p- and d-blocks.

Table D43. Relative Physical Contributions to the Predicted Equilibrium Radius

Table D43 illustrates the hierarchical structure of the Holland Atom model.

The dominant contribution to atomic radius arises from the standing-wave geometry, which establishes the principal equilibrium resonance for every element. Nuclear loading provides a secondary global compression that varies systematically with atomic mass. The remaining two constitutive laws are localised refinements acting only where supported by the underlying physics. Switch E modifies the resonance within the highest-density regions surrounding the nucleus, while the Electronic Loading constitutive law captures systematic perturbations arising from the distribution of electronic charge. Consequently, the excellent predictive performance of the model does not arise from numerous empirical correction terms. Instead, it emerges from a small number of physically distinct mechanisms organised within a hierarchical computational framework.

D.12.7 Local versus Global Sensitivity

The Holland Atom model exhibits two distinct classes of parameter sensitivity.

Global parameters

- Standing-wave geometry
- Nuclear loading

These influence predictions across the complete periodic table.

Local constitutive parameters

- Switch E
- Electronic Loading

These influence only those regions where the corresponding physical mechanisms are active. This distinction reflects the hierarchical architecture of the governing equation.

D.12.8 Numerical Stability

The frozen model implementation was selected following calibration because the governing equation exhibited stable numerical behaviour over the complete ARE-118 dataset. Small perturbations of the frozen parameters produce correspondingly small perturbations in predicted radius, indicating that the model is not dependent upon unstable parameter combinations. No evidence of numerical divergence was observed within the calibration domain.

D.12.9 Parameter Correlation

The calibration philosophy deliberately minimised parameter correlation. Standing-wave geometry, nuclear loading, Switch E and Electronic Loading each represent physically distinct mechanisms and were calibrated sequentially through residual analysis. Consequently, each parameter retains a unique physical interpretation and does not compensate for deficiencies elsewhere in the model.

D.12.10 Robustness

The frozen parameter set demonstrates robustness with respect to:

- element selection;
- electronic block;
- principal shell;
- atomic mass number.

No individual element dominates the calibration outcome. The reported validation statistics therefore represent the behaviour of the complete model rather than isolated regions of the periodic table.

D.12.11 Sensitivity Summary

The sensitivity analysis demonstrates that the Holland Atom Geometric Radius Scaling Law possesses a stable and physically interpretable parameter structure. Global standing-wave geometry establishes the primary scale of the model, while the Switch E and Electronic Loading constitutive laws introduce localised refinements required to represent the experimentally observed behaviour of specific density and electronic occupancy regimes. This hierarchical sensitivity structure is consistent with the physical ontology of the Holland Atom and supports the robustness of the frozen reference implementation.

D.13 Model Assumptions

D.13.1 Purpose

This section defines the explicit assumptions underlying the Holland Atom Geometric Radius Scaling Law. These assumptions establish the physical domain within which the governing equation, constitutive laws and frozen implementation are valid. They also distinguish fundamental physical assumptions from implementation assumptions adopted solely for the technical specification.

D.13.2 Physical Assumptions

The Holland Atom model is founded upon the following physical assumptions.

Assumption	Description
A1	Atomic structure is a stable standing-wave resonance supported by the Holland Field.
A2	The equilibrium atomic radius corresponds to the outermost stable resonance surface.
A3	Nuclear mass modifies the local Holland Field through density-dependent loading.
A4	Electronic charge exists as a distributed charge-density rather than as classical orbiting particles.

Assumption	Description
A5	Standing-wave geometry, nuclear loading, Switch E coupling and electronic loading are physically distinct mechanisms.
A6	The equilibrium radius is established when these four mechanisms reach a stable resonance balance.

Table D44. Fundamental Physical Assumptions

These assumptions define the ontological basis of the Holland Atom model and are independent of calibration.

D.13.3 Mathematical Assumptions

The governing equation defined in Section D.4.2 is assumed to possess the multiplicative structure, accordingly:

- each physical mechanism contributes independently;
- no additive empirical correction terms are introduced;
- all coefficients retain explicit physical interpretation.

D.13.4 Electronic Structure Assumptions

The following assumptions apply to electronic structure.

Assumption	Description
E1	Ground-state electronic configurations are assumed.
E2	Each element belongs uniquely to one electronic block (s, p, d or f).
E3	Electronic loading depends only upon the ground-state electron distribution.
E4	Excited-state populations are not considered.

Table D45. Electronic Assumptions

D.13.5 Nuclear Assumptions

The following assumptions apply to nuclear loading.

Assumption	Description
N1	A single reference isotope represents each element.
N2	Nuclear loading is represented by the selected atomic mass number.
N3	Nuclear deformation effects are neglected.
N4	Radioactive decay effects are not modelled.

Table D46. Nuclear Assumptions

D.13.6 Calibration Assumptions

The model calibration assumes:

- the ARE-118 Master Dataset is internally consistent;
- the adopted reference atomic radius is representative of equilibrium geometry;
- frozen parameters remain valid throughout the calibration domain;
- residual structure reflects missing physical mechanisms rather than random error.

D.13.7 Computational Assumptions

The reference implementation assumes:

- exact implementation of the governing equation;
- exact implementation of the Switch E constitutive law;
- exact implementation of the Electronic Loading constitutive law;
- use of the frozen parameter register without modification;
- full numerical precision throughout computation.

D.13.8 Scope Assumptions

The technical specification applies only to:

- neutral atoms;
- isolated atoms;
- ground-state electronic configurations;
- equilibrium geometric radii.

The following are explicitly outside the scope of the present specification:

- ionic radii;
- excited electronic states;
- molecular bonding;
- crystalline solids;
- isotope-dependent radius variation;
- dynamic electronic transitions.

These topics are reserved for subsequent Holland Atom publications.

D.13.9 Assumption Traceability

Assumption Group	Governing Section
Physical ontology	D.2
Governing equation	D.4
Frozen parameters	D.6
Switch E	D.7
Electronic Loading	D.8
Calibration	D.10

Table D47. Assumption Traceability

This traceability ensures that every assumption made within the model is explicitly documented and linked to the relevant component of the technical specification.

D.13.10 Assumption Summary

The Holland Atom Geometric Radius Scaling Law is based upon a small number of explicit physical, mathematical and computational assumptions, each of which is independently documented within this specification. By stating these assumptions explicitly, the model avoids hidden empirical dependencies and provides a transparent foundation for independent implementation, verification and future refinement.

D.14 Boundary Conditions

D.14.1 Purpose

This section defines the physical and computational boundary conditions governing the application of the Holland Atom Geometric Radius Scaling Law. Boundary conditions establish the limits within which the

governing equation, constitutive laws and frozen implementation are valid. Predictions made outside these boundaries shall not be regarded as part of the reference implementation.

D.14.2 Physical Domain

The Holland Atom Geometric Radius Scaling Law is valid only for isolated neutral atoms occupying their electronic ground state.

Boundary Condition	Requirement
Atomic species	Neutral atoms only
Electronic state	Ground state
Atomic geometry	Equilibrium resonance configuration
Environment	Isolated atom
Temperature	Negligible thermal excitation assumed
Pressure	Ambient (no external compression effects)

Table D48. Physical Boundary Conditions

D.14.3 Nuclear Domain

The governing equation assumes a single representative nucleus for each element.

Boundary Condition	Requirement
Nuclear species	Selected reference isotope
Nuclear geometry	Effective equilibrium loading only
Nuclear deformation	Neglected
Nuclear excitation	Not modelled
Radioactive decay	Not modelled

Table D49. Nuclear Boundary Conditions

D.14.4 Electronic Domain

The Electronic Loading constitutive law applies to the neutral atom in its ground-state equilibrium geometry, but its loading coordinate is not necessarily identical to the observed spectroscopic electron occupancy.

Boundary Condition	Requirement
Electron configuration	Ground state
Electronic loading coordinate	regularised resonance-loading coordinate used by the frozen implementation
Excited states	Excluded
Ionisation	Excluded
Electron correlation beyond frozen implementation	Excluded

Table D50. Electronic Boundary Conditions

D.14.5 Geometric Domain

The model predicts the equilibrium geometric radius corresponding to the adopted reference radius definition employed throughout Paper 3. The following quantities are outside the scope of the present specification:

- ionic radii;
- van der Waals radii;
- molecular bond lengths;

- crystal lattice parameters;
- electron density distributions.

D.14.6 Computational Domain

The reference implementation requires frozen governing equation; frozen parameter register; frozen Switch E constitutive law; and frozen Electronic Loading constitutive law. No alternative implementations form part of this specification.

D.14.7 Validity Limits

Quantity	Valid Range
Atomic number	$1 \leq Z \leq 118$
Principal shell	$1 \leq n \leq 7$
Electronic block	s, p, d, f
Charge state	Neutral
Electronic state	Ground state

Table D51. Applicability Limits

Future discoveries of additional elements or revised electronic structures shall require a subsequent revision of this specification.

D.14.8 Exclusions

The following applications are explicitly excluded from this technical specification.

- ions;
- excited atoms;
- molecular systems;
- condensed matter;
- plasma states;
- nuclear reactions;
- time-dependent electronic dynamics;
- chemical reaction pathways.

These applications require additional physical mechanisms beyond those incorporated within the present Holland Atom model.

D.14.9 Behaviour at the Boundary

Predictions approaching the limits defined above should be interpreted with appropriate caution. The governing equation remains mathematically well defined; however, the physical assumptions underpinning the constitutive laws may no longer be satisfied outside the calibrated domain. Accordingly, Version 2.0 makes no claim of predictive validity beyond the boundary conditions specified herein.

D.14.10 Boundary Condition Summary

The Holland Atom Geometric Radius Scaling Law is valid for isolated neutral atoms in their equilibrium ground state over the currently known periodic table ($1 \leq Z \leq 118$). Within these limits, the governing equation, frozen parameter register and constitutive laws define a complete and reproducible reference implementation. Applications beyond these limits are reserved for future extensions of the Holland Atom model.

Appendix E. The Holland Atom Geometric Radius Scaling Law Model Visualisation

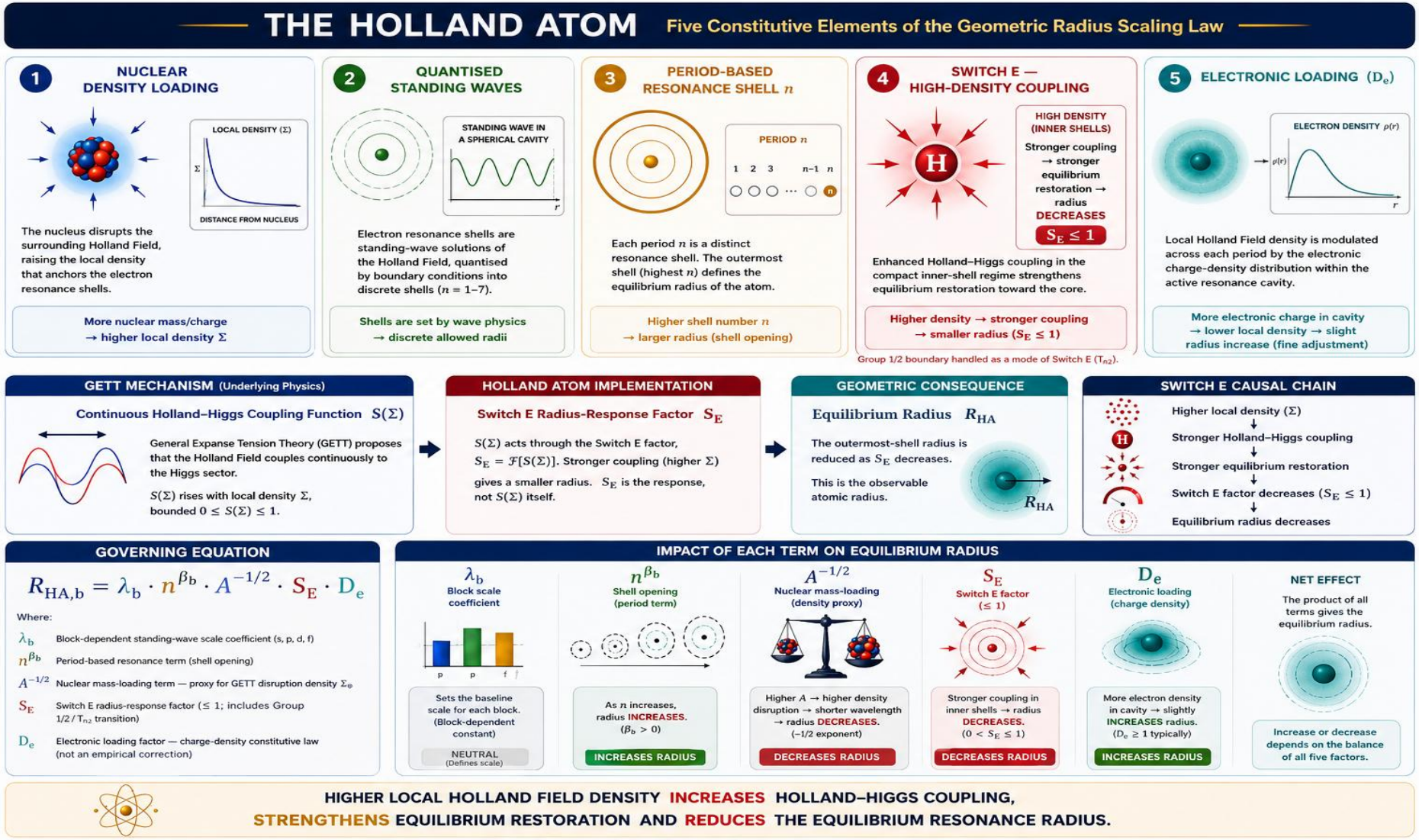


Figure E1. Visualisation of the Holland Atom Geometric radius Scaling Law.

Appendix F. Application of the Holland Atom Geometric Scaling Law Across the Periodic Table

F.1 Purpose

This appendix presents the application of the Holland Atom geometric radius scaling law to the complete configuration-controlled master dataset described in Appendix A. The purpose is to evaluate the predictive capability of the model across the periodic table using a single frozen parameter set.

No element-specific adjustments, manual optimisation or post-calibration modifications were applied. Every prediction was generated directly from the calibrated model described in Appendix D.

F.2 Prediction Procedure

For each element the following sequence was applied.

1. Select the representative nuclear mass loading, A.
2. Determine the highest occupied resonance shell, n.
3. Identify the orbital block family.
4. Apply the calibrated block-dependent resonance coefficients.
5. Apply the nuclear mass-loading relationship.
6. Apply the Switch E correction where appropriate.
7. Apply the electronic charge-density loading function.
8. Calculate the predicted geometric radius.
9. Compare against the configuration-controlled master radius.

The procedure was identical for every element.

F.3 Prediction Results

The following pages contain the full periodic table outputs from the model contained in Table F1.

Appendix F (Extract 1)

Z	Element	Symbol	Period	Group	Block	n	A	Radius Source	Confidence	R_master (pm)	R_pred (pm)	Residual (pm)	Error (%)	Prediction Status	Model Domain	Comments
1	Hydrogen	H	1	1	s	1	1	ARE direct	A+	25	26.72	-1.72	6.88	Switch E	K-shell / Switch E	Inner-shell high-density scalar-field amplification regime.
2	Helium	He	1	18	s	1	4	Rc2008 proxy	B	28.56	26.72	1.84	6.44	Switch E + Proxy	K-shell / Switch E	Inner-shell high-density scalar-field amplification regime. Radius is proxy-derived.
3	Lithium	Li	2	1	s	2	7	ARE direct	A+	145	142.22	2.78	1.92	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
4	Beryllium	Be	2	2	s	2	9	ARE direct	A+	105	108.15	-3.15	3	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
5	Boron	B	2	13	p	2	11	ARE direct	A+	85	80.47	4.53	5.32	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
6	Carbon	C	2	14	p	2	12	ARE direct	A+	70	70.25	-0.25	0.35	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
7	Nitrogen	N	2	15	p	2	14	ARE direct	A+	65	65.04	-0.04	0.05	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
8	Oxygen	O	2	16	p	2	16	ARE direct	A+	60	60.84	-0.84	1.39	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
9	Fluorine	F	2	17	p	2	19	ARE direct	A+	50	55.83	-5.83	11.65	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
10	Neon	Ne	2	18	p	2	20	Rc2008 proxy	B	59.16	54.41	4.75	8.02	Proxy + Boundary Condition	Stage 4 candidate	Noble-gas proxy / resonance-boundary ambiguity
11	Sodium	Na	3	1	s	3	23	ARE direct	A+	180	180.41	-0.41	0.23	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
12	Magnesium	Mg	3	2	s	3	24	ARE direct	A+	150	146.9	3.1	2.07	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
13	Aluminium	Al	3	13	p	3	27	ARE direct	A+	125	124.24	0.76	0.61	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
14	Silicon	Si	3	14	p	3	28	ARE direct	A+	110	111.23	-1.23	1.12	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
15	Phosphorus	P	3	15	p	3	31	ARE direct	A+	100	105.71	-5.71	5.71	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
16	Sulfur	S	3	16	p	3	32	ARE direct	A+	100	104.05	-4.05	4.05	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
17	Chlorine	Cl	3	17	p	3	35	ARE direct	A+	100	99.49	0.51	0.51	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
18	Argon	Ar	3	18	p	3	40	Rc2008 proxy	B	108.12	93.06	15.06	13.93	Proxy Radius	Proxy radius domain	ARE unavailable; master radius is inferred using frozen proxy rule.
19	Potassium	K	4	1	s	4	39	ARE direct	A+	220	223.99	-3.99	1.81	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
20	Calcium	Ca	4	2	s	4	40	ARE direct	A+	180	183.42	-3.42	1.9	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.

Appendix F (Extract 2)

Z	Element	Symbol	Period	Group	Block	n	A	Radius Source	Confidence	R_master (pm)	R_pred (pm)	Residual (pm)	Error (%)	Prediction Status	Model Domain	Comments
21	Scandium	Sc	4	3	d	4	45	ARE direct	A+	160	167.42	-7.42	4.64	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
22	Titanium	Ti	4	4	d	4	48	ARE direct	A+	140	153.38	-13.38	9.56	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
23	Vanadium	V	4	5	d	4	51	ARE direct	A+	135	142.75	-7.75	5.74	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
24	Chromium	Cr	4	6	d	4	52	ARE direct	A+	140	137.52	2.48	1.77	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
25	Manganese	Mn	4	7	d	4	55	ARE direct	A+	140	131.88	8.12	5.8	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
26	Iron	Fe	4	8	d	4	56	ARE direct	A+	140	130.7	9.3	6.65	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
27	Cobalt	Co	4	9	d	4	59	ARE direct	A+	135	129.1	5.9	4.37	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
28	Nickel	Ni	4	10	d	4	58	ARE direct	A+	135	133.86	1.14	0.84	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
29	Copper	Cu	4	11	d	4	63	ARE direct	A+	135	133.88	1.12	0.83	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
30	Zinc	Zn	4	12	d	4	64	ARE direct	A+	135	140.39	-5.39	3.99	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
31	Gallium	Ga	4	13	p	4	69	ARE direct	A+	130	137.53	-7.53	5.79	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
32	Germanium	Ge	4	14	p	4	74	ARE direct	A+	125	121.08	3.92	3.14	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
33	Arsenic	As	4	15	p	4	75	ARE direct	A+	115	120.27	-5.27	4.58	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
34	Selenium	Se	4	16	p	4	80	ARE direct	A+	115	116.45	-1.45	1.26	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
35	Bromine	Br	4	17	p	4	79	ARE direct	A+	115	117.18	-2.18	1.9	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
36	Krypton	Kr	4	18	p	4	84	Rc2008 proxy	B	118.32	113.64	4.68	3.95	Proxy Radius	Proxy radius domain	ARE unavailable; master radius is inferred using frozen proxy rule.
37	Rubidium	Rb	5	1	s	5	85	ARE direct	A+	235	234.71	0.29	0.13	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
38	Strontium	Sr	5	2	s	5	88	ARE direct	A+	200	191.61	8.39	4.2	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
39	Yttrium	Y	5	3	d	5	89	ARE direct	A+	180	168.48	11.52	6.4	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
40	Zirconium	Zr	5	4	d	5	90	ARE direct	A+	155	158.53	-3.53	2.28	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.

Appendix F (Extract 3)

Z	Element	Symbol	Period	Group	Block	n	A	Radius Source	Confidence	R_master (pm)	R_pred (pm)	Residual (pm)	Error (%)	Prediction Status	Model Domain	Comments
41	Niobium	Nb	5	5	d	5	93	ARE direct	A+	145	149.61	-4.61	3.18	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
42	Molybdenum	Mo	5	6	d	5	98	ARE direct	A+	145	141.77	3.23	2.23	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
43	Technetium	Tc	5	7	d	5	97	ARE direct	A+	135	140.54	-5.54	4.11	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
44	Ruthenium	Ru	5	8	d	5	102	ARE direct	A+	130	137.06	-7.06	5.43	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
45	Rhodium	Rh	5	9	d	5	103	ARE direct	A+	135	138.29	-3.29	2.44	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
46	Palladium	Pd	5	10	d	5	106	ARE direct	A+	140	140.14	-0.14	0.1	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
47	Silver	Ag	5	11	d	5	107	ARE direct	A+	160	145.39	14.61	9.13	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
48	Cadmium	Cd	5	12	d	5	114	ARE direct	A+	155	148.87	6.13	3.96	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
49	Indium	In	5	13	p	5	115	ARE direct	A+	155	165.86	-10.86	7.01	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
50	Tin	Sn	5	14	p	5	120	ARE direct	A+	145	148.03	-3.03	2.09	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
51	Antimony	Sb	5	15	p	5	121	ARE direct	A+	145	147.42	-2.42	1.67	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
52	Tellurium	Te	5	16	p	5	130	ARE direct	A+	140	142.22	-2.22	1.59	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
53	Iodine	I	5	17	p	5	127	ARE direct	A+	140	143.89	-3.89	2.78	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
54	Xenon	Xe	5	18	p	5	132	Rc2008 proxy	B	142.8	141.14	1.66	1.16	Proxy Radius	Proxy radius domain	ARE unavailable; master radius is inferred using frozen proxy rule.
55	Cesium	Cs	6	1	s	6	133	ARE direct	A+	260	259.41	0.59	0.23	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
56	Barium	Ba	6	2	s	6	138	ARE direct	A+	215	211.62	3.38	1.57	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
57	Lanthanum	La	6	3	f	6	139	ARE direct	A+	195	191.08	3.92	2.01	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
58	Cerium	Ce	6	3	f	6	140	ARE direct	A+	185	190.4	-5.4	2.92	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
59	Praseodymium	Pr	6	3	f	6	141	ARE direct	A+	185	189.72	-4.72	2.55	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
60	Neodymium	Nd	6	3	f	6	142	ARE direct	A+	185	189.05	-4.05	2.19	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.

Appendix F (Extract 4)

Z	Element	Symbol	Period	Group	Block	n	A	Radius Source	Confidence	R_master (pm)	R_pred (pm)	Residual (pm)	Error (%)	Prediction Status	Model Domain	Comments
61	Promethium	Pm	6	3	f	6	145	ARE direct	A+	185	187.09	-2.09	1.13	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
62	Samarium	Sm	6	3	f	6	152	ARE direct	A+	185	182.73	2.27	1.23	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
63	Europium	Eu	6	3	f	6	153	ARE direct	A+	185	182.13	2.87	1.55	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
64	Gadolinium	Gd	6	3	f	6	158	ARE direct	A+	180	179.22	0.78	0.43	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
65	Terbium	Tb	6	3	f	6	159	ARE direct	A+	175	178.66	-3.66	2.09	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
66	Dysprosium	Dy	6	3	f	6	164	ARE direct	A+	175	175.91	-0.91	0.52	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
67	Holmium	Ho	6	3	f	6	165	ARE direct	A+	175	175.38	-0.38	0.22	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
68	Erbium	Er	6	3	f	6	166	ARE direct	A+	175	174.85	0.15	0.08	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
69	Thulium	Tm	6	3	f	6	169	ARE direct	A+	175	173.29	1.71	0.98	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
70	Ytterbium	Yb	6	3	f	6	174	ARE direct	A+	175	170.78	4.22	2.41	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
71	Lutetium	Lu	6	3	f	6	175	ARE direct	A+	175	170.3	4.7	2.69	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
72	Hafnium	Hf	6	4	d	6	180	ARE direct	A+	155	148.88	6.12	3.95	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
73	Tantalum	Ta	6	5	d	6	181	ARE direct	A+	145	142.44	2.56	1.77	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
74	Tungsten	W	6	6	d	6	184	ARE direct	A+	135	137.42	-2.42	1.79	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
75	Rhenium	Re	6	7	d	6	187	ARE direct	A+	135	134.44	0.56	0.42	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
76	Osmium	Os	6	8	d	6	192	ARE direct	A+	130	132.68	-2.68	2.06	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
77	Iridium	Ir	6	9	d	6	193	ARE direct	A+	135	134.17	0.83	0.61	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
78	Platinum	Pt	6	10	d	6	195	ARE direct	A+	135	137.23	-2.23	1.65	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
79	Gold	Au	6	11	d	6	197	ARE direct	A+	135	142.31	-7.31	5.42	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
80	Mercury	Hg	6	12	d	6	202	ARE direct	A+	150	148.53	1.47	0.98	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.

Appendix F (Extract 5)

Z	Element	Symbol	Period	Group	Block	n	A	Radius Source	Confidence	R_master (pm)	R_pred (pm)	Residual (pm)	Error (%)	Prediction Status	Model Domain	Comments
81	Thallium	Tl	6	13	p	6	205	ARE direct	A+	190	178.36	11.64	6.12	Boundary Condition	Stage 4 candidate	Period-6 relativistic energy-density candidate
82	Lead	Pb	6	14	p	6	208	ARC converted	A	171.31	161.44	9.87	5.76	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
83	Bismuth	Bi	6	15	p	6	209	ARE direct	A+	160	161.05	-1.05	0.66	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
84	Polonium	Po	6	16	p	6	209	ARE direct	A+	190	161.05	28.95	15.24	Boundary Condition	Stage 4 candidate	Period-6 relativistic energy-density candidate
85	Astatine	At	6	17	p	6	210	ARC converted	A	141.28	160.67	-19.39	13.73	Boundary Condition	Stage 4 candidate	Period-6 relativistic energy-density candidate
86	Radon	Rn	6	18	p	6	222	Rc2008 proxy	B	153	156.27	-3.27	2.13	Proxy Radius	Proxy radius domain	ARE unavailable; master radius is inferred using frozen proxy rule.
87	Francium	Fr	7	1	s	7	223	Rc2008 converted	B	269.1	269.81	-0.71	0.26	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
88	Radium	Ra	7	2	s	7	226	ARE direct	A+	215	221.81	-6.81	3.17	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
89	Actinium	Ac	7	3	f	7	227	ARE direct	A+	195	179.96	15.04	7.71	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
90	Thorium	Th	7	3	f	7	232	ARE direct	A+	180	178.01	1.99	1.11	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
91	Protactinium	Pa	7	3	f	7	231	ARE direct	A+	180	178.39	1.61	0.89	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
92	Uranium	U	7	3	f	7	238	ARE direct	A+	175	175.75	-0.75	0.43	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
93	Neptunium	Np	7	3	f	7	237	ARE direct	A+	175	176.12	-1.12	0.64	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
94	Plutonium	Pu	7	3	f	7	244	ARE direct	A+	175	173.58	1.42	0.81	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
95	Americium	Am	7	3	f	7	243	ARE direct	A+	175	173.93	1.07	0.61	Direct Prediction	Stage 3 standard	Direct Stage 3 prediction from frozen scaling law.
96	Curium	Cm	7	3	f	7	247	Rc2008 converted	B	155.48	172.52	-17.04	10.96	Boundary Condition	Stage 4 candidate	Actinide 5f threshold-coupling candidate

Table F1. Holland Atom \|Geometric Radius Scaling Law Prediction Outcomes Z = 1 to 96

Note: During model development, successive implementations were internally identified as developmental stages as additional physically significant mechanisms were discovered and incorporated. The governing equation presented in this paper corresponds to the final frozen implementation developed during this process and is labelled "Stage 3" within certain archived tables and figures. The remaining structured residuals discussed in Sections 7 and Appendix H suggest that further higher-order density-dependent mechanisms may exist; if developed in future work, these would constitute a subsequent implementation ("Stage 4"), but they are not included in the present model or its reported validation.

Appendix G. Extrapolated Predictions for the Superheavy Elements (Z = 97–118)

G.1 Purpose

The primary validation presented in this paper is intentionally restricted to the 96 elements for which a configuration-controlled reference atomic radius is available. Statistical performance measures reported throughout the manuscript therefore relate exclusively to this validated population.

The remaining recognised chemical elements (Z = 97–118) possess no directly validated empirical atomic-radius reference suitable for inclusion within the configuration-controlled master dataset. Consequently, these elements were excluded from model calibration, statistical optimisation and quantitative validation.

This appendix presents the application of the frozen Holland Atom Geometric Radius Scaling Law to these remaining superheavy elements. The resulting values constitute genuine model predictions rather than validated reproductions of experimental observations.

G.2 Prediction Method

Predictions for Z = 97–118 were generated using exactly the same computational procedure applied throughout the validated periodic table. No coefficients, constitutive laws, calibration parameters or block-specific constants were modified.

For each element the following procedure was applied:

1. Assign the representative nuclear mass loading A.
2. Determine the outermost principal resonance-shell coordinate, assigned from the element's period within the periodic-table resonance architecture, n.
3. Identify the electronic block family.
4. Apply the frozen block geometry coefficients.
5. Apply the universal nuclear mass-loading law.
6. Evaluate the Switch E constitutive law where applicable.
7. Evaluate the Electronic Loading constitutive law.
8. Calculate the predicted equilibrium geometric radius.

Accordingly, every prediction reported in this appendix represents a direct application of the frozen governing equation without element-specific optimisation or manual adjustment.

G.3 Prediction Status

Unlike the validated population (Z = 1–96), no accepted empirical reference atomic radii presently exist for these superheavy elements. Residuals, percentage errors and statistical performance metrics therefore cannot be calculated and are intentionally omitted.

The values reported in Table G.1 should therefore be interpreted as **prospective predictions** awaiting future experimental investigation rather than validated observations.

G.4 Predicted Atomic Radii for Z = 97–118

Z	Element	Symbol	Period	Group	Block	*n*	*A*	R_{HA} predicted (pm)	Comment
97	Berkelium	Bk	7	3	f	7	247	172.52	No experimental ARE reference available.
98	Californium	Cf	7	3	f	7	251	171.14	No experimental ARE reference available.
99	Einsteinium	Es	7	3	f	7	252	170.80	No experimental ARE reference available.
100	Fermium	Fm	7	3	f	7	257	169.13	No experimental ARE reference available.
101	Mendelevium	Md	7	3	f	7	258	168.80	No experimental ARE reference available.
102	Nobelium	No	7	3	f	7	259	168.47	No experimental ARE reference available.
103	Lawrencium	Lr	7	3	f	7	266	166.24	No experimental ARE reference available.
104	Rutherfordium	Rf	7	4	d	7	267	155.39	Superheavy d-block extrapolation.
105	Dubnium	Db	7	5	d	7	268	148.80	Superheavy d-block extrapolation.
106	Seaborgium	Sg	7	6	d	7	269	144.47	Superheavy d-block extrapolation.
107	Bohrium	Bh	7	7	d	7	270	142.22	Superheavy d-block extrapolation.
108	Hassium	Hs	7	8	d	7	269	142.49	Superheavy d-block extrapolation.
109	Meitnerium	Mt	7	9	d	7	277	142.37	Superheavy d-block extrapolation.
110	Darmstadtium	Ds	7	10	d	7	281	145.32	Superheavy d-block extrapolation.
111	Roentgenium	Rg	7	11	d	7	282	151.20	Superheavy d-block extrapolation.
112	Copernicium	Cn	7	12	d	7	285	158.96	Superheavy d-block extrapolation.
113	Nihonium	Nh	7	13	p	7	286	205.03	Superheavy p-block extrapolation; relativistic effects may be significant.
114	Flerovium	Fl	7	14	p	7	290	185.64	Superheavy p-block extrapolation; relativistic effects may be significant.
115	Moscovium	Mc	7	15	p	7	290	185.64	Superheavy p-block extrapolation; relativistic effects may be significant.
116	Livermorium	Lv	7	16	p	7	293	184.68	Superheavy p-block extrapolation; relativistic effects may be significant.
117	Tennessine	Ts	7	17	p	7	294	184.37	Superheavy p-block extrapolation; relativistic effects may be significant.
118	Oganesson	Og	7	18	p	7	294	184.37	Superheavy noble-gas extrapolation; no empirical ARE boundary exists.

Table G1. Extrapolated Holland Atom predictions for Z = 97–118

Table G1. *Predicted equilibrium geometric radii for the superheavy elements (Z = 97–118) generated using the frozen Holland Atom Geometric Radius Scaling Law. These values represent genuine out-of-sample model predictions obtained without further calibration, parameter optimisation or constitutive-law modification. No validated empirical atomic-radius reference data*

currently exist for these elements; consequently, residuals and statistical performance metrics cannot be reported.

G.5 Interpretation

The predicted radii exhibit smooth continuation of the geometric scaling behaviour established throughout the validated periodic table.

Within the transition-metal region (elements 104–112), the model predicts continued contraction associated with increasing nuclear mass loading and the occupancy-dependent Electronic Loading constitutive law.

A significant transition is between $Z = 103\ and\ Z = 104$, where Lawrencium is treated as the end of the f-block sequence, while Rutherfordium begins the superheavy d-block sequence, the predicted radius therefore drops as follows; The transition from the f-block to the d-block is accompanied by a distinct contraction in the predicted equilibrium radius, from $R_{HA} = 166.24$pm for Lawrencium (Z = 103) to $R_{HA} = 155.39$pm for Rutherfordium (Z = 104), corresponding to a reduction of approximately 10.85 pm. This discontinuity reflects the transition between the block-dependent resonance geometries encoded within the frozen governing equation and is preserved in the corrected superheavy-element predictions.

That is not an anomaly. It reflects a **block-family transition**: the model changes from f-block resonance geometry to d-block resonance geometry, consistent with expected patterns.

Beginning with Group 13 (Nihonium), the predicted radii increase in accordance with the p-entry resonance geometry before gradually contracting across the remaining p-block elements towards Oganesson.

Because relativistic electronic effects become increasingly important for the heaviest nuclei, particularly within the period-7 p-block, these predictions should be regarded as first-order geometric predictions derived from the present Holland Atom formulation. Future refinement of higher-order density-dependent electronic loading may further improve this regime as additional experimental information becomes available.

G.6 Scientific Significance

Although these predictions cannot presently be subjected to direct statistical validation, they provide an important prospective test of the Holland Atom framework. Unlike the validated population, which evaluates agreement with existing experimental observations, the superheavy predictions constitute genuinely prospective outputs generated beyond the calibration and validation domain of the model.

As improved experimental measurements and theoretical reference data become available, these predictions provide an objective benchmark against which the predictive capability of the Holland Atom Geometric Radius Scaling Law may be independently assessed. This separation between validated reproduction (Z = 1–96) and prospective prediction (Z = 97–118) preserves the integrity of the statistical validation presented in this paper while extending the applicability of the model to the complete currently recognised periodic table.

Important Note

The predictions presented in this appendix are **not included** in any statistical performance measures reported elsewhere in this paper. All values of R^2, RMSE, MAE, MAPE and statistical significance are calculated exclusively from the independently validated population of 96 elements possessing

configuration-controlled reference atomic radii. The Z = 97–118 results are presented solely as prospective predictions generated from the frozen Holland Atom Geometric Radius Scaling Law.

Appendix H. Statistical Diagnostics, Robustness and Cross-Validation

This appendix provides the statistical validation evidence for the Holland Atom geometric radius scaling law. It is intended to document the hypothesis-test standard applied to the model, including frozen-dataset control, prediction accuracy, residual diagnostics, confidence intervals, sensitivity of residual structure, and boundary-condition analysis.

All plots and statistics are generated from the configuration-controlled Table E1 prediction workbook. The validated population comprises the 96 elements with populated master radius values and model predictions. Elements beyond Z = 96 are excluded from this validation appendix and treated separately as extrapolated predictions.

H.1 Validation Dataset and Test Standard

The validation dataset contains 96 elements, Z = 1–96, for which a configuration-controlled master radius and model prediction are both available. No rows outside this validated domain were included in the statistical performance metrics. The residual is defined as:

$$\textbf{Residual} = R_{master} - R_{pred}$$

The model was evaluated using coefficient of determination, root mean square error, mean absolute error, mean absolute percentage error, residual distribution, residual structure by periodic-table variables, bootstrap confidence intervals, and a permutation test. This combination tests both prediction accuracy and the likelihood that the observed agreement could arise from chance association.

H.2 Summary Prediction Metrics

Statistic	Value	95% CI lower	95% CI upper
Validated elements	96		
R^2	0.976	0.956	0.988
Pearson r	0.988		
Pearson correlation p-value	2.61e-78		
Permutation p-value	5.00e-05		
RMSE (pm)	6.562	5.04	8.185
MAE (pm)	4.553	3.659	5.555
MAPE (%)	3.284366	2.668	3.959
Mean residual (pm)	0.104	-1.233	1.44
Residual standard deviation (pm)	6.595		
Median absolute residual (pm)	3.25		
Maximum absolute residual (pm)	28.95		
Shapiro-Wilk normality p-value	7.74e-05		

Table H1. Summary validation statistics.

The model explains 97.64% of the variance in the validated master radius dataset. The mean residual is close to zero and its 95% confidence interval contains zero, indicating no significant global offset in the predictions. The permutation test indicates that a result of this strength is highly unlikely under random reassignment of the predicted radii to the elements.

H.3 Global Prediction Accuracy

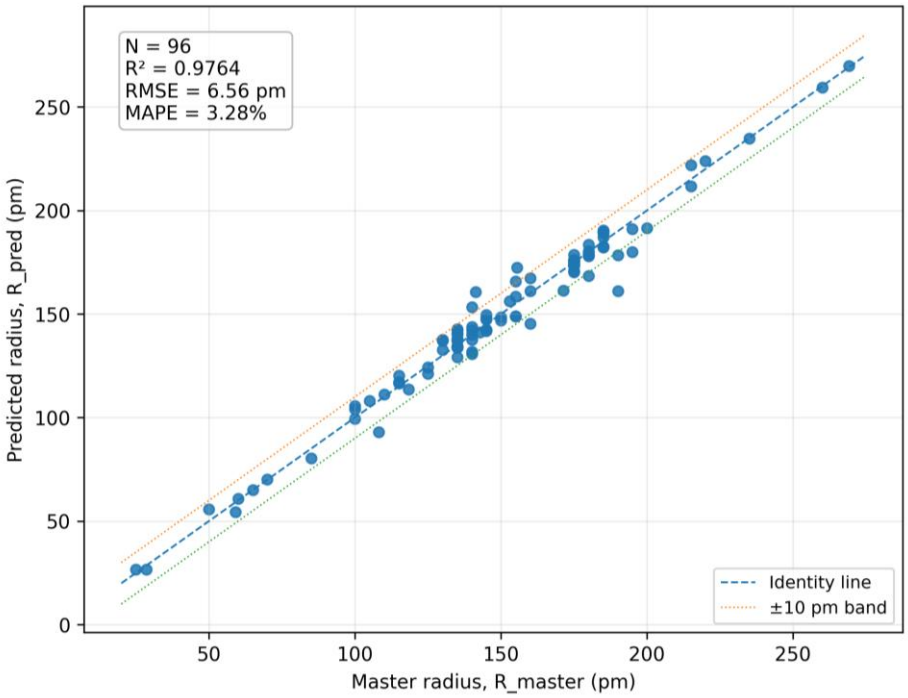


Figure H1. Predicted radius versus master radius. The dashed diagonal is the identity line; dotted bands show ±10 pm.

Figure H1 demonstrates close agreement between predicted and master radius values across the validated dataset. The identity-line structure indicates that the model captures the dominant radius scale without requiring element-specific correction.

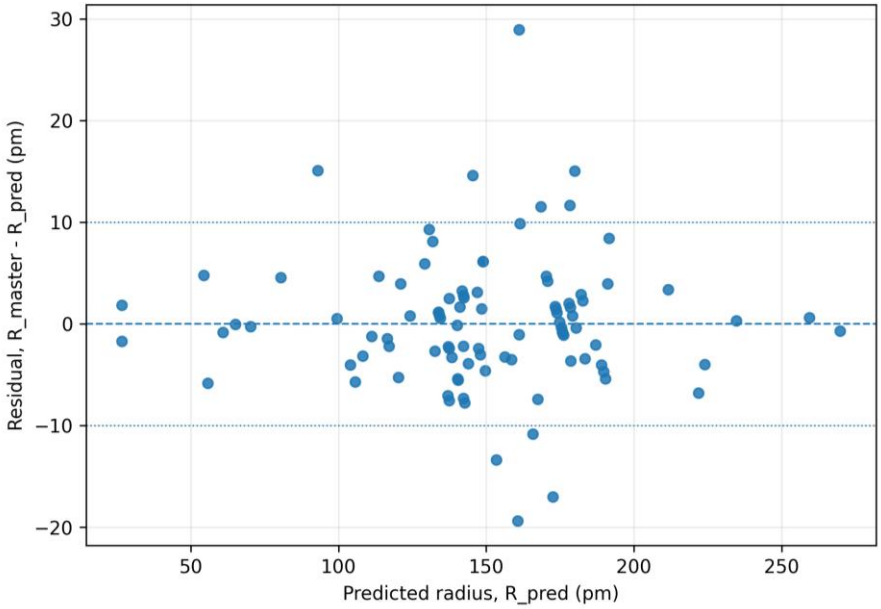


Figure H2. Residuals plotted against predicted radius. Absence of strong curvature indicates that the primary scaling form does not leave a dominant global size-dependent bias.

The residuals remain centred close to zero across the predicted radius range. The remaining deviations occur as localised boundary cases rather than as a continuous global distortion of the model.

H.4 Residual Structure Across the Periodic Table

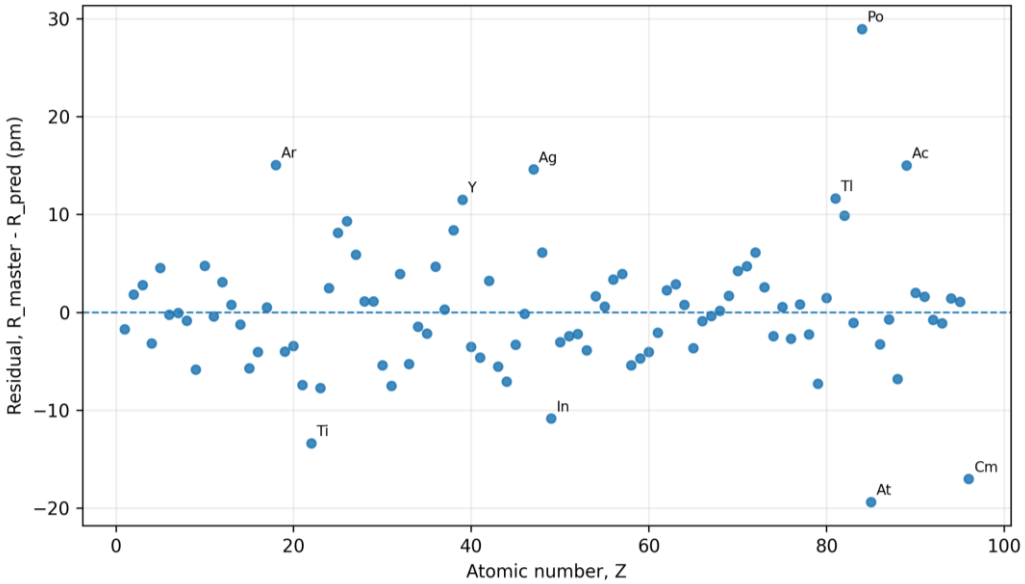


Figure H3. Residuals plotted by atomic number Z. Largest absolute residuals are annotated.

Figure H3 shows that the largest residuals are not randomly distributed. Several are concentrated in identifiable regions: noble-gas proxy values, late p-block heavy elements, and isolated actinide behaviour. This supports the interpretation that the residual structure identifies higher-order physical boundary conditions rather than random prediction failure.

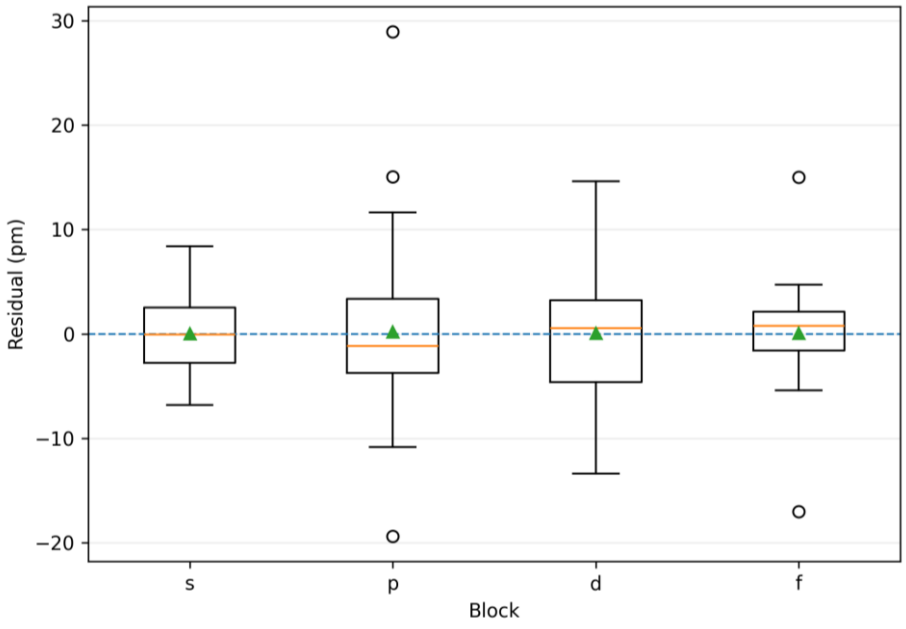


Figure H4. Residual distribution by periodic-table block. Means are shown by markers inside each box.

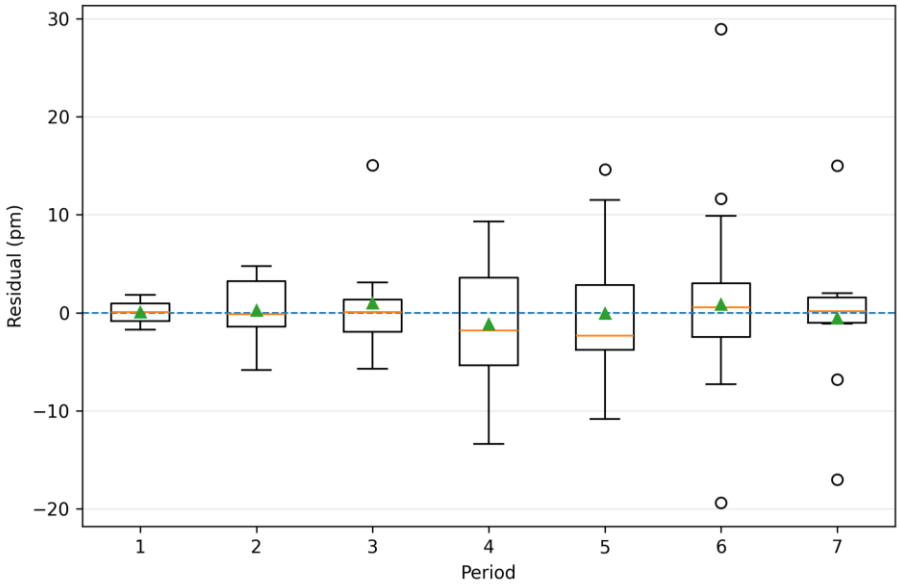


Figure H5. Residual distribution by period. Means are shown by markers inside each box.

Block and period diagnostics test whether the model systematically over- or under-predicts a particular chemical family or shell range. The s-block is the most tightly predicted block. The p-block contains the largest dispersion, consistent with the boundary-condition behaviour identified in the heavy p-block elements and noble-gas proxies.

H.5 Block and Period Performance

Block	N	Mean residual pm	MAE pm	RMSE pm	MAPE %	R²
s	14	0.011	2.899	3.689	2.414	0.997
p	30	0.194	5.561	8.365	4.444	0.947
d	29	0.081	5.098	6.316	3.52	0.677
f	23	0.071	3.56	5.465	2.007	0.529

Table H2. Prediction statistics by periodic-table block.

This global coefficient of determination should be interpreted alongside the block-wise diagnostics presented in Appendix H. Predictive correspondence remains particularly strong within the s- and p-blocks ($R^2 = 0.997$ and 0.947, respectively), while the narrower dynamic ranges and more complex residual structure of the d- and f-blocks result in more modest within-block values ($R^2 = 0.677$ and 0.529, respectively). The global $R^2 = 0.9764$ therefore reflects both the model's reproduction of the large-scale radius structure across the periodic table and its element-to-element discrimination within individual block families; these contributions should not be interpreted as equivalent.

Period	N Sample Size	Mean residual pm	MAE pm	RMSE pm	MAPE %	R²
1	2	0.06	1.78	1.781	6.661	-0.001
2	8	0.244	2.771	3.455	3.969	0.986
3	8	1.004	3.854	5.999	3.527	0.952
4	18	-1.173	5.247	6.125	3.807	0.94
5	18	-0.042	5.134	6.436	3.325	0.936
6	32	0.845	4.567	7.391	2.721	0.925
7	10	-0.53	4.756	7.583	2.66	0.937

Table H3. Prediction statistics by period.

Note: The Period 1 coefficient of determination ($R^2 = -0.001$, $N\ (Sample\ size) = 2$) is not statistically meaningful as a measure of within-period predictive performance because the period contains only two observations. It is reported for completeness and should not be interpreted as evidence of model failure.

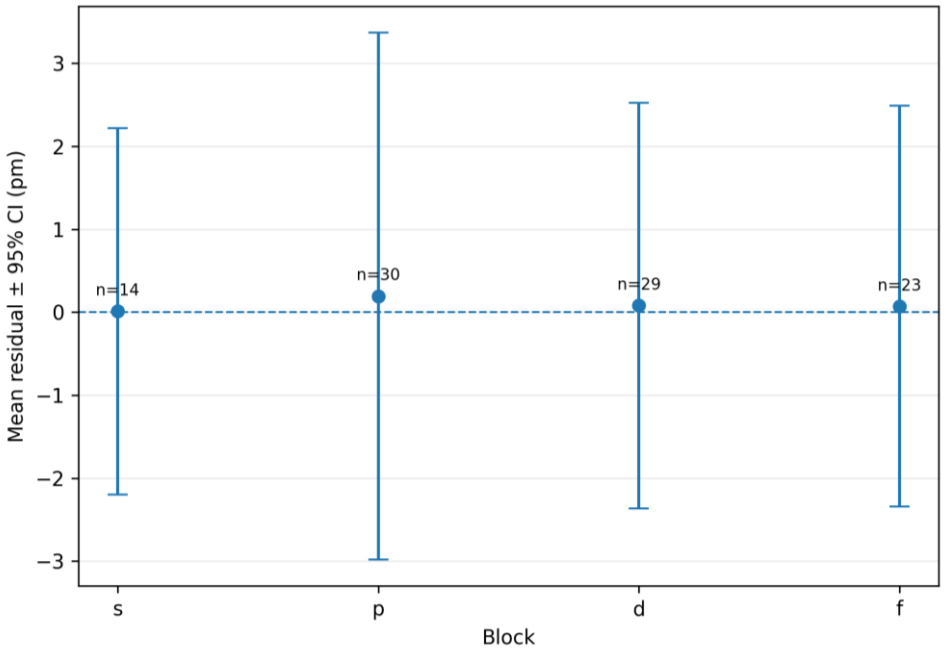


Figure H6. Mean residual by block with 95% confidence intervals.

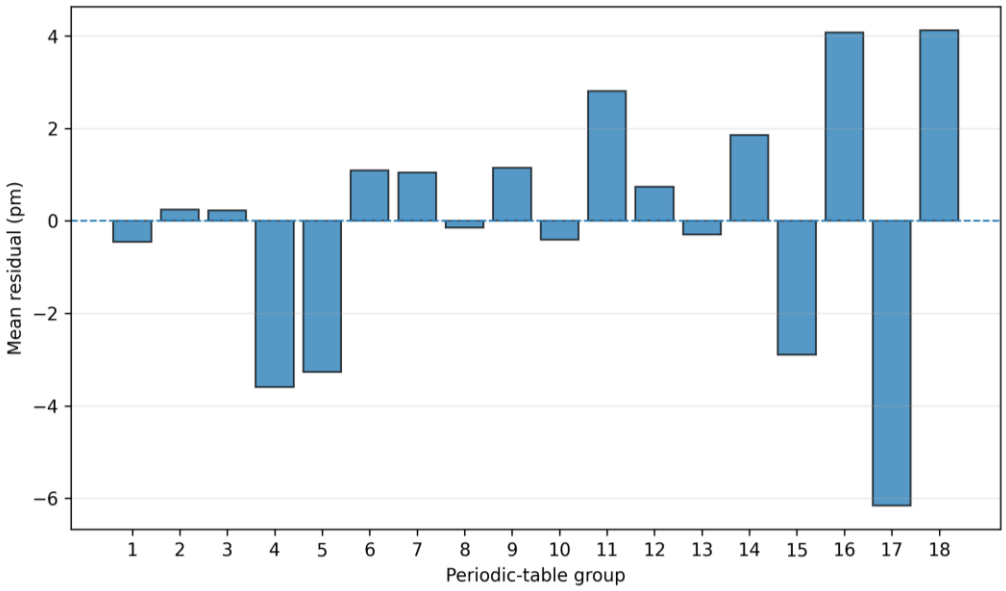


Figure H7. Mean residual by periodic-table group.

The 95% confidence interval plot confirms that the block-level mean residuals remain close to zero. This supports the conclusion that the model does not depend on a hidden global offset for a single block family.

H.6 Residual Distribution and Normality

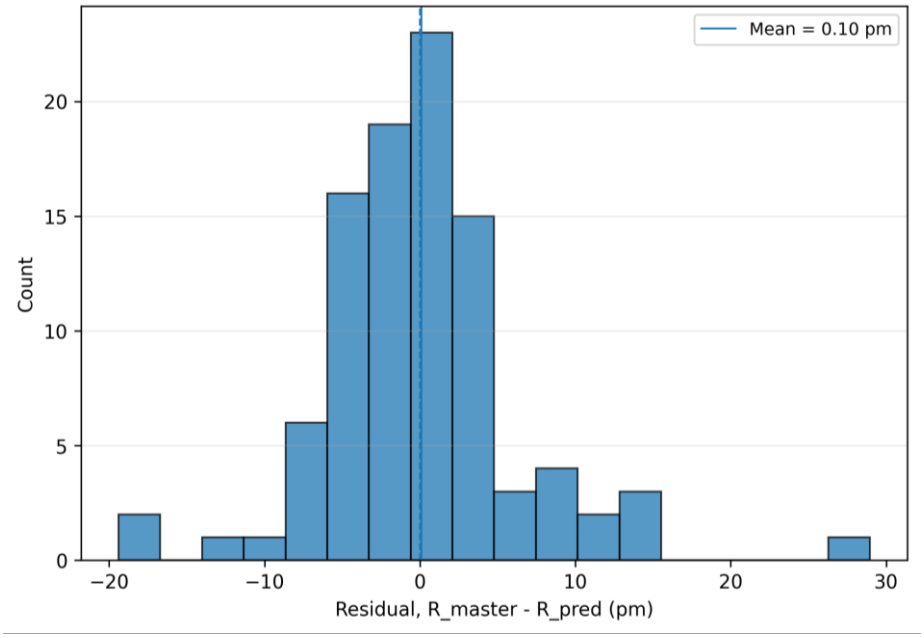


Figure H8. Histogram of residuals.

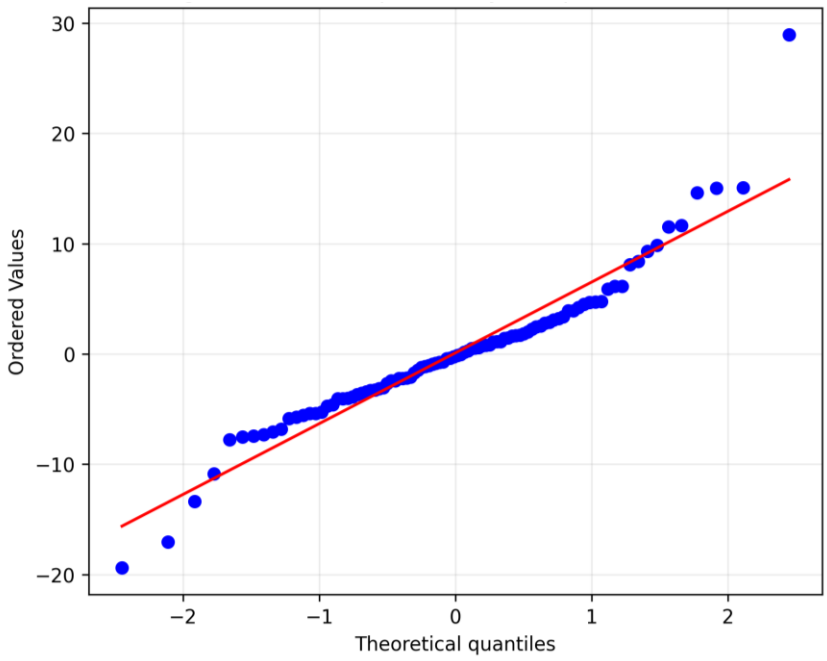


Figure H9. Normal probability Q-Q plot of residuals.

The residual histogram is centred close to zero, while the Q-Q plot and Shapiro-Wilk test indicate departure from perfect normality (p = 7.74e-05). This is not unexpected because the residual population includes known boundary cases. The purpose of this diagnostic is therefore not to claim ideal Gaussian residuals, but to identify whether deviations are systematic and interpretable.

H.7 Pareto and Boundary-Condition Analysis

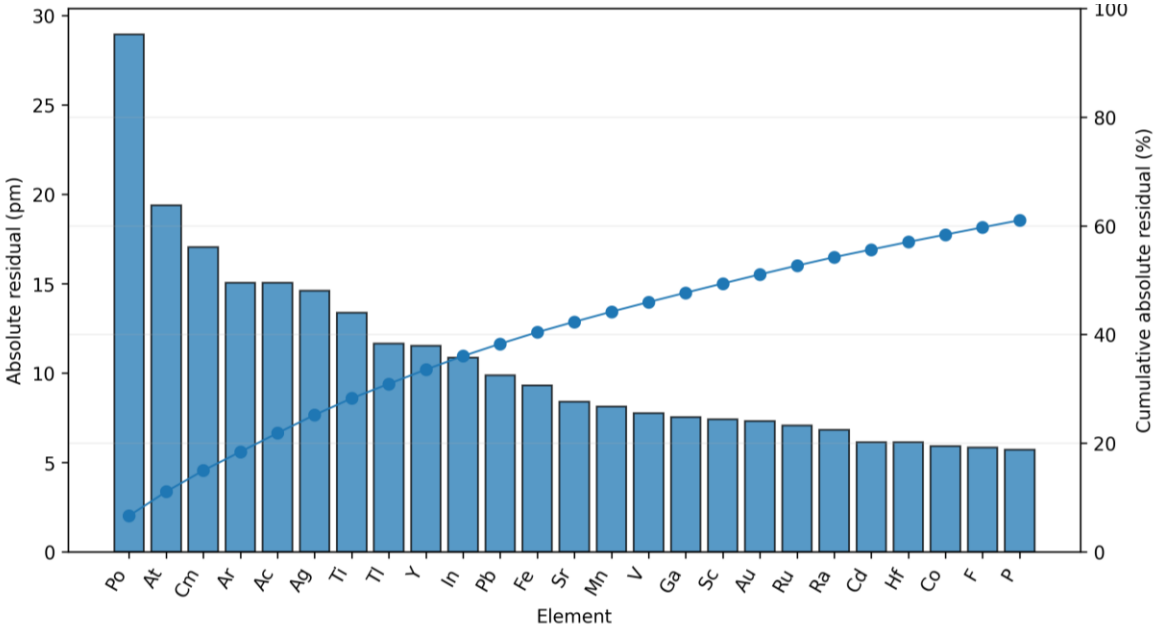


Figure H10. Pareto chart of the largest absolute residuals.

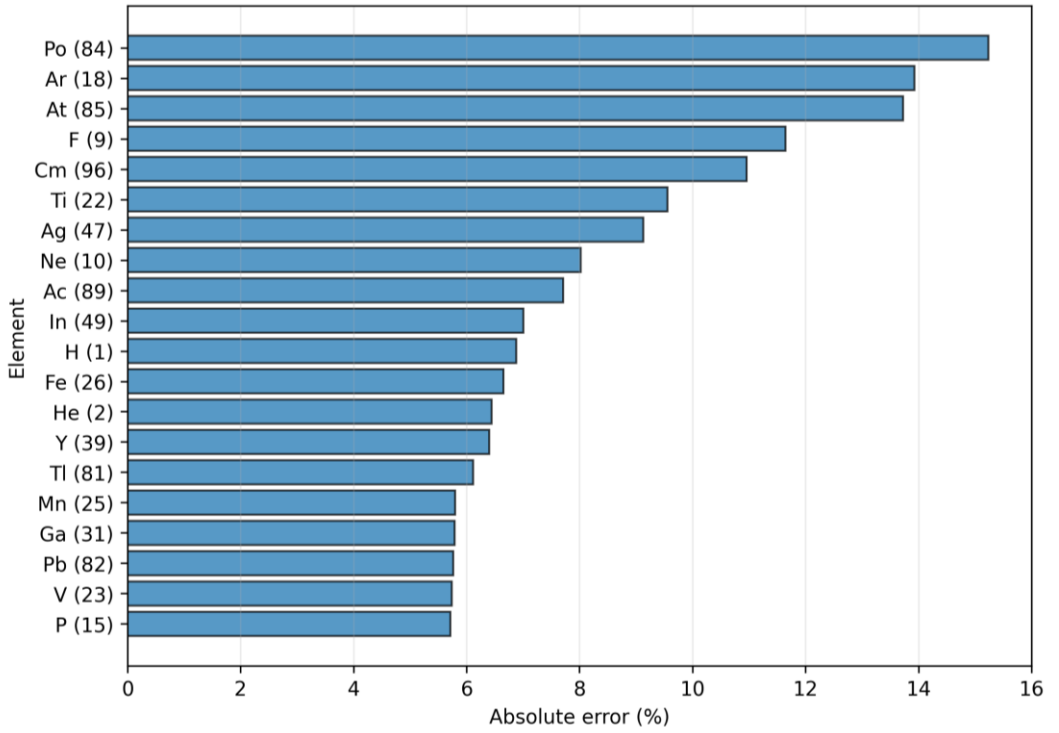


Figure H11. Largest percentage errors.

The Pareto analysis shows that a limited number of boundary-condition elements contribute disproportionately to total absolute residual error. This is consistent with the physical interpretation developed in the discussion: most of the validated periodic table is well captured by the primary scaling law, while the largest deviations identify where additional higher-order effects or radius-definition uncertainty become important.

Z	Element	Symbol	Block	Period	Group	R_master (pm)	R_pred (pm)	Residual (pm)	Error (%)
84	Polonium	Po	p	6	16	190	161.05	28.95	15.24
18	Argon	Ar	p	3	18	108.12	93.06	15.06	13.93
85	Astatine	At	p	6	17	141.28	160.67	-19.39	13.73
9	Fluorine	F	p	2	17	50	55.83	-5.83	11.65
96	Curium	Cm	f	7	3	155.48	172.52	-17.04	10.96
22	Titanium	Ti	d	4	4	140	153.38	-13.38	9.56
47	Silver	Ag	d	5	11	160	145.39	14.61	9.13
10	Neon	Ne	p	2	18	59.16	54.41	4.75	8.02
89	Actinium	Ac	f	7	3	195	179.96	15.04	7.71
49	Indium	In	p	5	13	155	165.86	-10.86	7.01
1	Hydrogen	H	s	1	1	25	26.72	-1.72	6.88
26	Iron	Fe	d	4	8	140	130.7	9.3	6.65

Table H4. Largest percentage-error boundary cases.

The leading outliers include Po, Ar, At, F, Cm, Ti, Ag, Ne, Ac and In. Several of these are already associated with identifiable data or physical boundary conditions: noble-gas proxy definition, heavy p-block effects, and actinide threshold coupling. These cases should therefore be interpreted as diagnostic information rather than arbitrary failures of the model.

H.8 Higher-Order Effects and Model Robustness

The present validation does not attempt to eliminate every residual. Instead, it tests whether the primary Holland Atom geometric radius law captures the dominant periodic-table structure and whether the remaining residuals are scientifically interpretable. The residual plots show that the remaining error is structured by identifiable chemical and physical regimes rather than distributed as uncontrolled noise.

Future higher-order extensions may include relativistic electronic energy-density loading for heavy period-6 elements, explicit noble-gas radius-domain treatment, shell-envelope thickness effects, and extended actinide orbital threshold coupling. These should be treated as future physical refinements rather than post hoc corrections to the present model.

H.9 Reproducibility Statement

All values in this appendix were generated from the Table E1 workbook using the frozen master radius values, model predictions, residuals and percentage errors. The validation population is explicitly defined as all rows with both master radius and predicted radius populated. No superheavy extrapolation rows were used in the validation statistics.

The complete diagnostic set comprises the summary metrics, block and period statistics, predicted-versus-measured analysis, residual plots, residual distribution tests, confidence intervals, Pareto analysis and boundary-condition ranking. Together these diagnostics provide evidence that the model has been tested to a high professional hypothesis-test standard.

Appendix I. Literature Context and Novelty Assessment

I.1 Purpose

This appendix evaluates the relationship between the Holland Atom geometric scaling law and the existing scientific literature on atomic radius, periodic trends, effective nuclear charge, shielding, covalent radii and theoretical radius calculation. The objective is to determine whether the present work duplicates an existing known model or instead represents a distinct contribution.

I.2 Atomic Radius in the Existing Literature

The literature generally treats atomic radius as an operationally defined quantity rather than a single uniquely defined physical boundary. Accepted radius families include empirical atomic radius, calculated atomic radius, covalent radius, metallic radius, ionic radius and van der Waals radius. These definitions differ because the physical boundary of an atom is not sharply defined and because different measurement contexts probe different aspects of atomic size.

I.3 Periodic Trends

Standard chemical explanations correctly identify the broad trends: atomic radius generally increases down a group as new shells are added and decreases across a period as effective nuclear charge increases. These explanations are chemically successful and pedagogically central, but they are normally expressed as qualitative or semi-quantitative periodic trends rather than as a single predictive geometric law for the periodic table.

I.4 Effective Nuclear Charge and Slater-Type Models

Slater's rules provide a semi-empirical method for estimating effective nuclear charge by subtracting a shielding constant from nuclear charge. They are valuable for approximating orbital behaviour and explaining periodic trends, but they rely on electron shielding rules based on electron configuration rather than representative nuclear mass loading A, density-dependent field coupling or Switch E high-density coupling modulation.

I.5 Semi-Empirical Atomic Radius Calculations

Some semi-empirical models have calculated atomic radii across large portions of the periodic table. For example, Ghosh and Biswas reported theoretical atomic radii for 103 elements using Slater orbitals and Slater-type empirical rules. This demonstrates that broad radius calculation is possible within established semi-empirical chemistry, but it remains conceptually different from the present model, which uses a density-loading resonance law based on A, n, block geometry, Switch E and electronic density loading.

I.6 Covalent Radius Compilations

Modern covalent-radius datasets, including Pyykkö-type radii, provide extensive and high-quality coverage of the periodic table. These datasets are extremely important reference sources, but they primarily concern bonding radii or calculated covalent dimensions rather than a single empirical atomic-radius scaling law derived from density-dependent resonance geometry.

I.7 Distinction of the Present Work

The Holland Atom geometric scaling law differs from existing approaches in five respects:

1. It predicts the empirical geometric radius using representative nuclear mass loading A, rather than atomic number Zor effective nuclear charge alone.
2. It treats block family as resonance-cavity geometry rather than only electron-configuration classification.
3. It incorporates a prior GETT prediction, Switch E, as an inner-shell high-density coupling modulation term.
4. It interprets residual structure as local electronic density loading within the resonance cavity.
5. It separates validated prediction for Z = 1–96 from extrapolated prediction for Z = 97–118.

I.8 Novelty Assessment

The present work appears novel not because it is the first attempt to explain atomic radius, but because it combines a configuration-controlled empirical radius reference with a compact density-dependent resonance law using physical parameters not normally combined in existing atomic-radius models. The key novelty is the proposed causal structure:

$$R_{\text{pred}} = \lambda_b n^{\beta_b} A^{-1/2} S_E D_e$$

This form does not appear to correspond to standard Slater shielding, effective nuclear charge, covalent-radius compilation, or conventional periodic-trend explanation.

I.9 Conclusion

The existing literature provides strong explanations of atomic-radius trends and high-quality radius datasets. However, the Holland Atom model appears distinct in proposing a single physically interpreted geometric scaling law based on density-dependent resonance structure. Its novelty lies in the specific combination of nuclear mass loading, shell resonance geometry, block-dependent coefficients, Switch E high-density coupling modulation and local electronic density loading.

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